schemata of assimilation, the greater the interest in novelty in general. New events have the more opportunities of animating at least one particular schema according as the ensemble of the schemata formed is large. For instance, interest in changes of perspective and displacements of objects, throwing, rolling, etc., takes root in many secondary circular schemata (shaking, swinging, rubbing, etc.) to which the new schemata are analogous but not at all identical. In this first sense, the progress of assimilation brings with it that of accommodation. Accommodation becomes an end in itself, separate from assimilation but complementary. We have already stated something similar in connection with vision: The more objects the child sees the more new ones he wishes to see. But, in the latter case, accommodation only forms one entity with the generalizing extension of as assimilatory schema, whereas henceforth accommodation exists before every true assimilation, and this accommodation is simply set in motion by earlier assimilations without being directly derived from them.

With regard to interest in the external result of acts, characteristic of the secondary circular reactions, it is also sooner or later a source of accommodation for the sake of accommodation. In effect, as we shall emphasize in connection with the concept of object and of causality, the very progress of assimilatory utilization of material objects results in substantiating them. For example, a hanging object which one can shake, swing, strike and finally drop, becomes little by little an independent center of forces and ceases to be simply an element of a self-enclosed cycle circumscribed by the schema of assimilation. Now at the time when causality thus becomes objectified and the universe becomes stocked with the centers of forces, it is apparent that the child's effort will no longer only consist in making things enter into known schemata but also, in the case of thwarting of the immediate assimilation, in discovering which are the properties of these centers of forces. For example, in Observation 145, it can clearly be seen how the attempts to "cause rolling" engender attitudes of expectation, surprise, and almost of anxiety and great astonishment [as when the stick rolls at 1;4 (0)] which reveals the progressive spontaneity that the child concedes to things. This is not yet the place to speak of people, to whom the

child naturally attributes a still greater spontaneity. In short, the objectification of causality is the source of experimentation. Here again, assimilation is protracted in accommodation and the latter becomes differentiated from the initial tendency which gave rise to it.

Thus can be explained how the growing complexity of assimilation brings with it the advent of an interest in novelty as such, that is to say, of an experimentation composed of accommodation henceforth differentiated. But must the assertion be made that this liberated accommodation will remain antagonistic to assimilation or that it will become more and more complementary? In studying the invention of new means through active experimentation we shall see how assimilation and accommodation are reconciled when there is a question of attaining a certain end. In such cases accommodation realizes what assimilation assigns as the goal of the action. But, from now on, when accommodation seems to remain in its pure state under the form of "experiment in order to see" it is possible to discern what close correlation it in fact maintains with assimilation.

The procedure of accommodation, in the case of the experiment in order to see, is that of groping. Now there is a number of distinct types of groping as we shall see when discussing Claparède's theory (see conclusions §4). Regarding the present case, let us limit ourselves to the following remark. Far from constituting pure groping which would furnish the prototype of accommodation without assimilation, the experiment in order to see consists in a sort of cumulative groping in the course of which each new attempt is directed by the preceding ones. So it is that when Jacqueline varies the perspectives of an object, throws or rolls what she holds in her hands, she doubtless experiments blindly at first, but directs increasingly her subsequent attempts; this is particularly apparent when she tilts up the box in Observation 146. Thereafter it can be maintained that, if the ensemble of the behavior pattern is due to a need for accommodation, the successive attempts are assimilated to each other as they occur. In this respect tertiary reaction is certainly a "circular" reaction despite the search for novelty which characterizes

it. Thereafter, if there is differentiated accommodation, it immediately evokes assimilation.

In the last analysis, in order to contrast these behavior patterns with the earlier ones, it must simply be said that in the "experiment in order to see" accommodation becomes differentiated from assimilation while directing it at each moment, while in the secondary circular reactions and the behavior patterns which are derived from them, it is the attempt at assimilation which controls and precedes accommodation. Moreover, in the earlier cases, accommodation remains simultaneously undifferentiated from and partly antagonistic to assimilation, whereas henceforth accommodation begins to become complementary to the assimilatory tendency from which it dissociates itself.

Finally, let us recall, in order to avoid ambiguity, that, even while in a sense preceding assimilation, the accommodation peculiar to the "experiments in order to see" is always the accommodation of a schema, and that the act of accommodating an earlier schema of assimilation consists in differentiating it according to the variations of the actual experiment. There is never, in effect, a "pure experiment." Even when he gropes in order to discover something new, the child only perceives and conceives of the real as a function of his assimilatory schemata. Besides, groping when confronted by a new experiment is always only an accommodation—but henceforth one which is desired and sought for its own sake—of these earlier schemata. Varying the perspectives, dropping or throwing, rolling, floating, etc., is, at the point of departure, a simple differentiation of secondary schemata such as displacing, swinging, etc. While preceding and henceforth directing new assimilations, accommodation always prolongs earlier assimilations. We shall see this still more clearly in connection with the following behavior patterns.

§2. THE DISCOVERY OF NEW MEANS BY EXPERIMENTATION. I. THE "SUPPORTS," THE "STRING" AND THE "STICK."—The "discovery of new means through active experimentation" (an abstraction drawn from the speed of de-

velopment) is to the tertiary circular reactions what the "application of familiar means to new situations" is to secondary circular reaction. The behavior patterns which we shall study thus constitute the highest forms of intellectual activity before the advent of systematic intelligence which implies deduction and representation. Furthermore, contrary to the acts of intelligence described in Observations 120 to 130, those acts which we shall now examine constitute inventions or at least real discoveries, already manifesting the constructive element peculiar to human intelligence. All the more reason, consequently, to examine these facts closely. Moreover, we shall analyze each of them separately, and shall ony group afterward the conclusions obtained.

The first manifestation of inventive intelligence which we observed in our children consisted in bringing distant objects closer by drawing to oneself the supports on which they were placed. We shall call this behavior pattern the "behavior pattern of the support" in contradistinction to that of the string or that of the stick. This kind of a behavior pattern, being at the same time the simplest of those of the fifth stage, will enable us from the outset, like all cases of transition, to understand the difference between the behavior patterns of the fourth stage and those of the present one.

In principle, nothing would prevent the "behavior pattern of the support" from arising during the fourth stage and, in fact, it sometimes appears sporadically during this period in the capacity of a simple coördination of schemata. But, as we shall see, its systematization requires more than such a coördination: It presupposes a special accommodation whose function we shall precisely try to understand. To do this, let us take as our point of departure the episodic situations in which the behavior pattern of the support is made manifest in the fourth stage. In such a case the child, trying to attain a too-distant objective, satisfies an ungratified need by applying the schema of "grasping" to the first object which appears and, when this object happens to be the support of the objective, he thus draws the latter to him. The schema of prehension of the objective is thus momentarily coordinated with the prehension of another object, just as, in Observation 121 repeated, the act of striking a doll attached to a

string is coördinated with the schema of striking a parrot attached to the other end, or again, as in Observation 127, the action exerted upon the hand of another person is coördinated with that which the child wishes to apply to the objective itself. But, if such an episodic coördination can give rise to some fortuitous success when the support is particularly mobile, it could not insure the formation of a stable procedure, and this is why. In the examples cited of behavior patterns of the fourth stage, the connections established by the child between the objects in play are always on a par with the coördination of the schemata, given the apparent or real simplicity of these relationships. So it is that, in order to push back an obstacle or utilize someone else's hand as intermediary, etc., the child has no connections to understand except those which are given either in the familiar schemata envisaged separately (the hand of another, for instance, is assimilated to his own), or in the very fact of their coördination (the connection implied in the act of removing the obstacle presupposes nothing more than the comprehension of an incompatibility between the presence of this obstacle and the action which the child wishes to exert on the objective). To put it more simply, the coördination of the schemata peculiar to the fourth stage does not involve any invention or construction of new "means." On the contrary, the relation which exists between an object and its support is a relation unfamiliar to the child at the time that the behavior pattern which we are going to describe makes its appearance.[2] At least the thing took place in this way with our children and that is why we class this behavior pattern in the fifth stage: If the relation "placed upon" had already been known to them (which could doubtless happen with other subjects), the behavior pattern of the support would only have been a matter of coördination of schemata and we would have classed it in the fourth stage. This relation thus being new to the child he will only come to utilize it systematically (in contrast to the fortuitous and episodic successes which have been under study) by understanding it, and he

2 The relation "placed upon" or relation between an object and its support could not, in effect, be discovered except through "tertiary circular reaction" (see Vol. II, Chap. II, §§3 and 4).

will understand it due only to active experimentation, analogous to that of "tertiary circular reaction." It is precisely in this respect that the behavior pattern which we are about to examine is new and differs from a simple coördination of schemata. But it rests upon that kind of coördination and it is even under the influence of this coördinating activity that the child applies himself to searching for new means by accommodating in the process of coördination to the unknown data of the problem.

In a general way, the "discovery of new means through active experimentation" involves, therefore, not only a coördination of familiar schemata (like the behavior patterns of the fourth stage that the present behavior thus protracts), but also a construction of new relations, obtained by a method similar to that of tertiary circular reaction.

Here are the facts:

Observation 148.—Until 0;10 (16) it can be said that Laurent has not understood the relation "placed upon," hence the relation existing between an object and its support. This is what we shall try to demonstrate at greater length in Volume II (Chap. II, Obs. 103) when studying the concept of space characteristic of the fourth stage.

1. With regard to the "behavior pattern of the support," numerous experiments repeated between 0;7 (29) and 0;10 (16) reveal that Laurent, until the latter date, has remained incapable of utilizing it systematically. At 0;7 (29) he has succeeded, once in four attempts, in drawing a cushion toward him in order to grasp a box placed upon it; at 0;8 (1) he behaves in the same way, as well as at 0;8 (7) etc. But there it is still only a question of a coördination of schemata, analogous to that of the fourth stage. Being unable to grasp the box directly, the child instead takes possession of the first object encountered, while subordinating this act to the persistent desire to attain the objective. The proof that this is the case is the existence of the following reactions: (1) When the support (the cushion, for example) is not within immediate reach of the child's hand (when it is 15 or 20 cm. away), Laurent does not try to reach it in order to draw the objective toward him, but tries to grasp the objective directly, and subsequently grasps the objects situated on this side of the support (for instance he pulls his covers or his sheets). (2) When I hold the objective in the air, 20 cm. above the support, Laurent pulls the latter toward him, as though the object were placed upon it. (3) When the support is placed obliquely, thus being at the child's disposition, not exactly in front of him but a little to the side (20 cm. from his waist), Laurent does nothing to reach it, trying to grasp

the objective directly or, failing this, the objects interposed between himself and the objective (the sheets, for example). For more details on these preliminaries, see Volume II, Observation 103.

II. At 0;10 (16) on the other hand, Laurent discovers the true relations between the support and the objective and consequently the possibility of utilizing the first to draw the second to him. Here are the child's reactions:

(1) I place my watch on a big red cushion (of a uniform color and without a fringe) and place the cushion directly in front of the child. Laurent tries to reach the watch directly and not succeeding, he grabs the cushion which he draws toward him as before. But then, instead of letting go of the support at once, as he has hitherto done, in order to try again to grasp the objective, he recommences with obvious interest, to move the cushion while looking at the watch. Everything takes place as though he noticed for the first time the relationship for its own sake and studied it as such. He thus easily succeeds in grasping the watch.

(2) I then immediately attempt the following counterproof. I put two colored cushions in front of the child, of identical form and dimensions. The first is placed as before, directly in front of the child. The second is placed behind, at an angle of 45°, that is to say, so that a corner of the cushion is opposite the child. This corner is placed on the first cushion but I manage to flatten the two cushions at this place, where one is partially superposed on the other, so that the second does not protrude and is not too visible. Finally I place my watch at the other extreme end of the second cushion.

Laurent, as soon as he sees the watch, stretches out his hands, then grasps the first cushion which he pulls toward him by degrees. Then, observing that the watch does not move (he does not stop looking at it), he examines the place where the one cushion is superposed on the other (this is still the case despite the slight displacement of the first one), and he goes straight to the second one. He grasps it by the corner, pulls it toward him over the first cushion, and takes the watch.

The experiment, when repeated, yields the same result the second time.

(3) I now place the two cushions next to each other, the proximal side of the second parallel to the distal side of the first. But I superpose the first on the second on a strip about 20 cm. wide (the watch of course being at the extremity of the second cushion). Laurent immediately pulls the first cushion but, observing that the watch is not displaced, he tries to raise this cushion to reach the second one. At a certain moment he has succeeded in raising up the first cushion, but without removing it, and he holds it against his chest with his left hand while trying to pull the second one with his right hand. He finally succeeds and takes

possession of the watch, thus revealing his perfect comprehension of the role of the support.

(4) Finally I place the second cushion as in (2) but sideways, the proximal corner of the second superposed on one of the distal corners of the first. Laurent does not make a mistake and at once tries to reach the second cushion.

These four reactions combined reveal that the relation between the objective and its support has been acquired.

Observation 148 repeated.—During the weeks that follow, Laurent rediscovers the same schema every time there is a question of drawing an object toward himself, and displaces the support following a rectilinear trajectory. On the other hand, the supports necessitating a rotary movement give rise to a new apprenticeship.

In Observation 143 it has been shown how Laurent, at 1;1 (24) had unsuccessfully tried to pull toward him one of the circular tiers of a table. The tier pivoting around an axis then began to turn slightly, instead of coming nearer. In order to strengthen the child's interest I immediately placed an interesting toy beyond his reach. The "experiment in order to see" was thus transformed into an effort entering the present group of behavior patterns.

At first Laurent looks at the toy without moving but never tries to reach it directly. Then he grasps the tier of the table and tries to draw it toward him in a straight line. The tier again turns by chance (a few degrees only). Laurent lets it go again, then recommences, and so on a certain number of times. That is only a series of attempts without being linked to each other which the child apparently considers failures. But he suddenly seems to perceive that the desired object comes closer. He again grasps the tier, lets it go and grasps it again until he has attained success. But the child's behavior does not yet give the impression that he has understood the role of rotation; he simply repeats a movement which was once efficacious, without turning the tier intentionally.

At 1;2 (6) Laurent is again confronted by the tier of the table and looks at a pebble which I place at the opposite extremity. He immediately tries to pull the tier toward him in a straight line, but only succeeds in making it turn on itself a few degrees. He then grasps it in the same way a certain number of times until he is able to reach the pebble. However one does not always have the impression that the child is turning the tier intentionally.

At 1;2 (7) on the other hand, Laurent only tries to pull the table once; he then definitely pivots the tier. From 1;2 (10) he tries at the outset to move the tier circularly in order to get hold of the objects out of reach. The schema appropriate to the situation accordingly is definitely acquired.

Observation 149.—As early as 0;9 (3), Jacqueline discovers by chance the possibility of bringing a toy to herself by pulling the coverlet on which it is placed. She is seated on this coverlet and holds out her hand to grasp her celluloid duck. After several failures she grasps the coverlet abruptly, which shakes the duck; seeing that she immediately grasps the coverlet again and pulls it until she can attain the objective directly.—Two interpretations are possible. Either she perceives the duck and the coverlet as a solidified whole (like a single object or a complex of connected objects) or else she simply satisfies her need to grasp the duck by grasping no matter what and so discovering by chance the possible role of the coverlet.

Until 0;11 Jacqueline has not again revealed analogous behavior. At 0;11 (7), on the other hand, she is lying on her stomach on another coverlet and again tries to grasp her duck. In the course of the movements she makes to catch the object she accidentally moves the coverlet which shakes the duck. She immediately understands the connection and pulls the coverlet until she is able to grasp the duck.

During the weeks that follow Jacqueline frequently utilizes the schema thus acquired but too rapidly to enable me to analyze her behavior. At 1;0 (19) on the other hand, I seat her on a shawl and place a series of objects a meter away from her. Each time she tries to reach them directly and each time she subsequently grasps the shawl in order to draw the toy toward herself. The behavior pattern has consequently become systematic; but it seems that it does not yet involve conscious foresight of the relationships since Jacqueline only utilizes this schema after having attempted direct prehension of the object.

Observation 150.—In the case of Lucienne, the same behavior pattern appeared at 0;10 (27). Seated on her bed Lucienne tried to grasp a distant toy when, having by chance moved the folded sheet she saw the object sway slightly. She at once grasped the sheet, noticed the object shake again and pulled the whole thing toward her. But as this reaction was too rapid to be properly analyzed, I devise the following:

At 1;0 (5) Lucienne is seated on her folding chair and in front of her a small tier A erected on the tier B of the table attached to the chair. The tier A only covers part of tier B. I spread a handkerchief on tier B in such a way that the frontal edge of the handkerchief is under tier A and cannot be grasped directly. Then I place a small bottle on the handkerchief. Lucienne then grasps the handkerchief at once, without any hesitation, and draws the bottle to her. The same occurs five or six times whether I place the bottle or my watch on the handkerchief. But the reaction again being so rapid, it is not even possible to know whether Lucienne has tried to grasp the object or whether the handkerchief itself attracted her. I repeat the experiment, but in the following way:

I place a handkerchief in the same way as before but instead of putting the object upon it I place it next to it, about 5 cm. away from the left edge of the handkerchief, also on tier B. Lucienne at once pulls the handkerchief, then tries to reach the bottle. Not succeeding, she then searches for the handkerchief, holds it for one or two seconds and then rejects it. Same reactions to my watch during a second attempt but she rejects the handkerchief still more quickly after having sought it.

I now increase the distance between the object and the handkerchief. I place the bottle 10–15 cm. to the side of the handkerchief. Lucienne then limits herself to trying to reach the object directly and no longer bothers with the handkerchief. When I bring the object nearer she looks alternately at the bottle and the handkerchief and finally, when I place the bottle on the handkerchief, she at once grasps the latter. It therefore seems that she has grasped its signification.—I repeat the experiment, again gradating the distances from 15 cm. to direct contact: same reactions.

This time I place my watch 15–20 cm. away from the handkerchief: Lucienne tries to grasp it directly. Then I stretch the chain between the watch and the handkerchief, leaving the watch at a distance of 15 cm. and putting the end of the chain on the handkerchief: Lucienne, who at first did not notice what I was doing, begins by trying to reach the handkerchief, then perceives the chain and then pulls the handkerchief![3] This last behavior pattern well shows that the prehension of the handkerchief is not a mechanical act.

Observation 150 repeated.—The same day, seeing a green bottle which is inaccessible but placed on a cover which is within reach, she at once pulls the cover in order to grasp the bottle (see Obs. 157).

Observation 151.—At 1;0 (16) Lucienne is seated before a big square cushion C, placed on the ground. Beyond cushion C is a second identical cushion D, so that Lucienne has two cushions facing her. I place my watch on D, the farthest from the child. Lucienne looks at the watch but does not try to grasp it directly. She takes cushion C and removes it at once, then pulls cushion D toward her and takes the watch.

At 1;1 (4) Lucienne is seated on an adult's bed in front of a cleaning rag placed on the sheet. As soon as I put my eyeglasses on the rag, Lucienne pulls it. When I place my glasses beyond the rag, she removes it at once and pulls the sheet toward her.

Observation 152.—At 1;0 (5), that is to say, right after the attempts described in Observation 150, Lucienne is confronted with a solid support and no longer a soft one (such as coverlets, shawls, or handkerchiefs. On tier B of her table I place a box with upturned edges (the cover of a large box, turned upside down), in such a way that the

[3] It is necessary to know that since 1;0 (3) Lucienne knows how to use the chain to draw the watch to her.

frontal side of the box is wedged under tier A and I place on the box, as far away as possible, the bottle or the watch which were used with the handkerchief (Obs. 150). I was thus able to observe seven sequential reactions:

(1) Lucienne at first tries to grasp the box, but she goes about it as though the handkerchief were still involved. She tries to pinch it between two fingers, in the center, and tries this for a moment without being able to grasp it. Then, with a rapid and unhesitating movement she pushes it at a point on its right edge (apparently Lucienne, unable to grasp it at the center, tried to roll it up or unstick it or simply to displace it slightly, and that is why she pushed it to the edge). She then notes the sliding of the box and makes it pivot without trying to lift it; as the box revolves, she succeeds in grasping the bottle.

(2) This time I place the watch at the extremity of the box. Lucienne again tries to grasp the box in the middle. Being unsuccessful, she gives up more quickly than in (1) and displaces the box by pushing it on the right edge.

(3) She no longer tries to grasp it at the center and immediately makes the support pivot.

(4) I place a new doll on the box in order to revive her interest: Lucienne again tries at first to make the box pivot. But not having brought it sufficiently near to her, she cannot grasp the object. She then returns to the right edge and pushes it more.

(5) Same play, with corrections at the center.

(6) She tries, doubtless for the sake of greater speed, to raise the box right away by grasping it in the same place as before, but by pulling it toward her instead of sliding it. Failing in this (the box is held by tier A), she gives up and resumes the pivoting.

(7) Same reactions, but Lucienne resumes the pivoting more promptly.

At 1;0 (11), that is to say, six days later, I repeat the same experiment with another box without edges (a simple box and no longer a cover). I also wedge this under tier A, and place different objects on the farthest extremity from Lucienne. She then manifests three sequential reactions:

In the first place, she tries to pinch the box in the center, as though it were a piece of material.

In the second place, she tries to lift the box by the right edge and so to draw it directly to her. This second attempt lasts several minutes because she always thinks she is near success.

In the third place, she finally returns to sliding. By gently pushing the right edge of the box, she makes it slide over tier B with the part wedged under tier A as center of the pivoting, and so succeeds in grasping the objects. In the subsequent attempts she adopts this last method right away.

The first examples show us at once in what the behavior pattern which we call "discovery of new means through active experimentation" consists. The whole situation is exactly the same as with respect to Observations 120 to 130, that is to say, the "application of familiar means to new circumstances": The child tries to attain a goal but obstacles (distance, etc.) prevent him. The situation is therefore "new" and the problem is to discover appropriate means. But, reversely from the behavior patterns mentioned (Obs. 120–130), no familiar method presents itself to the child any more. It is therefore a question of innovating. It is then that a behavior pattern intervenes which is analogous to that of the tertiary circular reactions, that is to say, an "experiment in order to see": the child gropes. The only difference is that, now, the groping is oriented as a function of the goal itself, that is to say, of the problem presented (of the need anterior to the act) instead of taking place simply "in order to see."

In the particular case, and without yet wishing to discuss the ensemble of the general problems raised by these observations, it is apparent that the groping leading to the discovery of new means presupposes an accommodation of familiar schemata to the present experiment. The accommodation as such is groping, but only the earlier schemata give meaning to what this groping discovers. When, for instance, Jacqueline, unable to catch her duck, grasps the coverlet instead and sees the duck shake, she would not understand this phenomenon at all if she were not accustomed to seeing objects move when a string is pulled (secondary schemata). But, knowing that intermediates can make it possible to act upon objects which cannot be directly grasped, she at once perceives a connection between the coverlet and the duck: impelled by the need to grasp the latter, she then haphazardly pulls the support and success ensues. In such a behavior pattern, then, there exists on the one hand a groping directed by the schema of the goal (to grasp the duck) and on the other hand an ensemble of significations attributed to intermediate events as a function of the earlier schemata and as a function of this very goal.

So also, when Lucienne tries to grasp an object on a box (Obs. 152) and discovers the possibility of making the box pivot,

it is certainly due to groping that the child comes to push the box by its edge, but the groping is double directed. It is directed at first by the schema assigning an end to the action. Wanting to draw to herself the object placed on the box and treating this like the handkerchief to which she is accustomed, Lucienne tries to grasp the box. Not succeeding immediately, she gropes, that is to say, tries to accommodate the schema to the present situation. It is then that she goes so far as to touch the edge of the box. In the second place, the groping is directed by the earlier schemata which give a meaning to the events arising by chance and again as a function of the goal of the action. Having touched the edge of the box, Lucienne sees it move and at once assimilates it to an object which can be displaced; she then pushes it in order to grasp the desired object.

Such is the nature of groping: As in the case of the tertiary circular reactions, it is an accommodation of earlier schemata which become differentiated as a function of the present experiment. But, in the particular case, accommodation, instead of being an end in itself, is only a means at the service of the pursuit of the goal.

Furthermore, groping, in which this accommodation consists, is cumulative, that is to say, each successive attempt constitutes a schema of assimilation with regard to the following attempts. When Lucienne discovered that it was necessary to push the box in order to draw the object to her, this means is rediscovered more rapidly each time in the course of the subsequent attempts. In this there is apprenticeship. Accommodation is therefore directed not only from the exterior (by earlier schemata), but also from within (due to this apprenticeship): It is therefore dually united to assimilation.

A second example of "discovery of new means through active experimentation" is the "behavior pattern of the string," which was so well studied by Karl Bühler: drawing an object to oneself by using its extension string, chain), etc.[4]

[4] Ch. Bühler and H. Hetzer (Kleinkindertests, in *Testing Children's Development from Birth to School Age*, translated by H. Beaumont, London, G. Allen & Unwin, 1935) believe this behavior pattern appears during the eleventh and twelfth months.

Observation 153.—It has been shown. in Observations 121 and 121 repeated, how Jacqueline made use of the strings hanging from the hood of her bassinet in order to pull desired objects to her. But it is not yet possible to compare these efforts to the behavior pattern which consists in pulling an object by means of a string. In the latter case the string is, in effect, considered as being the extension of the object, whereas in the first case the object is simply assimilated to the objects which can be swung by means of a string.

The true behavior pattern of the string began in Jacqueline's case at 0;11 (7). She was playing with a brush when, before her eyes, I attached this object to a string. Afterward I placed the brush at the foot of the armchair on which Jacqueline was seated, so that she was no longer able to see it (but the child could follow all my movements) and I left the end of the string on the arm of the chair. As soon as my preparations were finished, Jacqueline leaned toward the brush while stretching forth her hands. But not observing anything other than the string, she grasps it and pulls. The end of the brush then appears; Jacqueline at once drops the string in order to try to grasp the object directly. Of course the brush falls again, Jacqueline bends to look for it, rediscovers the string, pulls it again and lets it go once more when she perceives the desired object. The same series of operations ensues three or four times; each series ends in failure because Jacqueline lets go of the string as soon as she perceives the brush. However, when Jacqueline pulls the string she definitely looks in the direction of the brush and so expects to see it.

It is accurate to add that the child is still unaware of the role of gravity (see Obs. 144) and so, when he lets go of the string to grasp the brush he acts as though both were on a horizontal plane. The only real accommodation to the situation has been this: At a certain moment Jacqueline pulled the string with one hand when she perceived a very obvious knot not far from the brush (10–15 cm. away). The fraction of string between the knot and the brush appeared to her to constitute the extension of the brush. In effect, while she was still trying to pull the end of the string with her right hand, she tried to catch the knot with her left hand. As soon as the knot was grasped it served to pull the brush.

During the subsequent series of efforts the behavior pattern seems to have been acquired. I detach the brush and replace it, before Jacqueline's eyes, with a parrot; then I place this at the foot of the armchair and leave the other end of the string next to the child. Jacqueline grasps the string and, hearing the sound of the rattle inside the parrot, immediately pulls while looking beforehand at the place where the latter must appear. When she sees the toy she tries to grasp it with one hand while still pulling the string with the other. During the subsequent attempts, same reactions and same success.

Third series: I replace the parrot with a book. Jacqueline pulls the string while staring at the place where the object will appear. As soon as she sees it she manages to grasp it. Same reactions with a clothespin and a safety pin.

Observation 154.—At 1;0 (7) Jacqueline is seated in her bassinet whose handle is supported by a table facing the child. I show Jacqueline her swan whose neck has a string attached to it, then I put the swan on a table while leaving the string in the bassinet. Jacqueline grasps it immediately and pulls it while looking at the swan. But as the string is long she does not stretch it out but is limited to waving it. Each shake of the string makes the swan move but it comes no nearer.

After many attempts of the same kind I move the swan farther away which results in stretching the string. Jacqueline still shakes it, without really pulling it. The swan falls; Jacqueline holds onto the string, pulls it, but as the swan does not come at once, she resumes shaking the string.

New attempt: Jacqueline shakes harder and harder which results in making the swan move forward a little. But she tires of this and gives up.

At 1;0 (8), the next day, I resume the experiment. At first Jacqueline shakes the string, then pulls it. When the swan is near enough she tries to reach it directly with her hand. When she does not succeed she gives up instead of resuming pulling. The following days, same reactions, but it seems that she shakes the string less each time and pulls it more.

Finally, at 1;0 (19) Jacqueline draws the object to her correctly by pulling the string but she never does it without shaking it beforehand as though that were necessary. Only ten days later does she pull it right away.

Observation 155.—At 1;0 (26) in Jacqueline's presence I place my watch on the floor, beyond her field of prehension. I put the chain in a straight line in Jacqueline's direction but place a cushion on the part which is nearest the child. Jacqueline at first tries to grasp the watch directly. Not succeeding, she looks at the chain. She notes that the latter is under the cushion. Then Jacqueline removes the latter at one stroke and pulls the chain while looking at the watch. The movement is adapted and quick. As soon as the watch is within reach, Jacqueline lets go of the chain in order to grasp the object directly. There is, consequently, no interest in the chain itself; it is the watch that is wanted.

Same reaction several times in succession in a variety of conditions.

In the same way, Lucienne, at 1;0 (3) and the days following, looks at the watch as soon as she perceives the chain around her mother's neck. When she finds a necklace she confines herself to grasping it,

whereas sight of the chain always sets in motion search for the watch and the action of pulling.

Observation 156.—In a single day Laurent acquired the "behavior pattern of the string" but he only succeeded in this through "active experimentation" and not through immediate comprehension or mental construction.

It is noteworthy that until 0;11 despite his earlier utilizations of hanging strings I was not able to cause Laurent to reveal the existence of any tendency to make use of extensions of the objective in the capacity of intermediates or of "strings." Thus at 0;8 (1) after having played with my detached watch chain he does not have the idea of using it to draw to him the watch he wants once the chain is again attached. He stretches out his hand toward the watch only and neglects the chain which I place between him and the watch. Various analogous attempts either with the same chain or with strings attached to several objects yielded nothing until 0;10.

On the other hand, at 0;11 (16) Laurent manifests the following behavior pattern: He is seated on a dark rug. I show him a red object (a shoehorn) hanging from a string, then I place this objective about a meter from him while making the string describe a winding trajectory culminating next to the child. But Laurent, instead of using the string to attain the objective, confines himself to stretching out his hands toward it. I displace the string several times to draw Laurent's attention to it but each time avoid stretching it out in a straight line between the child and the objective. Each time Laurent looks at the string but without utilizing it, again trying to grasp the object directly.

Then I stretch the string in a straight line but making it end next to Laurent and not yet in front of him. He nevertheless reacts as before; that is to say, he still does not pay attention to the string and tries to reach the objective directly (it must be said that I displace the latter slightly, at each new attempt, in such a way as to revive the child's interest).

Finally I give the string once more a sinuous shape but this time making it end in front of Laurent. Laurent, after trying twice to grasp the objective directly, takes possession of the string. He does not try to stretch it but confines himself to looking at it while shaking it slightly. He therefore took it simply for its own sake because of having been unable to reach the shoehorn he wanted; he does not yet understand at all the relations which exist between the two. But, in shaking the string he perceives that the shoehorn moves. He shakes the string more and more vigorously while attentively observing the movements of the shoehorn.

This last behavior pattern still does not differ in any way from the secondary circular reactions such as those in Observations 94–104:

pulling a string in order to shake the bassinet hood, etc., when the child has just discovered by chance the effect thus produced. But, having discovered the possibility of acting upon the shoehorn by means of the string, Laurent returns to his initial desire, which is to attain the object. Instead of shaking the shoehorn in every direction, he then seems to pull the string intentionally and thus gradually brings the objective nearer to him. Once he has grasped it, I repeat the experiment several times in succession. Every time Laurent at first grasps the string, shakes it a moment and then pulls it more or less systematically.

But this last behavior pattern does not, it seems to us, yet constitute an authentic example of "behavior pattern of the string." In effect, while already attaining his ends, Laurent still believes himself to be obliged, before pulling the string, to shake it for a moment, and he contrives all the transitions between the act of shaking and that of pulling. In other words, he utilizes an already acquired schema toward a new end. The action thus carried out therefore still remains at the level of the actions of the fourth stage, that is to say, the stage of the coördination of schemata. At 0;6 (1) Laurent has already revealed a very similar behavior pattern (Obs. 120).

How does the child transcend this stage of simple coördination of schemata to arrive at the discovery of the role of the string? During subsequent attempts, I make the string describe an increasingly winding trajectory so that Laurent, in shaking one end of it, does not at first succeed in moving the shoehorn. He nevertheless still tries, once or twice, to shake the string. But each time he applies himself more promptly to pulling it. It is difficult to describe in detail, without the aid of a film, how the movement of traction is learned. But, in the main, one can say that there was groping by progressive correction; the child eliminates from his earlier schemata the movements consisting of shaking and develops those which result in pulling. Very rapidly, despite the complications I introduce into the experiment, Laurent succeeds in finding the best procedure. He pulls the string with each hand alternately and so attains the objective with only a few movements.

Observation 156 repeated.—An hour later I put Laurent on a sofa and place the same red shoehorn on a chair, facing him. The string to which this objective is attached hangs from the chair onto the floor, then goes up the sofa next to the child. He looks at the objective for a moment, then his eyes follow the string, he grasps it and pulls it first with one hand, then with the other. When the shoehorn disappears from his visual field he nevertheless continues his maneuver until he has been completely successful.

I then present several objects to him (books, toys, etc.) but out of reach and attached to ribbons, cords, etc. (different from the string hitherto involved). Moreover, I vary the trajectories of these inter-

mediates in such a way as to avoid any visual suggestion. Laurent succeeds, however, in all these attempts with almost no groping; the "schema of the string" has therefore been acquired.

The following days I regulate the thing with several new objects. Laurent immediately uses both hands to draw them to him by means of the cords to which they are attached. At first he looks at the objective, then seeks the proper intermediate.

Let us resume, in connection with this "behavior pattern of the string," the discussion of "supports." The child's behavior, which once more consists in finding a procedure to draw distant objects to him is constituted here also by a groping accommodation doubly directed by the schemata of assimilation. It is important to determine exactly the role of this accommodation and that of assimilation. It is the problem of the relations between experience and intellectual activity of which we shall find a particular aspect once more.

Accommodation is necessarily the adjustment of earlier, already constituted schemata to new circumstances. It is in this sense, at first, that it is directed by assimilation. It is directed by the schema assigning an end to the present action as well as by certain schemata serving here as means and which accommodation will differentiate. Confronted by a brush attached to a string, for example, Jacqueline wishes to grasp this brush and, to do so, utilizes once more the schema of objects hanging from the hood and from which hangs a string. We recall, in effect, that she has already made use of these strings to grasp objects attached to them (Obs. 121 repeated). Hence she pulls the string in order to attain the brush. But, in acting thus, Jacqueline still only considers the string as being a magic-phenomenalistic procedure and not at all an extension of the object [see Obs. 153, at 0;11 (7)].

In effect, when she sees the brush appear, she forgets about the string, tries to grasp the object directly and fails. It is then that actual accommodation and groping begin. The experiment shows the child that his earlier schema is not adequate and Jacqueline finds herself obliged to find the true relations joining the string to the object attached. The same applies exactly to Observation 154: Jacqueline shakes the string as though a string

hanging from the hood were involved, then, observing her failure, she must accommodate herself to the new situation.

How does this accommodation work? Through tertiary circular reaction. In Observation 153, Jacqueline tries new combinations; at first she grasps the string at a visible knot and so succeeds in catching the brush, or else she pulls the string more and more until she is able to attain the parrot and hand it over. In Observation 154 she shakes the string less and less and stretches it more and more, etc. Hence there is experimentation and utilization of this experimentation. But how can this dual capacity be explained?

As far as accommodation in the capacity of experimental contact with the given reality is concerned, there is nothing to explain if not that in searching, the child hits upon facts. This happens by chance and the facts obtrude to the extent that they refute the expectation due to earlier schemata. All that we have said regarding interest in novelty in connection with tertiary circular reaction easily applies here. Lying in wait for the new experience, the child encounters it to the extent that he no longer tries to force reality to enter into his earlier schemata.

With regard to what is, on the other hand, utilization of the experience, groping accommodation must once again be directed by assimilation, but in a second sense. This time it is directed by the schemata capable of supplying a meaning to the events arising by chance, these schemata being themselves subordinated to the schema which assigns an end to the ensemble of the action. The events which arise in the course of the experiment could not be apprehended by the subject's consciousness except as a function of the earlier schemata of assimilation. For example, when Jacqueline discovers that by pulling and stretching the string she draws to her the object attached to it, she necessarily assimilates this fact, however new to her, to familiar schemata. She "understands" that the string is a "means for bringing," that is to say, she classes it among the other "means for bringing" such as the "supports," etc. The vicissitudes of searching, therefore, only acquire meaning as functions of the schema of the goal pursued and of the schemata which were earlier in relationship with this precise goal.

In short, accommodation is directed by two kinds of assimilation: by the "initial" schemata (the schema of the end and those of the means) which it is precisely a question of adjusting to the new situation, and by the schemata evoked on the way (let us call them "auxiliary" schemata) which give meaning to the products of experience or accommodation, and that again as function of the goal of the action. But then, do these products of accommodation present no new aspect to the child's eyes? In other words, by dint of being interpreted, does the new experiment seem from the outset as though it were already familiar? There is of course nothing to this, precisely since accommodation upsets and differentiates all the schemata which direct it, as we have noted in connection with tertiary circular reaction.

How should this acquisition be regarded? Here apprenticeship intervenes, that is to say, the cumulative element of groping. While being directed or oriented by the earlier schemata of assimilation, accommodation (hence experience) makes them flexible, differentiates them and so precedes, this time directing a new attempt at assimilation. This assimilation, interior or immanent in successive acts of accommodation, is the apprenticeship: Each attempt constitutes a mold for the next one, hence the embryo of an assimilatory schema. So it is that after having learned to pull the string by stretching it, Jacqueline pulls it better and better. The three series of Observation 153 and the successive series of attempts described in Observation 154 demonstrate this progress very well.

Now it is not a play upon words again to speak of assimilation to characterize this progress immanent in accommodation. Apprenticeship is nothing else but a circular reaction proceeding by reproductive, recognitory and generalizing assimilations. As we have seen at the beginning of this chapter, it is only because the complexity of the schemata of assimilation henceforth permits intentional searching for novelty for its own sake, that circular reaction is "tertiary," that is to say, oriented toward accommodation as such.

In short, one understands how extremely complicated is that which associational empiricism considered a primary factor: the contact with experience. The contact, that is to say, the ac-

commodation, is always inserted between two (or even three) series of assimilatory schemata which frame it: the schemata (initial or auxiliary) which give direction to accommodation and those which register its results by thus letting themselves be directed by it.

Let us note, finally, that once the new schema is acquired, that is to say, once the apprenticeship is finished, this schema is applied from the outset to analogous situations. Thus it is that in Observation 155 the "behavior pattern of the string" is without any difficulty applied to the watch chain. Thus at each acquisition we fall back on the application of "familiar means to new situations" according to a rhythm which will extend to the beginning of systematic intelligence (Chap. VI).

A third "discovery of new means through active experimentation" will permit us to make this analysis still more precise. This is the "behavior pattern of the stick." The string is not an instrument; it is the extension of the object. On the contrary, the "stick" is an instrument. How is the command of this first tool acquired? Possibly this occurs by a sudden mental construction when the child only discovers the stick late, at the level of systematic intelligence (see Chap. VI, §1). Or else it might occur by means of groping and active experimentation. Lucienne and Jacqueline have furnished us with the example of the latter procedure, Lucienne by acting in an entirely spontaneous way, Jacqueline aided by imitation. We shall emphasize here the case of Lucienne, that of Jacqueline merely serving as supplementary means of analysis:[5]

Observation 157.—At 1;0 (5) Lucienne already possesses the "behavior pattern of the support," as has been seen in Observations 150 and 152. I try to determine, the same day, if she is capable of that of the stick. One will see that she is not.

The child is playing with a very elongated cover which can fulfill a stick's function; with it she hits the tiers of her table, the arms of her chair, etc. Then I place before her, out of reach, a small green bottle for which she immediately has a strong desire. She tries to grasp it with outstretched hands, struggles vigorously, wails, but does not have the idea of using the cover as a stick. I then place the cover between

[5] According to Ch. Bühler and H. Hetzer (*op. cit.*, p. 63) the behavior pattern of the stick normally appears during the second half of the second year.

her and the bottle: same lack of comprehension. Then I place the bottle at the end of the cover: Lucienne pulls the cover to her and grasps the bottle as we have observed in Observation 150 repeated. Then I again put the bottle out of reach, but this time I place the cover next to the object and at the child's disposal; nevertheless it does not occur to Lucienne to use it as a stick.

At 1;2 (7), on the other hand, Lucienne happens to make a notable discovery: While playing at hitting a pail with a stick she is holding (all this without preliminary goals) she sees the pail move at each blow and then tries to displace the object. She strikes it more or less obliquely to augment the movement and does this many times; but she does not utilize this finding to bring the pail nearer to her nor to move it in a certain direction.

Observation 158.—At 1;4 (0) Lucienne is seated opposite a sofa on which is placed a small aluminum flask. Next to it lies the same stick as before with which she amused herself of recent weeks, using it to hit objects and the floor, but without progress since 1;2 (7). At first, she tries to grasp the flask directly, with her right hand. Not succeeding, she takes the stick. This behavior constitutes an important novelty: The stick is no longer only utilized when it is already in hand, it is sought for its own sake. Moreover, having grasped it by the middle and observing, after trying it out, that it is not long enough, Lucienne switches it to her other hand, then takes it again in her right hand, this time by the end. But the rest of the observation shows that the stick is not yet grasped with the purpose of pushing the flask; Lucienne merely hits the object and this does not authorize us to envisage that as prevision of its falling. The flask, however, falls and Lucienne picks it up. It is clear that the desire to attain the flask aroused the schema of striking by means of a stick, but one cannot therefore see in this behavior pattern a procedure already adapted to the particulars of the situation.

A moment later, on the other hand, I place the flask on the floor, 50 cm. away from Lucienne. She begins by wanting to grasp it directly, then she takes the stick and hits it. The flask moves a little. Then Lucienne, most attentively, pushes it from left to right, by means of the stick. The flask is thus brought nearer. Lucienne again tries to grasp it directly, then takes the stick again, pushes it once more, this time from right to left, always bringing the object towards her. Delighted, she grasps it, and succeeds in all the subsequent attempts.

Observation 159.—It has been seen before (Obs. 139) how Jacqueline, at around 8 months, made objects swing by "derived circular reaction." In her case this behavior fortuitously prepared the behavior pattern of the stick. At 1;0 (13) Jacqueline holds an elongated rattle when she perceives the tail of a leather donkey which hangs in front of her. She immediately tries to make it swing. But, having the rattle in her hand, it

is not her hand but the rattle which she directs toward the donkey. She thus shakes its tail and repeats the experiment many times. One cannot yet speak of the behavior pattern of the stick in this connection. The rattle has not been grasped in order to act upon the object but was used by chance in the capacity of a chance extension of the hand. As this behavior did not recur during the following days, I tried to reconstruct an analogous situation, making use of imitation, not in order to study the latter but in order better to analyze the mechanism of the acquisition. At 1;0 (28) Jacqueline tries to attain a cork placed before her at eye level but out of reach. In her right hand she holds a stick but does not use it and tries to grasp the cork directly with her left hand. I then take the stick and make the cork fall and Jacqueline at once grasps it. Then I put the cork back in place and return the stick. Jacqueline, who has watched me most attentively, at once repeats my movement with precision. She directs the stick toward the cork and makes it fall.

At this point of the experiment two explanatory hypotheses present themselves to us and it is in order to decide between them that we forced things by causing the factor of imitation to intervene: either when imitation set in motion a sort of already prepared "structure," the child would henceforth apply the latter without any groping, or else when imitation limited itself to setting an example, the child would subsequently grope to rediscover it, in the way in which Lucienne groped when confronted only by objects. The rest of the observation shows that this second solution is the right one.

I put the cork on the edge of the bassinet. The stick is placed next to the child. Jacqueline stretches her arms toward the cork, groans with disappointment and ends almost by crying, but always without thinking of grasping the stick. I show it to her, raising it and resting it in front of her, before her eyes, but she does not take it and continues to try to grasp the cork directly.

New attempt: I offer her the stick; she grasps it and immediately directs it toward the cork which she causes to fall and grasps. The fact that she has the stick in her hand makes her reproduce, by circular reaction, the movement imitated before, but the capacity to perform this movement does not suffice to enable the child to rediscover it and utilize it when he does not hold the stick and he sees it lying before him.

During the three following attempts the same result ensues. Jacqueline continues to wish to attain the cork directly and only uses the stick when I offer it to her. I then interrupt the experiment for a moment.

On resuming it she reveals progress. Jacqueline again tries to grasp the object, she still does not look for the stick which is, however, in front of her within reach and in her visual field, but when I point to

it with my finger, she grasps and uses it. Same reaction five times in succession.

A final series: She still tries to reach the cork with her hand (although the cork remains in the same place) but, after a moment's wailing, seeks the stick by herself, in order to use it immediately.

It is noteworthy that, during these attempts, Jacqueline has revealed a sustained interest, that she constantly wailed or even cried in the event of failure (when her hand did not attain the cork) and that each time her expression changed and her lamentations ceased when she understood the role of the stick; in the beginning, when I put it in her hand, then when I pointed to it with my finger and finally when she recalled its use by looking at it. Thus it may be seen that the dynamic schema outlined by the initial imitation only incorporated optical factors little by little, that is to say, it was slow to confer a signification to the visual spectacle of the stick.

Observation 160.—The next day, at 1;0 (29) I present the same cork to Jacqueline, at the same place and putting the stick in front of her, without hesitation she grasps the latter and directs it toward the cork. But she observes while doing this that the stick is too short (she grasped it three quarters of the way down). She transfers it to her other hand by grasping its end. Still without hesitation, she holds it out toward the cork and grasps it and it falls within reach.

After playing with the cork for a moment, I take it back from her and again place it out of reach of her hands. Jacqueline at once looks for it on the floor (she is seated), but, instead of grasping the stick which she sees, she takes her picture book (made of cloth) and directs it toward the cork. The book folds and does not reach the goal. Jacqueline wails but perseveres in about ten attempts. Afterward she puts it down, tries with her hand alone, then with the stick (grasping it so that it is too short); she rejects it and takes a rubber banana. As this is still too short, she removes it after several fruitless attempts and returns to the stick. She finally succeeds.

It may therefore be seen that the behavior pattern of the stick has been acquired and immediately generalized even with respect to flexible objects.

Observation 161.—At 1;1 (0) Jacqueline tries to attain a plush cat located on the wood of her bassinet outside her field of prehension. She gives up after a series of fruitless attempts and without thinking of the stick. I then put my finger 20 cm. above the latter. She perceives the stick, grasps it at once and makes the cat fall. At 1;1 (28) she is seated on the floor and tries to reach the same cat, this time placed on the floor. She touches it with her stick but without trying to make the cat slide to her, as though the act of touching it sufficed to draw it to her.

Finally, at 1;3 (12) she discovers the possibility of making objects slide on the floor by means of the stick and so drawing them to her; in order to catch a doll lying on the ground out of reach, she begins by striking it with the stick, then, noticing its slight displacement, she pushes it until she is able to attain it with her right hand.

These observations make it possible, it would seem, to progress a step further in the analysis of accommodation. But first let us emphasize what they have in common with the preceding ones.

The behavior pattern of the stick, like the behavior patterns of the support and of the string, arises by differentiation of earlier schemata. The desire to strike or swing objects fortuitously reveals to the child the power of the stick when by chance the latter extends the action of the hand. Observation 157 and the beginning of Observation 159 thus show us what makes ready the behavior pattern of the stick. Thereafter, when the child aims to reach an object situated outside his field of prehension, it is natural that his desire should arouse the schemata in question (due to the mechanism of the coördination of schemata, existing from the fourth stage); the beginning of Observation 158 shows us this. At its point of departure, accommodation is therefore directed by the schema of the goal (to grasp the distant object) in the same way as by the schemata coördinated with it (striking, etc.) and serving as "means." But it is a question of accommodating these schemata to the present situation. It is not enough to strike an object with a stick in order to draw it to oneself and it is necessary to discover how to give an object an appropriate movement. It is then that accommodation begins.

Let us again note that this accommodation, as in the case of the supports and of the string, is conditioned by a series of earlier schemata which give a signification to the successive discoveries. So it is that the child, as soon as he sees the object being displaced a little under the influence of the stick's blows, understands the possibility of utilizing these displacements with the view of drawing the object in question to him. This comprehension is not only due to the "initial" schemata which are at the root of the subject's searching (schema of grasping and that of striking) and of which the present accommodation constitutes

a differentiation, but it is also due to the "auxiliary" schemata which join themselves to the former. It is doubtless because he already knows how to displace objects by means of the supports and the string that the child understands the signification of little displacements due to blows of the stick.

But how does this accommodation work, that is to say, this differentiates for that purpose (striking); this conjunction conmulative process will give rise to a new assimilation? Here the observations relating to the stick enable us to transcend the conclusions obtained by analyzing the behavior pattern of the support and that of the string. We have seen, in connection with the string, that the acquisition of novelties, that is to say, apprenticeship, consisted in a tertiary circular reaction itself operating through reproductive, recognitory and generalizing assimilations. Accommodation to old schemata thus gives rise to new schemata capable, as such, of assimilation. But how is that possible? The observation of the behavior pattern of the stick will now show us.

Three solutions are conceivable. Either the differentiation of the old schemata, which accordingly constitutes accommodation (in the particular case, the transformation of the schema of "striking" into a new schema of "displacing with the stick") consists in a sort of dislocation of this schema; that is to say, there would be simple, undirected groping, entailing by chance variations on the general theme of the schema. In this first solution, attainment of the end would have to be conceived as being a selection after the event of chance variations. The second solution would consist, on the contrary, of assuming an immediate reorganization of the schemata: the schema of "striking with a stick" coördinated with that of "grasping" or "bringing to oneself" would, at a given moment, suddenly give rise to the system of "bringing to oneself with a stick." This sudden crystallization would therefore be comparable to those reorganizations of the ensemble of the field of perception which Gestalt psychology makes the essence of intellectual invention. In the third place one could posit an intermediate solution which would not at all consist of a mixture of the two others or a compromise but which would introduce a factor of directed activity. The schema serv-

ing as means (striking, swinging, etc.) would become differentiated as a function of the final schema (bringing to oneself) and consequently would be directed by it, but this conjunction of two schemata instead of giving rise to a sudden reorganization would simply engender a series of cumulative efforts, that is to say, a progressive accommodation whose every term would be assimilated to the preceding terms while being oriented by the ensemble of the conjunction. The originality of this third solution, in relation to the second one, would therefore be that the new schema would not be structured at the outset but would remain in a state of structuring activity until the time when it would have assimilated to itself the ensemble of the situation.

These three solutions having been separately delineated, it is clear that the third is the only one which conforms to Observations 157–161, as well as to the preceding ones. The first solution must be withdrawn because in such cases the child's groping never consists in a series of acts performed by chance. On the one hand, accommodation is enclosed in the schema of the goal (to bring to oneself) and that which serves as means which it differentiates for that purpose (striking); this conjunction consequently reduces chance to small proportions. On the other hand, each attempt conditions the attempts that follow and depends on the preceding ones. Doubtless chance can sometimes play a role in discovery. So it is that in Observation 157, Lucienne perceives that by hitting a pail she displaces it. But this discovery which characterizes a pure tertiary circular reaction (but which we class here because it could just as well occur in the course of seeking to bring things to oneself) is immediately assimilated and at once conditions the subsequent attempts. Chance, therefore, in the accommodation peculiar to sensorimotor intelligence, plays the same role as in scientific discovery. It is only useful to the genius and its revelations remain meaningless to the unskilled. In other words, it presupposes a directed search and is incapable of orienting it by itself.

The second solution is more satisfactory. But it encounters the difficulty that, in our observations, accommodation is not immediate. What is essential appears as being not the structure in which this accommodation results, but the structuring activity

which makes possible its culmination. In this respect Observation 159 is very instructive. By presenting Jacqueline with an already structured example to imitate with regard to the behavior pattern of the stick I must have, it would seem, evoked in the child's mind an immediate comprehension of the use of that instrument. Jacqueline imitates me unhesitatingly, with interest and precision, giving the impression that she would now be able to repeat the same behavior pattern indefinitely. Now the rest of the observation shows that the schema aroused by imitation remains simply in the state of tendency or dynamism and does not give rise at the outset to a reorganization of perception. During the attempts immediately following the imitation, the sight of the stick does not suffice to set in motion its utilization and Jacqueline must already hold it in her hands in order to rediscover its signification which she can then do without any difficulty. Subsequently, on the other hand, visual elements are incorporated into this dynamic schema, very slowly and progressively. First I must point to the stick with my finger in order that the stick be used, then the sight of it alone suffices to produce this result.

We can conclude from such observations that accommodation characteristic of the discovery of new means operates due not to a sudden reorganization but due to a series of cumulative attempts being assimilated to each other and thus entailing the formation of a schema which assimilates to itself the ensemble of the situation (including, little by little, visual elements). It may thus be seen how, as we glimpsed in connection with the "supports" and the "string," accommodation is directed not only from the outside by the coördination of the final schema (the schema assigning an end to the action) and of the initial schemata serving as means, schemata that accommodation differentiates, not only by the auxiliary schemata which give meaning to the discoveries made by this accommodation, but that it is also and primarily directed by an assimilation immanent in accommodation and resulting from it just as circular reaction results from the novelties which give rise to it.

It is noteworthy that, as in connection with the "string" and the "supports," the new schema, as soon as it has been ac-

quired, is applied by generalization to analogous situations, the behavior pattern thus entering the group which we have called "applications" of familiar means to new situations. So it is that, in Observation 160, Jacqueline, knowing how to use a stick without hesitation, also uses a book and a banana as instruments.

§3. THE DISCOVERY OF NEW MEANS THROUGH ACTIVE EXPERIMENTATION. II. OTHER EXAMPLES.—The analysis we have just attempted to make concerning the accommodation characteristic of the discovery of new means can now be extended by the study of more complex behavior patterns. We shall first try to find out how the child goes about drawing objects to him through the bars of his pen. This kind of an experiment is such as to permit us to examine the relations between the dynamic schemata and visual perception or representation.

Observation 162.—At 1;3 (12) Jacqueline is seated in her playpen, that is to say, in a square enclosure whose four sides are formed by vertical bars connected at base and summit by a horizontal bar. The bars are 6 cm. apart. I place outside the pen, parallel to the side where Jacqueline is, a stick 20 cm. long which takes up the distance of about 3 spaces between the bars. We shall call these three spaces *a*, *b* and *c*, space *b* corresponding to the middle part of the stick and spaces *a* and *c* to the end parts. The problem is to transfer this stick from outside to inside the pen.

1. Jacqueline begins by grasping the stick through space *b*, she raises it along the bars but holds it horizontally and parallel to the frame so that the harder she pulls the less it moves. She then extends her other hand through *c*, but holds the stick horizontally and does not succeed in making it come through. She finally lets go of the object which I put back in its initial position.

2. Jacqueline at once begins over again, by again grasping the stick at *c*. But, in raising it, she tilts it up a little, by chance, and so makes it slightly oblique. She immediately takes advantage of what she perceives and, passing her hand through *c*, she tilts the stick until it is sufficiently vertical to pass through. She then brings it into the pen through *b*.—Why did she tilt it up? Was it through foresight or did she simply extend the movement which was due to chance so as to see what would happen? The rest of these attempts rather substantiate this second interpretation.

3–4. This time Jacqueline grasps the stick through space *c*, that

is to say, at one of its ends (doubtless because she tilted it up at *c* during the preceding attempt). She draws it horizontally against the bars but encountering resistance from them she quickly makes it vertical and pulls it through without difficulty. The speed of this adaptation is due to the fact that the stick was grasped by one of its ends; the subsequent attempts show that nothing systematic yet exists.

5. Jacqueline again grasps the stick by the middle, at *b*. She raises it, puts it horizontally against the bars, as in 1. She pulls and seems very surprised by her failure. It is only after a while that she tilts it up (this time, it appears, intentionally) and succeeds in bringing it in.

6–10. Same reactions. At each new attempt, she begins by trying to make it penetrate horizontally, parallel to the frame. It is only after this preliminary failure that she tilts up the stick, still quite slowly.

11. This time Jacqueline turns the stick more rapidly because she grasped it at *c*.

12–15. She again grasps it at *b* and recommences to try to bring it through horizontally, as in 5–10. Then she tilts it up, more slowly than in 11, and succeeds.

16. She continues to take it at *b* and to try to pull it through horizontally but, this time she does not persist and tilts it up immediately.

17. For the first time Jacqueline tilts the stick before it touched the bars and no longer tries to bring it in horizontally. However, she grasped it at the middle (at *b*).

18–19. She again begins by trying to bring it through horizontally but it seems that this was due to automatism and she tilts it up immediately afterwards.

20 et seq. She finally turns it systematically before it touches the bars (see 17).

Observation 163.—At 1;3 (13) we resume the same experiment with Jacqueline, but by complicating it in the following way: the stick henceforth used is too long to pass through horizontally. The bars of the pen are 50 cm. high (with a space of 46 cm. between the lower bar and the upper one) and the stick given to the child is 55 cm. long. We shall call the middle of the stick *A*, and *B* and *C* the two points situated at one third and two thirds of the distance between the middle and the end. The stick is again placed on the ground parallel to the side of the frame which Jacqueline is seated facing. Ten attempts were enough to enable her to solve the problem:

1. Jacqueline grasps the stick at *B*. She raises it horizontally and thus pulls it against the bars. She pulls it against the bars with all her strength and then displaces it unsystematically, raises it and all at once brings it through by chance without having understood how she did it.

2. This time she grasps the stick at *A*, puts it horizontally against the bars and pulls as hard as she can. She then tilts it up systematically

but the stick, touching the ground at its lower end, remains oblique. She again pulls very hard, then gives up.

3–4. She still begins by pulling horizontally, then raises it, pulls again and finally tilts it in such a way as to bring it through correctly. Both times she grasped it at *B*.

5. Jacqueline grasps the stick at C, pulls it horizontally, then raises it. But she tilts it up so that it is higher than the frame at the top and remains caught at the bottom. She then shakes it and ends by bringing it through accidentally.

6. Same beginnings. The stick is held at the top by the edge of the frame and at the bottom by Jacqueline's dress which is pressed against the lower edge of the pen. Jacqueline then watches both ends of the stick attentively, then raises it gently to disengage it from her dress. She then brings it in slowly by the bottom part and then pulls until she has been completely successful.

7. First she grasps the stick at *A*, applies it horizontally and pulls. Then she grasps it with the other hand at *C* (still holding it firmly at *A*, tight against the bars) and brings it through by raising it first and then pulling it by the lower part, as before (in 6).

8. Jacqueline succeeds at once this time, almost without pulling the stick against the bars. She grasps it, tilts it up and brings it through by the lower end.

9. She grasps it with the wrong hand (too high to be able to pull it through by the lower part). She immediately switches to the other hand and succeeds at once.

10. Immediate success, without groping and without first touching the bars: she barely grazes them when bringing the stick through.

At 1;3 (15) Jacqueline fails at the first attempt and again pulls horizontally but, during this second attempt, she rediscovers the combined two acts of tilting up the stick and bringing it in by the lower part. At 1;4 (0) after an interruption of several days, she relapses into her former mistakes, then succeeds.

Observation 164.—The following facts will help us to clarify the powers and limitations of visual perception. At 1;3 (13) Jacqueline tries to bring in a case for eyeglasses: she succeeds at once. However, she grasped it but turned it to the vertical position before it touched the bars. Same success subsequently with a stick of sealing wax.

Then I place a book on its edge outside the frame (the spine of the book upward and parallel to the frame). She grasps it and pulls the whole width of it against the bars. Afterward she pulls it with the spine of the book placed horizontally against the bars, then, thirdly, she tilts up the book and brings it through vertically, spine first, without any difficulty.

A half hour later Jacqueline begins again by pulling the whole width of the book against the bars in order to bring it inside in this

way, and she again pulls with all her strength. Afterward, she places it on the floor on its edge, parallel to the frame, then she grasps it with her other hand by the spine, raises it up vertically before it touches the bars and brings it through.

In the course of a final experiment she tilts it up right away before another attempt and brings it in unhesitatingly.

At 1;4 (21), on the other hand, she tried to take out of her pen cylindrical, wooden, Russian dolls, too wide to push through the bars. She does not understand her failure and pushes them anyway. She does not succeed in inventing the procedure which would consist in sliding them along the bars in order to bring them over the top.

Observation 165.—At 1;3 (14) Jacqueline received a cardboard rooster by means of which I try the following experiment. I place it lying on the ground, outside the frame but introducing the head and tail of the cock in the direction of the child. In other words, the head passes through a space between two bars, the tail passes through the next space and the rooster's back is held back by the bar separating these two spaces. If the child wishes to pull the rooster to him he must first push it away, then tilt it up and finally bring it through head or tail first.

During this first experiment Jacqueline confines herself to merely pulling the rooster by the head or by the tail but without previously pushing it away or tilting it up; she consequently fails completely.

At 1;3 (16), on the other hand, I simplify things a little by putting the rooster back a little; it still faces a bar but, instead of being in contact with it, it is 5 cm. behind it. Here is the series of attempts:

1. Jacqueline pulls the rooster toward her and it gets caught in the bars. She pulls hard for a while, then switches hands. While she is doing this, the rooster happens to fall quite far away so that when picking it up again, she tilts it to a vertical position without difficulty. She then sees it in profile and has only to turn it full face to bring it through. These acts of raising the recumbent animal, then of bringing it in full face only constitute, of course, the application of discoveries made in connection with the stick (Obs. 162 and 163) and primarily in connection with the book (Obs. 164), that is to say, the discoveries made the preceding days.

2. In grasping the rooster Jacqueline pushes it back slightly and so again succeeds in tilting it up without difficulty. She does this systematically and without hesitation.

3. The rooster gets caught this time. Jacqueline nevertheless pulls it without thinking of pushing it back. After vain attempts she changes hands and begins over again. Then she takes it in her right hand and pulls it with renewed strength. At last she tries to tilt it up, but still without pushing it back. She then gives up and lets it go.

4–6. The rooster again gets caught at each attempt. Jacqueline recommences to pull with each hand alternately. But each time the rooster finally falls sufficiently far away to enable her to tilt it up without difficulty. She therefore knows how to tilt the object up but still does not know how to push it away for this purpose; it is only chance that allows her to do it.

7. The rooster remains caught for a long time. She pulls it with both hands. It falls but she gets it caught while trying to tilt it up; she does not understand and pulls harder. At last it falls far enough away to permit her to tilt it up and pull it in without difficulty.

8. This time the rooster, caught at first, falls six times a short distance away and each time it would have sufficed to pull it back even a little bit to tilt it up. However, she got it caught at each new attempt and resumed pulling without understanding.

9–10. Same reactions. She fatigues and we pause.

The afternoon of the same day, around 1 o'clock, we resume the experiment: complete failure.

That evening, around 6 o'clock, new attempts which this time are successful. Here is the series of attempts:

1. Failure: she pulls, switches hands, and gives up.

2. She succeeds by chance in tilting it up before it touches the bars and doubtless before it becomes caught. It goes through without difficulty.

3. It becomes caught and she pulls it for a moment, then lets it drop, perhaps intentionally, after which she tilts it up before it touches the bars.

4–9. Same beginning, but this time it is certain that she lets it fall intentionally and sooner after the beginning of the attempts. Then she tilts it up very well, taking care to do so ahead of time (before pulling) and at last she pulls it to her. The game amuses her to such an extent that as soon as the rooster is inside the frame, she puts it outside by herself in order to begin all over again.

10. I now wedge it ahead of time as at the beginning of our experiment [at 1;3 (14)]. Jacqueline then pulls at first and is surprised by her failure; she still does not know how to push the object back beforehand. On the other hand, when she observes the failure, she knows how to drop the rooster intentionally. It then falls 3 cm. away from the bars, she tilts it up and brings it in without difficulty.

11–12. This time, a noteworthy novelty: The rooster gets caught, she pulls it for a moment, then, without letting it fall, she places it on the floor (while holding it by its head) tilts it up and pulls it to her. She has not definitely pushed it away but by placing it on the floor she had sufficient leeway to enable her to tilt it up without difficulty.

13. She pulls, then lets it fall again (intentionally), as in 4–10. Then she tilts it up and brings it in.

14. Jacqueline pulls, then once more tilts the rooster up, on the floor, without letting it fall or letting it go (as in 11–12).

15–16. She again lets it fall but then returns it very carefully, while watching its tail which was in danger of catching in the bars.

17. This time, Jacqueline definitely pushes the rooster back before pulling it and tilts it up without letting it go.

18. Same reaction, but in addition, she takes the rooster outside the bars by herself, as soon as she has brought it in, in order to repeat the experiment because she is so delighted at her last discovery. This play goes on until satiety.

Observation 166.—At 1;3 (17) the day after the preceding series, I resume the experiment of pulling the rooster through the bars. It is worthwhile to describe in a new observation the results of these repeated attempts after the discovery of the correct procedure, for they are of a kind to clarify the relations of visual representation with the dynamic schema.

Here is the series of these new attempts:

1. Jacqueline pulls the rooster to her, as if the bar located between the head and the tail were not going to catch the animal's back. She pulls as she did the first day, perseveringly and recommencing with vigor after short rests. Then the rooster falls by chance and she is then able to tilt it up without difficulty and to pull it to her.

2. Same reaction, but she lets it go quickly, perhaps intentionally, and tilts it up on the ground.

3. She begins again by pulling, then, without letting it go, lowers it to the ground, pushes it back intentionally, tilts it up and brings it in.

4. Same reaction, very definite, but she pushes it back by making it slide along the floor and turns it so much that it catches on the other side.

5–7. She pushes it back almost immediately, but commences each time by pulling it directly.

It may thus be seen how much more rapid is the discovery of the correct procedure than it was the day before but one observes that, nevertheless, progress always takes place through motor assimilation and not through representation. A quarter of an hour later, I resume the experiment and observe the ten following attempts:

1–4. Jacqueline first pulls the rooster, then pushes it back soon after, and tilts it up without letting it go.

5–6. She pushes it back, this time, right away. She even pushes it back the second time 15 cm. along the floor, without letting it go, and once it is tilted up, brings it in victoriously.

7. She begins again by pulling first, then pushes it back and tilts it up.

8–10. Correct procedure as in 5–6.

At 1;3 (21), that is to say, four days later, Jacqueline again pulls the rooster directly twice and then pushes it back at the third attempt. At 1;3 (27) same sequence. At 1;4 (0), she pushes it back beforehand at the second attempt but simply from habit because, at the fifth attempt, when the rooster gets caught by chance, she pulls it again with all her strength without knowing how to correct it. Then at the sixth attempt, she again pushes it back beforehand. At 1;4 (20) I make the same observation. But it suffices to show her a new rooster to make her succeed in bringing it in right away and push it away and tilt up ahead of time both the old and the new rooster.

This new series of facts warrants our resuming the discussion of the mechanism of accommodation. As we have seen in connection with the "stick," there are three possible solutions for interpreting such facts; chance and selection, the hypothesis of "structures," and that of a structuring assimilatory activity which is not structured from the outset.

The first of these solutions seems at first very likely. These observations, still more than those pertaining to the stick (see Obs. 159 during which Jacqueline learns little by little to utilize an imitated example) seem to speak in favor of a sort of training, the incorrect procedures being gradually replaced by correct methods. But, on examining things closely, one sees that this is only apparent and that the gradual victory of the correct procedures is not at all due to an automatic selection: It is simply a question of progressive comprehension analogous to that which we observe in ourselves when we only grasp little by little the factors of a problem and only after much groping arrive at clear and unified vision. In such cases, we begin by anticipating, in a way, the right solution. In other words, an accommodation of familiar schemata to the new situation permits us to differentiate them in a relatively adequate schema but the latter remains in the state of intention or of simply structuring outline; that is to say, it orients searching without yet being sufficiently strong to eliminate false solutions, it coördinates the progress of groping without itself being structured, and finally it utilizes fortunate chance circumstances without being yet able to do without

their coöperation (but it is never derived from them). The same is true of our Observations 162 to 166. The child tries to bring the object to him (there is the schema assigning an end to the action and thus directing the groping), and quickly understands, when he fails, that certain displacements of the object become necessary (there are the schemata serving as means which accommodation will differentiate). With regard to the origin of these latter schemata, it is to be sought in the tertiary circular reactions relating to changes of position (Obs. 141–142 and 144–145) and primarily in the many experiments which the child makes each day in order to grasp a cumbersome object, to tilt it up, to dissociate it from those in which it is inserted, etc. (see Obs. 146). This outline of a solution, obtained by directed differentiation of earlier schemata, then gives rise to a series of attempts in the course of which chance certainly intervenes unceasingly, but which are not dominated by chance. If, thereafter, the false solution continually reappears (drawing the object directly to oneself), that simply means that the outline of the correct solution is too weak to counterbalance the influence of a procedure having the force of habit and the allurement of apparent evidence; that does not mean at all that the correct solution is obtained by purely automatic training founded upon chance and selection. The correct solution, once glimpsed, grows progressively stronger, not in the manner of a phenomenon that statistically outweighs another as function of a selection taking place in time, but in the manner of a cumulative experience or comprehension. For example, in Observation 165, the first ten attempts reveal no progress because the solution (pushing the object back in order to tilt it up) has not been glimpsed; but as soon as it has been (beginning of the second series) it becomes consolidated (attempts 3–10), then becomes explicit (attempts 11–14), and finally is definitively established (attempts 17–18). This development is therefore not that of a series of blind gropings in the course of which propitious acts become fixed due to confirmation; it is that of a directed apprenticeship, analogous to the example of a student trying to solve the same problem in arithmetic twenty times, knowing the final result but without

having understood the connection, which he guessed at, between the different operations to be performed.

These remarks make us realize at the same time the differences existing between the cumulative assimilation presupposed by such accommodations and the ready-made "structures" of Gestalt psychology. Again referring to Observation 165, there are three operations to perform: pushing back the rooster, tilting it up, and bringing it in full face. Jacqueline knows how to perform the two latter ones due to her recent acquisitions (Obs. 162–164). It only remains for her to discover the necessity for the first in order to intercoördinate the other two. Now she anticipates this solution as soon as she lets the rooster fall and thereafter acquires the possibility of tilting it up without difficulty. The schema thus outlined (beginning of the second series) becomes consolidated, explicit and finally established, as we have just recalled. How can this development be explained? It cannot be a question of an immediate structurization precisely because, in the course of attempts 1 to 16, there is no correct solution but simply progress towards a solution. All that remains is to assert the existence of a cumulative assimilation, analogous to that of the tertiary circular reactions and according to which the new motor schema, outlined by accommodation, like every assimilatory schema, is developed by repetition, recognition and generalization. Once again we find a structuring assimilation and not at the outset a structured coördination. It is by functioning that a schema structures itself and not before functioning. It is true that, in order to function, that is to say, in order to assimilate the real situation, a schema has need of at least a *minimum* of structure; but this structure is nothing independently of the act of assimilation and so only becomes crystallized in the course of this act.—Regarding Observations 162–164, the same is true: The solution outlined at the beginning becomes consolidated and explicit through reproductive, generalizing and recognitory assimilation.

In connection with this dynamism it remains to speak of the respective roles of visual representation and simple motor assimilation. How does it happen that, in the course of Observations 165 and 166, Jacqueline tries again and again to bring

a rooster through the bars when she sees that one of these bars holds it back systematically and when she has already discovered the correct solution beforehand and several times in succession? How does it happen (Obs. 162 and 163) that she even tries to bring in a long stick held back by 2, 3 or 4 bars at once, as though the stick were going to cut the bars or cross them as a wire cuts butter? Or why (Obs. 164) does she persist in trying to bring through the space between two bars a doll which is wider than this space? Is it because visual perception only plays a secondary role in such behavior patterns and that they are purely a matter of motor searching; or because this perception is different and does not take account of the solidity of objects? In fact, it seems to us that both solutions amount to the same thing. Everything occurs as though, to the child, the bars constituted pure images without depth or solidity (pictures and not substances) and as though these images could be traversed through and through without difficulty. But why is this so? Precisely because a sensorimotor elaboration has not yet conferred upon them the qualities of resistance and substantiality which they lack. Here again it is difficult to speak with the Gestalt psychologists of a sudden reorganization of the perceptual field independently of structuring assimilatory activity: It is the action that fashions the field of perception and not the reverse.

In short, the theory of pure groping makes of the discovery of new procedures a simple accommodation, thus neglecting the formal coördination belonging to assimilation: hence this theory is analogous to an empiricism ascribing invention to experience alone and neglecting the activity of the mind. The theory of "structures," on the contrary, emphasizes the existence of formal coördinations but neglects accommodation, in this being comparable to an apriority which disdains experience. For us, accommodation is necessarily on a par with a cumulative assimilation, structuring and not structured from the outset. The schema of assimilation thus reconciles the necessary role of experience, that is to say, of accommodation, with the no less necessary role of formal coördination.

Before concluding, let us cite a certain number of mixed observations in which the discovery of new means through ac-

tive experimentation simultaneously raises all the problems discussed hitherto:

Observation 167.—At 1;3 (12) Jacqueline throws a plush dog outside the bars of her playpen and she tries to catch it. Not succeeding, she then pushes the pen itself in the right direction! By holding onto the frame with one hand while with the other she tried to grasp the dog, she observed that the frame was mobile. She had accordingly, without wishing to do so, moved it away from the dog. She at once tried to correct this movement and thus saw the pen approach its objective. These two fortuitous discoveries then led her to utilize movements of the playpen and to push it at first experimentally, then systematically. There was a moment's groping, but it was short.

At 1;3 (16), on the other hand, Jacqueline right away pushes her playpen in the direction of the objects to be picked up.

Observation 168.—This last observation during which the child displaces himself in order to attain the objective, leads us to the situation in which the subject is obliged to withdraw part or all of his own body in order not to interfere with the movements of the object. For instance, at 1;6 (15) Jacqueline is standing on a rag (50 x 30 cm.) which she is trying to pick up. She pulls, is surprised at the resistance, but it does not occur to her to move. Finally, she gives up.

At 1;7 (0), on the contrary, she is standing on a handkerchief and, after having pulled it, she moves her feet until it is free. At the second attempt, she moves away beforehand, but at the third, she continues to pull a long time before taking away the foot that was impeding her.

Observation 169.—Here is a behavior pattern intermediate between the preceding ones and those that consist in utilizing the relationships of contents and container. At 1;3 (14) Jacqueline tries to open a jewel case (3 x 5 cm.). With one hand she holds it and without knowing it tightens the cover on the box, and with the other hand she tries to raise this cover, of course without succeeding! However, by dint of transferring the object from one hand to the other without the possibility of noting all the vicissitudes, she finally draws back her right hand (which holds the case) as much as possible while she pulls at the lid with the other. But there is not yet any systematic procedure.

At 1;3 (15) on the other hand, after two attempts during which Jacqueline recommences trying to hold the cover with one hand while she tries to displace it with the other, she places the box on the ground and opens it without difficulty. This act of placing it on the ground was not the result of invention, properly so called. She simply removed her right hand and, unable simultaneously to hold the box and open it with her left hand, she placed it on the ground.

That evening, she tries to open a pipe case (same kind of fastening:

two valves applied against each other). She tries indefinitely to open it with one hand while holding it closed with the other. But then the case happens to fall and opens: Jacqueline opens it and then closes it several times in succession on the ground and with one hand. Then she takes it up again in one hand and recommences to try to open it with the other: complete failure. She then rests it on the ground, this time intentionally, and opens it without difficulty.

After a new attempt with both hands, she rests it again on the floor and only tries this way.

At 1;3 (16) same reaction. On the one hand, Jacqueline well knows how to open the case when she rests it on the ground. She searches for the slit with one finger and raises one of the valves without touching the other. When her finger again covers both, she lowers it very attentively until she feels the slit and then opens the case easily. But, on the other hand, when she holds the case in both hands, she is incapable of proceeding. While trying to raise a valve, she holds it firmly with the other hand. In the latter case, she rests the case on the table and succeeds in opening it by means of one hand only. Finally, she only attempts this second procedure and gives up all efforts with both hands.

Observation 170.—Here is an analogous observation concerning Lucienne. At 1;1 (23) Lucienne puts (by chance?) a cake in the form of a ring or a torus in a circular wooden box. She immediately tries to take it out. But she puts her thumb against the outside of the box while pulling the cake with her index and other fingers so that the very palm of her hand prevents the object from coming out. She finally attains her goal after extensive efforts. She then at once begins over again and so on twenty times in succession through an apparent need of assimilation. From time to time she does it by simple empirical groping, by shaking the box or turning it upside down (not deliberately), but, on the whole, there is definite progress. Little by little she manages to hold the box in one hand while she pulls the cake with the other without obstructing the cake with her thumb.

There exists in this case, besides cumulative assimilation, a process of progressive dissociation. The child detaches three objects from one another: box, cake and hand. At first she does not see that the hand is an obstacle as well as an instrument. Then, due to her directed gropings, she understands the exact interrelations of the objects and succeeds in solving the problem she herself raised.

Observation 171.—These last behavior patterns lead us to the acts relating to the connection between container and contents. Here is the simplest of those we have observed:

At 1;3 (28) Jacqueline receives a toy consisting of hollow blocks which fit into each other and which we separate before her eyes and

scatter about. The problem is to know how she will learn to put the little blocks into the large ones.

1. Jacqueline begins by manipulating eight blocks of different sizes by trying to put the little ones in the big ones and the big ones into the little ones, varying the combinations (see particulars in Vol. II, Chap. II).

Toward the end of these first gropings she seems more rapidly than before to abandon the project of putting a large block into a small one.

She finally grasps a large block with one hand, a small one with the other and looks for the opening of the former in order to place the second one systematically in it; hence the experiment is accompanied at this time by a sort of reflection or mental concentration.

2. At 1;3 (29), the following day, I again give the blocks to Jacqueline who is in her playpen. She begins by trying to make a block which is too big come through the bars. After having given up, she puts a small block into a big one and shakes the latter in order to make a block which is too big come through the bars. After having given up, she puts a small block into a big one and shakes the latter in order to make a noise. She is not interested in anything else so I take the game away from her.

3. At 1;4 (0) she tries at first to put one block into another which is slightly larger. Then she recommences trying to place a large one in a small one but corrects herself very quickly.

4. From 1;4 (5) Jacqueline's attempts lead to satisfying results on the whole. She no longer tries to put large blocks inside small ones, she takes account of the position of the angles and succeeds in extricating the enclosed blocks by sliding them out with her index finger. These three behavior patterns have consequently been acquired due to directed gropings and the progressive correction of the initial schemata.

Observation 172.—Here is a slightly more complex example. At 1;1 (3) I present Lucienne with a wooden pail (10 cm. in diameter) to which she is accustomed and place my watch chain next to it. At first Lucienne tries to put the chain into the pail, being accustomed to putting different objects into it. She grasps the middle of the chain between thumb and index finger and places it on the edge of the pail. But of course the greater part of the chain remains outside the pail. Lucienne at once grasps the outside end in order to make it all go inside, but she grasps it so high up, as though the part already inside were not joined to the other, that the whole chain comes out and she has to begin all over again. This recurs many times because Lucienne, through lack of assimilation, each time puts the chain into the pail but always goes about it in the same way. However, little by little she manages to make her movements gentler and to pick up the hanging end of the chain without jostling the other. At last she succeeds once in making the whole chain enter the pail.

At 1;3 (13) she tries to solve the same problem by herself, but with a necklace and a watering can. First she puts the end of the necklace inside, then the rest of it by degrees without making the already introduced part fall out. She thus succeeds, after several failures, in making the whole necklace go inside the can twice.

Observation 173.—These last experiments now lead us to the analysis of a proof which has shown itself to be particularly fruitful: making a watch chain enter a narrow opening. This experiment, which in Lucienne's case took place after the preceding ones, yielded quite different results than with Jacqueline, results which were new both with respect to the chain and pail and to the necklace and watering can. Lucienne solved this problem by an act of true invention which we shall study in the course of Observation 179. With regard to Jacqueline, on the other hand, the behavior pattern was revealed to be exactly analogous to that of Lucienne in the preceding observation.

At 1;7 (25) Jacqueline holds a rectangular box, deep and narrow, whose opening measures 34 x 16 mm. (for this purpose I use the cover of a match box which is three quarters open), and she tries to put my watch chain into it (45 cm. long). During the first fifteen attempts, she goes about it in the following way: First she puts one end of the chain into the box (2 to 4 cm.), then she grasps the chain about 5 cm. from this end and thus puts a second segment into the box. She then gets ready to do the same with a third segment when the chain, no longer supported by the child's hand, slides out of the box and falls noisily. Jacqueline recommences at once and fourteen times in succession sees the chain come out as soon as it is put in. It is true that, around the tenth attempt, Jacqueline has tired of it and was about to give up; but I placed the chain in the box (without the child's seeing how) and then she regained hope by noting that such a result was not impossible.

At the sixteenth attempt, a new phenomenon: Jacqueline having grasped the chain nearer the middle, the chain no longer lengthened as before at the time when the child raises it but takes the form of two entwined cords. Jacqueline then understands the advantage she can take of this new presentation and tries to make the two ends enter the box together (more precisely, one immediately after the other, the second following shortly after the first). She no longer lets the chain go after putting one of the ends into the box, as was the case in attempts 1–15, but tries to put all of it in. But, as always occurs when a child of this age manipulates flexible objects, Jacqueline considers the chain as being rigid and lets go the whole of it when both extremities have been put in the box. The chain then comes out again somewhat, but Jacqueline gently reintroduces the part that hangs (the middle part).

Attempt 17: Jacqueline distinctly tries to repeat the preceding movement. At first she does not grasp the chain at one end but pulls it

together somewhat and grasps the middle part (without of course trying to find the actual middle). She again succeeds in putting both ends in together.

Attempt 18: resumes the initial procedure and fails.

Attempt 19: rediscovers the procedure of attempts 16 and 17.

Attempt 20: same reaction, but this time Jacqueline encounters some difficulty in putting the second end in. Not succeeding, she recommences trying to put in a single end first. But as the chain slides out, she resumes the procedure of attempts 16, 17 and 19.

Attempts 21–22: same hesitations, with ultimate success.

Observation 173 repeated.—An hour later I again present the box and chain to Jacqueline. Four interesting attempts ensue.

1. Jacqueline grasps the chain with both hands, probably by chance. Then she examines with curiosity the shape thus obtained: the chain being grasped simultaneously at about one third and two thirds of its length, these two ends hang parallel, about 15 20 cm. away, while the middle part is horizontal. But then Jacqueline, instead of putting both ends inside simultaneously by bringing them near each other, confines herself to putting one of them in the box with great delicacy, and lets the whole of the chain go as though this end would draw the rest after it: the chain collapsed.

2. She now grasps the chain toward the middle and tries to put both ends in at once. This is the procedure discovered during the preceding series and it succeeds again this time.

3. This time she begins by grasping the chain not far from one of its ends but she corrects this movement before letting it all go. Seeing that only a small part of the chain entered the box, she intentionally displaces her hand toward the middle of the chain in such a way as to acquire a better grip and to put both ends inside at the same time. But, experiencing some difficulty in this operation (the chain spreads out and becomes too wide) she corrects herself a second time and simultaneously invents a new procedure.

4. Seeing both parts of the chain separate, Jacqueline rolls the chain up and puts it into the box very easily.

This last procedure, which is the simplest, is only discovered after the stage constituted by attempts 16–22 of the preceding series, by attempt 1 of the present series and finally by correction of the beginning of attempt 3. Instead of inventing all at once the procedure of "rolling up," as Lucienne will do, Jacqueline forms it progressively by combined assimilation and accommodation.

At 1;8 (2) Jacqueline rediscovers the procedure of rolling up at the very beginning, then returns to the system of the chain hanging by a middle part. This last method is alone utilized at 1;9 (21) after failure due to a regression of the initial incorrect procedure.

Observation 174.—These kinds of apprenticeship can be further complicated by requiring the child to correct the position, not only of the object to be put in, but also of the container. So it is that at 1;1 (23) Lucienne sees me put a ring into half of a case for eyeglasses. She looks at the object inside the case, shakes the case and lets the ring fall out. She then tries immediately to replace it but the apprenticeship is accomplished on two occasions.

During a first phase, Lucienne tries four sequential maneuvers, all unsuccessful. (1) She first presses her three fingers holding the ring against the opening of the case and drops the ring. The ring falls to the side because her fingers prevent it from entering. (2) She presses the ring against the closed end of the case and lets it go. (3) She holds the case upside down and puts the ring into the opening but without tilting it up. The ring falls out at the first movement of the case. (4) She places the ring on the floor and presses both ends of the case against it, alternately, as though the ring would enter it by itself.

During a second phase, on the contrary, Lucienne learns to correct her attempts. First, she no longer places the case on the ring as though the latter would enter by itself. Then when she presses the ring against the wrong end of the case, she does not let it go but turns the case over in order to slide it into the opening. She holds the case almost vertically and, when it is too slanting, she straightens it before letting the ring go. Finally, she learns only to drop it inside the case by first sliding it to the end of her fingers instead of letting it fall when the fingers still obstruct an opening of the case.

At 1;1 (24) after having disengaged the ring from her thumb around which it fell by chance, Lucienne sees me put it around a stick. She then tries to draw it toward her without putting it along the wood. Then she shakes the stick and the ring falls. In order to put it around the stick again, she simply presses it at a certain place and lets it go. Same reaction six times in succession. Then she tries to put it on at the end, but lets it drop. That afternoon she succeeds twice in putting it on the stick but she merely presses it several times against the stick. The following days both reactions subsist without excluding each other, but the attempts to put it on the stick prevail increasingly above the others.

§4. THE DISCOVERY OF NEW MEANS THROUGH ACTIVE EXPERIMENTATION. III. CONCLUSIONS.—These last facts having thus completed the preceding ones, let us try to draw from all our data a conclusion relating to the present type of behavior patterns, beginning by trying to place them in the general picture of intelligence.

The behavior patterns peculiar to sensorimotor intelligence

can be divided into two big groups. First there are those whose goal is in some way imposed by the external environment. Such are the circular relations, secondary or tertiary, which consist in merely repeating or varying an interesting result obtained by chance. Such, too, are the facts of the comprehension of signs or of exploration in which an external factor obtrudes without having been chosen and requires assimilation by the subject. These different behavior patterns certainly constitute intelligent behavior since it is always a question of adjusting means to ends, and whether these ends consist in repeating, understanding, or foreseeing is of little importance. But they are intelligent in different degrees. It can be said, on the whole, that an act is the more intelligent the greater the number of schemata it subsumes and the more difficulties the latter present in intercoördinating. Thereafter, the operation requiring the least intelligence is that of secondary circular reaction: merely rediscovering the means that made it possible to obtain an interesting result. Concerning comprehension, prevision and exploration, they are the more "intelligent" the more complex they become, and they can reach a high degree of complexity. But as these behavior patterns are in a way directed from the outside by the facts which come to the child's attention, they do not give rise to actual inventions, that is to say, to the most complex systematizations of which sensorimotor intelligence is capable at its beginnings. Moreover, the first comprehensions, previsions and explorations simply consist in making an object or an event enter into one or more sequential schemata, intelligent search consisting not in coördinating the schemata, but in appropriately choosing between them.

A second group of intelligent behavior patterns is constituted by those whose purpose is, on the contrary, due to the subject's spontaneous intention. It goes without saying that this distinction is relative, since intention is always occasioned by encountering an external fact. But this fact does not here impose itself in the capacity of an external motor of thought; it is simply an opportunity for various projects and it is these projects which impose themselves on it. It goes without saying, thereafter, that obstacles arise between the intention and its realization and that a more or less large number of means must be put to work

to remove these difficulties. It is the subordination of means to end that constitutes the intelligent act. In such behavior patterns the principal schema which, by assimilating the data, gives a purpose to the action, and the secondary schemata which constitute the means and become coördinated with the former; a certain number of auxiliary schemata can intervene besides as the search goes on; the final schema is thus called upon to systematize the ensemble of these terms in a new unit. If behavior patterns such as secondary or tertiary circular reaction, the comprehension of signs or exploration constitute the sensorimotor equivalent of judgment, the more complex behavior patterns of which we now speak constitute reasonings. As we have already seen, the subordination of means to ends is, in effect, comparable to the subordination of premises to the conclusion. This situation explains why the behavior patterns of the first group enter ceaselessly in the capacity of elements into those of the second group. This distinction must not, however, be considered as being too absolute; just as judgment is virtual reasoning, so also there exist all intermediates between the two groups until the time when comprehension even becomes an end in itself and gives rise to the same complex and deductive steps as invention itself.

Regardless of this last point, this second group of intelligent behavior patterns admits of three distinct types: the "application of familiar means to new situations," the "discovery of new means through active experimentation," and the "invention of new means through mental combination." In order to understand the nature of the behavior patterns of the present type it is necessary to analyze them in relation to the two others: discovery through experimentation is essentially a transition between simple application of the familiar to the new and invention properly so called.

Two characteristics common to these varied behavior patterns should be noted here in order simultaneously to grasp the continuity and relative contrast of the three types under consideration. These are experimentation through directed groping, the source of acquisition, and the application of familiar schemata, source of systematization. In a word, it can be said that the first type is defined by the primacy of application, the second by

the primacy of groping, and the third by the unification of these two characteristics. But groping is not foreign to any of the three terms of the series, as we shall see, and application remains essential to the second one, although dominated by groping.

The relationship between the three types of behavior patterns therefore consists in that the subject finds himself confronted by a situation new to him and, in order to arrive at his ends, he must discover suitable means not immediately given. The simplest solution in such a case, it goes without saying, consists in searching in the stores of already acquired schemata for some known procedure which might solve the problem. It is this step which constitutes the first type of behavior patterns: the "application of familiar means to new circumstances." It goes without saying that application predominates in such a behavior pattern. But groping is not excluded from it since it is a question of adapting old schemata to the new situation and this adaptation presupposes, on the one hand, a search for the right schema and an elimination of the useless schemata and, on the other hand, an adjustment of this suitable schema. Consequently, either in the course of this search or of this adjustment, the subject will be seen to hesitate and correct himself, in short, behave in a way which presages the second type. Moreover all the intermediates exist between them: Observation 122 thus furnishes us with an example of evident groping. But as long as this groping simply results in rediscovering a familiar procedure and adjusting it without transforming it, we remain confronted by a behavior pattern of the first type. The behavior patterns of the second type begin in exactly the same way; but, after having tried an initial means (which was discovered by the child due to the assimilatory process belonging to the "application of familiar means to new situations"), the subject finds himself obliged to differentiate it. It is in the course of this operation that groping accommodation intervenes, as we have seen in connection with the "supports," the "string," and the "stick": from the time when the familiar means does not suffice to solve the problem, it is necessary to grope. This groping begins by attempts at simple adjustment, then, gradually as the child's experiments go on this adjustment is transformed so that out of the differentiation of

the initial schemata come new schemata, thus implying a real discovery. But such a behavior pattern which is increasingly distinguished from the simple "application of the familiar to the new" nevertheless conserves one of its essential characteristics. As we have remarked in connection with each of the examples analyzed, it is by constant application of earlier schemata to the present situation that groping is deserted and the events arising during this searching are interpreted. In short, if there is a difference between types I and II, there is nevertheless complete continuity.

Analogous comments apply, as we shall see, to the relations which connect types II and III. Groping, which has primacy in type II, does not disappear in the "invention of new means through mental combination" but is internalized and proceeds by means of representations instead of depending exclusively on external and immediate activity. In other words, effective experimentation becomes "mental experience." On the other hand, the application of earlier schemata which is not absent from behavior patterns of type II (effective groping) again assumes, in invention through mental combination, the importance it had in behavior patterns of type I (application of the familiar to the new). Invention thus synthesizes searching and deduction by merely protracting the two preceding types of behavior patterns.

These remarks enable us to understand the relations between assimilation and accommodation in empirical groping. As we have seen, it is only at the level of tertiary circular reactions that assimilation and accommodation begin to become truly differentiated. At the time of acquisition of the first habits through primary circular reaction, the two terms remain relatively undifferentiated. Every attempt at assimilation is simultaneously an attempt at accommodation. With the advent of secondary circular reaction a new fact appears: interest in the external results of acts. This interest surely marks progress toward differentiation, since the external result of acts, by differentiating the primitive schemata, thus compels them to incessant accommodation. But, as we have noted, this accommodation is still imposed and not yet sought after for its own sake. The interesting act which the child tried to conserve through

assimilation arises *ex abrupto* and, if it interests the subject, it is inasmuch as it is connected, by continuity or contrast, with the already existing schemata. On the contrary, with the advent of tertiary circular reaction, accommodation becomes an end in itself which certainly protracts the earlier assimilations (the subject only accommodates already formed schemata), but which precedes new assimilations and so intentionally differentiates the schemata from which it sprang. It is then that experience begins to be formed and is distinguished from simple utilization of the real with a view to fueling the internal functioning. Henceforth there exists interest in the new as such. But will this differentiated accommodation be antagonistic or complementary to assimilation? Study of empirical groping in the search for and discovery of new means furnishes in this respect an exact answer. Accommodation to experience and deductive assimilation henceforth alternate in a movement whose rhythm can vary but whose cyclical character attests to an increasingly close correlation between the two terms. It is, in effect, under the pressure of necessity (hence of the principal assimilatory schema) and of the schemata tried out in the capacity of initial means, that groping accommodation goes in quest of new means and results in the formation of new schemata capable of being coördinated with the old ones. The "discovery of new means through active experimentation" thus marks the beginning of a union of experience and assimilatory activity, a union which "invention through mental combination" will consecrate by raising it to the rank of interdependence.

However, we must not exaggerate. At the level of empirical groping, this union, however remarkable it may be with reference to the preceding behavior patterns, remains in a state of promise or outline, compared to its future developments. Primitive accommodation to experience and complementary assimilation of the datum which characterize empirical groping present the common aspect of being immediate and hence limited. "Experience" as practiced by sensorimotor groping, is immediate in the sense that it considers things as they appear to be instead of correcting them and elaborating them mentally. Assimilation, on the other hand, only bears upon direct perception, and not

yet on representation. These two characteristics constitute one and the same phenomenon, only envisaged from two different points of view.

As far as representation is concerned, we have already dealt with it in connection with the experiment of objects to be brought through the bars of the playpen. Observations 167–174 wholly confirm these conclusions: At the level of groping, representation does not precede action and does not even directly result from it. On the contrary, everything occurs as though the object seen were conceived as being identical to that which it appears to be in immediate perception. So it is that, in Observation 173, Jacqueline, in order to put a long chain into a small box, confines herself to putting one end of this inside, without rolling up the whole chain or foreseeing the object's flexibility and fall. Despite the first fruitless efforts she recommences indefinitely and sensorimotor groping alone corrects her vision of things. So also in Observation 174 Lucienne, in order to put a ring in a case or around a stick confines herself to placing the ring under the case or against the stick. Now it is again experience, and not representation, which permits her to transcend this initial level. In accounting for such facts one is in the habit of saying that "optical contact" prevails over the subject's every other preoccupation. But it is necessary to understand that, if the visual seems thus to predominate, this does not speak in favor of the primacy of representation. On the contrary, it shows that the subject's optics remains immediate and does not yet give rise to the mental constructions transforming the object as it appears to be into the object as it is. Furthermore, this transformation is effected as a function of the motor schemata in the very course of the experimental groping; therefore it could not be directed by representations precisely because it consists in preparing their elaboration. In short it can be said that, at the level of empirical groping, representation does not yet intervene and that progressive comprehension is assured by a purely sensorimotor assimilation.

With regard to accommodation, that is tantamount to saying that the subject's experience remains immediate and consequently victim of the most naïve phenomenalism. That is demonstrated not only by the primacy of the "optical contact" as we

have just interpreted it, but also by Observations 168–170. Two correlative conditions are necessary to replace the universe such as it is: the constitution of permanent objects (inserted in coherent groups of displacements and maintaining among themselves clear causal relations) and the elimination of illusions due to the particular point of view (by insertion of this point of view in an objective system of laws of perspective). These two processes are interdependent: In order to constitute "objects" in a system of spatial and causal relations, one must place oneself among these objects and, in order thus to come out of one's own perspective one must elaborate a system of spatial, causal, and objective relations. Now Observations 168–170 show us, in the most naïve and concrete form, how immediate experience does not consist in relating oneself to the object and how the constitution of the object first consists in detaching it from oneself. Jacqueline and Lucienne do not at first succeed in picking up rags, opening boxes or removing contents from the container because unknowingly they create an obstacle to their own efforts and because the most difficult obstacle to perceive in everything is oneself! It can readily be seen how such facts again reveal the primacy of action over representation and consequently the immediate character of the subject's experiences as well as his assimilatory activity.

On the other hand if, during this fifth stage, representation is not yet freed from perception, the system of signs due to the growing mobility of the schemata effects new progress toward prevision. We recall that from the time when the secondary schemata begins to be coördinated (fourth stage), this mobility allows the signals to become detached from the activity itself in order to constitute a prevision relating to the objects themselves. This capacity for prevision still develops during the fifth stage and, without resulting in actual representation, it gives rise to practical anticipations based upon generalization of earlier experiences.

Here are some examples:

Observation 175.—At 1;2 (30) Jacqueline is standing in a room which is not hers and examines the green wallpaper. Then she touches it gently and at once looks at her fingertips. This is evidently the generalization

of schemata due to the following secondary or tertiary circular reactions: touching food (jams, etc.) and looking at her fingers, or dipping her fingertips in soap lather while being washed and examining them afterward.

Another example. At 1;1 (23) Jacqueline, finding an orange peel, turns it upside down on the table to make it rock. She therefore immediately foresees the signification of this object.

At 1;3 (12) she is standing in her playpen and I place a clown, which she recently received, on the top of the frame, in different places in sequence. Jacqueline advances laboriously along the frame but, when she arrives in front of the clown, she grasps it very cautiously and delicately, knowing that it will fall at the slightest shake. She behaved in this way ever since the first attempt.

One observes that, in each of these three cases, Jacqueline foresees certain properties of the object which are independent of its action with respect to herself. The green wallpaper is conceived as though it ought to leave colored traces, the orange peel as being able to rock itself once it is placed in a suitable position and the clown as falling down at the first touch. These previsions, like those of the fourth stage, reveal an objectification of the signals into signs relating to the external processes themselves. In other words, the signification of perceived objects is not only that they are to be grasped, shaken, swung, rubbed, etc., but that they are the cause of phenomena external to the action itself. But, on the other hand, these signs are not limited, like those in Observations 132–135, to basing prevision on sequences already observed in the same form. It would seem that there exists, in the three cases of Observation 175, a generalization from analogous experiences and a generalization with present groping. In short, such signs add to the characteristics of those which are based on simple mobile schemata the advantage of the "experiment in order to see" or of the directed groping belonging to the present stage.

Before proceeding to the study of inventions through mental combinations and representations, it is fitting again to remember that the directed groping of active experimentation is essentially vicarious. As soon as this process results in the formation of new schemata, the latter can function either in invention through mental combination as we shall see, or in the capacity of familiar means which apply to new circumstances. In the second case we

refer to the situations studied in Chapter IV: that is quite natural and amounts to saying that "familiar means" can have been acquired just as well through active experimentation as through secondary circular reaction.

In a general way, the fact should be emphasized that the behavior patterns characteristic of the different stages do not succeed each other in a linear way (those of a given stage disappearing at the time when those of the following one take form) but in the manner of the layers of a pyramid (upright or upside down), the new behavior patterns simply being added to the old ones to complete, correct or combine with them. In this connection we can cite certain behavior patterns in the course of which the schemata in process of formation due to the process of tertiary circular reaction are applied to new circumstances, not through "active experimentation" nor even through "application of familiar means to new situations," but simply in the manner of the "procedures to make an interesting spectacle last" (studied in Chapter III, §4). Here is an example:

Observation 176.—At 1;6 (8) Jacqueline is seated on an adult's bed having in front of her a big slanting quilt. I place a little wooden lamb on the peak of this mountain and, striking the lower part of the quilt, I made the animal descend several centimeters at each shake. Jacqueline at once profits from this observation and brings the animal back to her each time I put it again on the summit of the quilt.

I then place the lamb on a table, 1 m. away from the bed, at the same height but separated by a corridor 50 cm. wide which Jacqueline notices; nevertheless she strikes the quilt as before while watching the lamb. I remove the latter after a moment and then put it back on the same table; Jacqueline recommences hitting the quilt.

A quarter of an hour later I begin the experiment over again with a celluloid fish. Jacqueline makes it fall by hitting the quilt and, when I place the fish on the table, she continues to hit the quilt. On the other hand, when I place the fish (or the lamb) on a window sill, further away and higher up, she abandons all efforts.

At 1;6 (12) Jacqueline spontaneously engages in behavior of the same kind but without any possible suggestion from the preceding experiment. She happens to shake a chair whose back touches the open window. She thus indirectly makes the window move and then recommences intentionally to shake the chair while watching the window. Then she walks in the room without seeming to think about the thing any more. But, knocking another chair 1.50 m. away, she immediately

shakes it in the same way and looks at the window. She continues for a while in spite of failure while attentively watching the window.

At 1;6 (20) Jacqueline brings down a watch chain from the top of a quilt by hitting the latter. I then place the chain on a chair, 50 cm. from the bed. She strikes the quilt three times while watching the chain but without conviction and as though "to see if" that can yield something.

Thus it may be seen how directed groping as well as the application to the experiment of schemata due to tertiary circular reaction can, in situations of which the child does not understand the particulars, be extended in "applications of familiar means to new situations" and even in "procedures to make an interesting spectacle last" reminiscent of the behavior patterns of the fourth and third stages.

In conclusion, the behavior patterns characteristic of the fifth stage constitute a homogeneous totality: The "tertiary circular reaction" marks the beginning of experimental behavior, whereas the "discovery of new means through active experimentation" utilizes the method thus found by the child for the solution of new problems. As we shall see in Volume II, moreover, this more advanced adaptation of intelligence to the real is accompanied by a structurization of the external environment into permanent objects and coherent spatial relations as well as by a correlative objectification and spatialization of causality and time.

CHAPTER VI

THE SIXTH STAGE:

The Invention of New Means Through Mental Combinations

The ensemble of intelligent behavior patterns studied hitherto—secondary circular reaction, application of familiar means to new situations, tertiary circular reaction and discovery of new means through active experimentation—characterizes a single, big period. To be sure, there is progress from one type to another behavior pattern and so one can consider the three main groups which we have delineated in the preceding chapters as forming three sequential stages (it being understood that the advent of each new stage does not abolish in any way the behavior patterns of the preceding stages and that new behavior patterns are simply superposed on the old ones). But the facts remain so complicated and their sequence can be so rapid that it would be dangerous to separate these stages too much. On the other hand, with the behavior patterns, which we are now going to describe, begins a new period which everyone will concur in considering as appearing tardily, much later than the preceding behavior patterns. We can therefore speak of a sixth stage which does not mean that the behavior patterns hitherto under study will disappear, but merely that they will henceforth be completed by behavior patterns of a new type: invention through deduction or mental combination.

This new type of behavior patterns characterizes systematic intelligence. Now it is the latter which, according to Claparède, is governed by awareness of relationships and no longer by em-

pirical groping. It operates, according to Köhler, by sudden structurizations of the perceptual field or, according to Rignano, is based on purely mental experience. In short, all writers, whether associationists like Rignano, believers in "structures" like Köhler or, like Claparède, believers in a more or less directed groping, agree that there exists an essential moment in the development of intelligence: the moment when the awareness of relationships is sufficiently advanced to permit a reasoned prevision, that is to say, an invention operating by simple mental combination.

We are consequently confronted by the most delicate problem which any theory of intelligence has to treat: that of the power of invention. Hitherto the different forms of intellectual activity which we have had to describe have not presented particular difficulties of interpretation. Either they consisted in apprenticeships during which the role of experience is evident, discovery consequently surpassing true invention, or else they consisted in simple applications of the familiar to the new. In both cases, thereafter, the mechanism of adaptation is easy to explain and the play of assimilations and of primitive accommodations suffices to explain all the combinations. On the other hand, as soon as real invention arises the process of thought baffles analysis and seems to escape determinism. Will the schemata to which the preceding facts have accustomed us fail in the task, or will the new facts which we are about to describe appear once more to be prepared by all the functional mechanism of earlier activities?

Let us observe at the outset, in this connection, but without wishing to find an explanation ahead of time, that real invention arises as a function of a sort of rhythm conditioned by the ensemble of the preceding behavior patterns. This rhythm determines the sequence of acquisitions and applications. With secondary circular reaction we are in the midst of acquisition: New schemata are constructed through reproductive assimilation and accommodation combined. With the application of familiar means to new situations, these same schemata give rise to some original applications (through generalizing assimilation) without actual acquisition being involved. With tertiary circular reac-

tion and the discovery of new means through apprenticeship, we are once again in a period of apprenticeship but, in this case, the very complexity of acquisition involves a constant intervention of all that has been acquired earlier. With invention through mental combination we can at last speak of a new process of application, for all invention presupposes a mental combination of already elaborated schemata, but an application on a par with acquisition since there is invention and consequently there are original combinations. Given this rhythm, invention is therefore comparable to the "application of familiar means to new situations" since, like the latter, it operates by deduction; but this deduction, being creative, also partakes of the processes of acquisition hitherto under study and, oddly enough, of the discovery of new means through active experimentation.

§1. THE FACTS.—First, here is a series of observations beginning with those most reminiscent of the discoveries due to directed groping. It happens that the same problem, such as that of the stick to be brought through the bars, can give rise to solutions through real invention as well as to solutions involving simple experimental groping. Analysis of such cases will enable us to see right away both the originality of the new behavior patterns and their relationship to the preceding ones. This relative contrast of solutions can be observed either in passing from one child to another or in the same child several months later.

Observation 177.—In order to explain the difference between the present and preceding behavior patterns it can be instructive to examine the way in which Laurent all at once discovered the use of the stick after not having known how to utilize that instrument for several months.

In contradistinction to Jacqueline and Lucienne whom we know were subjected to numerous experiments during which they had opportunity to "learn" to use the stick, Laurent only manipulated it at long intervals until the time when he knew how to use it spontaneously. It is therefore worth while, in order to characterize that moment, briefly to retrace the ensemble of Laurent's earlier behavior patterns relating to the stick.

As early as 0;4 (20), that is to say, at the beginning of the third stage, Laurent is confronted by a short stick which he assimilates to some object. He shakes it, rubs it against the wicker of his bassinet, draws

himself up, etc. In a general way he makes it the equivalent of the paper knife in Observation 104. But, at 0;4 (21), when Laurent is holding the stick, he happens to strike a hanging toy and immediately continues. But during the next hours Laurent no longer tries to reproduce this result even when I put the stick back into his hand.—This first situation, then, is not an example of the "behavior pattern of the stick." Laurent confined himself to momentarily inserting a new element in an already constructed schema (the schema of striking). But the fortuitous intervention of the latter gave rise to no immediate comprehension or even experimentation. The following days I give him the stick again and try to make him associate it to the activity of the various schemata. But Laurent does not react then or in the following weeks. The "behavior pattern of the stick," that is to say, the utilization of the stick in the capacity of intermediate or instrument, does not seem able to be acquired during the stage of the secondary circular reactions, even when chance has favored the momentary insertion of the stick in an already existing schema.

In the course of the fourth stage, characterized by the coördination of the schemata, the use of the stick makes no progress. However, during this stage, the child comes to use the hand of another person as an intermediate to act upon distant objects, thus succeeding in spatializing causality and preparing the way for experimental behavior. But when, at 0;8 or even 0;9 I give Laurent the stick, he only uses it to strike around him and not yet to displace or bring to him the objects he hits.

At 1;0 (0)—that is to say, well into the fifth stage (it is during this stage that Jacqueline and Lucienne succeeded in discovering the utilization of the stick)—Laurent manipulates a long wooden ruler for a long time, but only arrives at the three following reactions. In the first place, he turns the stick over systematically while transferring it from one hand to the other. Then he strikes the floor, his shoes and various objects with it. In the third place, he displaces it by pushing it gently over the floor with his index finger. Several times I place, at a certain distance from the child, some attractive objective to see whether Laurent, already holding the stick, will know how to use it. But each time Laurent tries to attain the object with his free hand without having the idea of using the stick. Other times I place the stick on the floor, between the objective and the child, in order thus to provoke a visual suggestion. But the child does not react to that either.—There does not yet exist, therefore, any trace of the "behavior pattern of the stick."

At 1;0 (5), on the other hand, Laurent is playing with a little child's cane which he handles for the first time. He is visibly surprised at the interdependence he observes between the two ends of this object. He displaces the cane in all directions, letting the free end drag along the floor, and studies the coming and going of this end as

function of the movements he makes with the other end. In short, he begins to conceive of the stick as a rigid entity. But this discovery does not lead him to that of the instrumental signification of the stick. In effect, having by chance struck a tin box with the cane, he again strikes it but without the idea either of making it advance in that way or of bringing it to him.—I replace the box with various more tempting objectives: the child's reaction remains the same.

At 1;2 (25) I give him back the stick because of his recent progress. He has just learned to put objects on top of one another, to put them into a cup and turn it upside down, etc.: the relationships which belong to the level of the behavior pattern of the stick (see Vol. II). He grasps the stick and immediately strikes the floor with it, then strikes various objects (boxes, etc.) placed on the floor. He displaces them gently but it does not occur to him to utilize this result systematically. At a given moment his stick gets caught in a rag and drags it for a few moments in the course of its movements. But when I put various desirable objectives 50 cm. or 1 m. away from Laurent he does not utilize the virtual instrument he holds.—It is apparent that, if I had repeated such experiments at this period, Laurent, like his sisters, would have discovered the use of the stick through directed groping and apprenticeship. But I broke off the attempt and only resumed it during the sixth stage.

At 1;4 (5) Laurent is seated before a table and I place a bread crust in front of him, out of reach. Also, to the right of the child I place a stick about 25 cm. long. At first Laurent tries to grasp the bread without paying attention to the instrument, and then he gives up. I then put the stick between him and the bread; it does not touch the objective but nevertheless carries with it an undeniable visual suggestion. Laurent again looks at the bread, without moving, looks very briefly at the stick, then suddenly grasps it and directs it toward the bread. But he grasped it toward the middle and not at one of its ends so that it is too short to attain the objective. Laurent then puts it down and resumes stretching out his hand toward the bread. Then, without spending much time on this movement, he takes up the stick again, this time at one of its ends (chance or intention?), and draws the bread to him. He begins by simply touching it, as though contact of the stick with the objective were sufficient to set the latter in motion, but after one or two seconds at most he pushes the crust with real intention. He displaces it gently to the right, then draws it to him without difficulty. Two successive attempts yield the same result.

An hour later I place a toy in front of Laurent (out of his reach) and a new stick next to him. He does not even try to catch the objective with his hand; he immediately grasps the stick and draws the toy to him.

Thus it may be seen how Laurent has discovered the use of the stick almost without any groping when, during the preceding stages, he

handled it without understanding its usefulness. This reaction is therefore distinctly different from that of his sisters.

Observation 178.—We recall Jacqueline's gropings at 1;3 (12) when confronted by a stick to be brought through the bars of her playpen (Obs. 162). Now it happens that the same problem presented to Lucienne at 1;1 (18) gives rise to an almost immediate solution in which invention surpasses groping. Lucienne is seated in front of the bars and I place against them, horizontally and parallel to the bars (half way up them) the stick of Observation 162. Lucienne grasps it at the middle and merely pulls it. Noticing her failure, she withdraws the stick, tilts it up and brings it through easily.

I then place the stick on the floor. Instead of raising it to pull it directly, she grasps it by the middle, tilts it up beforehand and presses it. Or else she grasps it by one end and brings it in easily.

I start all over again with a longer stick (30 cm. long). Either she grasps it by the middle and tilts it up before pulling it, or else she brings it in by pulling on one end.

Same experiment with a stick 50 cm. long. The procedure is obviously the same but, when the stick gets caught, she pulls it away briefly, then lets it go with a groan and begins over again in a better way.

The next day, at 1;1 (19), same experiments. Lucienne begins by merely pulling (once), then tilts up the stick and so rediscovers the procedures of the day before. At 1;2 (7) I resume the observation. This time Lucienne tilts up the stick before it touches the bars.

It may thus be seen how these attempts are reminiscent of Jacqueline's, taking place through groping and apprenticeship. Lucienne begins by merely pulling the stick and repeats this once the next day. But, in contrast to her sister's prolonged efforts, Lucienne at once profits from her failure and uses a procedure which she invents right away through simple representation.

Observation 179.—The example of the watch chain to be put into an aperture 16 x 34 mm. is more complex. Here again we remember Jacqueline's gropings (Obs. 173 and 173 repeated). But Lucienne has solved the problem by sudden invention:

At 1;4 (0) without ever having contemplated this spectacle, Lucienne looks at the box which I bring nearer and return without her having seen the contents. The chain spreads out on the floor and she immediately tries to put it back into the box. She begins by simply putting one end of the chain into the box and trying to make the rest follow progressively. This procedure which was first tried by Jacqueline, Lucienne finds successful the first time (the end put into the box stays there fortuitously), but fails completely at the second and third attempts.

At the fourth attempt, Lucienne starts as before but pauses, and after a short interval, herself places the chain on a flat surface nearby

(the experiment takes place on a shawl), rolls it up in a ball intentionally, takes the ball between three fingers and puts the whole thing in the box.

The fifth attempt begins by a very short resumption of the first procedure. But Lucienne corrects herself at once and returns to the correct method.

Sixth attempt: immediate success.

Thus one sees the difference between the behavior patterns of Jacqueline and of Lucienne. What was, in the former, the product of a long apprenticeship, was suddenly invented by the latter. Such a difference is surely a question of the level. So it is that at 2;6 (25) Jacqueline, with whom I repeat the experiment, solves the problem unhesitatingly. By grasping the chain in both hands she puts it in with her left hand while holding the remaining part in her right, to prevent it from falling. In the event that it gets caught, she corrects the movement.

Observation 180.—Another mental invention, derived from a mental combination and not only from a sensorimotor apprenticeship was that which permitted Lucienne to rediscover an object inside a matchbox. At 1;4 (0), that is to say, right after the preceding experiment, I play at hiding the chain in the same box used in Observation 179. I begin by opening the box as wide as possible and putting the chain into its cover (where Lucienne herself put it, but deeper). Lucienne, who has already practiced filling and emptying her pail and various receptacles, then grasps the box and turns it over without hesitation. No invention is involved of course (it is the simple application of a schema, acquired through groping) but knowledge of this behavior pattern of Lucienne is useful for understanding what follows.

Then I put the chain inside an empty matchbox (where the matches belong), then close the box leaving an opening of 10 mm. Lucienne begins by turning the whole thing over, then tries to grasp the chain through the opening. Not succeeding, she simply puts her index finger into the slit and so succeeds in getting out a small fragment of the chain; she then pulls it until she has completely solved the problem.

Here begins the experiment which we want to emphasize. I put the chain back into the box and reduce the opening to 3 mm. It is understood that Lucienne is not aware of the functioning of the opening and closing of the matchbox and has not seen me prepare the experiment. She only possesses the two preceding schemata: turning the box over in order to empty it of its contents, and sliding her finger into the slit to make the chain come out. It is of course this last procedure that she tries first: she puts her finger inside and gropes to reach the chain, but fails completely. A pause follows during which Lucienne manifests a very curious reaction bearing witness not only to the fact that she tries to think out the situation and to represent to herself through mental combination the operations to be performed, but also

to the role played by imitation in the genesis of representations. Lucienne mimics the widening of the slit.

She looks at the slit with great attention; then, several times in succession, she opens and shuts her mouth, at first slightly, then wider and wider! Apparently Lucienne understands the existence of a cavity subjacent to the slit and wishes to enlarge that cavity. The attempt at representation which she thus furnishes is expressed plastically, that is to say, due to inability to think out the situation in words or clear visual images she uses a simple motor indication as "signifier" or symbol. Now, as the motor reaction which presents itself for filling this role is none other than imitation, that is to say, representation by acts, which, doubtless earlier than any mental image, makes it possible not only to divide into parts the spectacles seen but also to evoke and reproduce them at will. Lucienne, by opening her mouth thus expresses, or even reflects her desire to enlarge the opening of the box. This schema of imitation, with which she is familiar, constitutes for her the means of thinking out the situation. There is doubtless added to it an element of magic-phenomenalistic causality or efficacy. Just as she often uses imitation to act upon persons and make them reproduce their interesting movements, so also it is probable that the act of opening her mouth in front of the slit to be enlarged implies some underlying idea of efficacy.

Soon after this phase of plastic reflection, Lucienne unhesitatingly puts her finger in the slit and, instead of trying as before to reach the chain, she pulls so as to enlarge the opening. She succeeds and grasps the chain.

During the following attempts (the slit always being 3 mm. wide), the same procedure is immediately rediscovered. On the other hand, Lucienne is incapable of opening the box when it is completely closed. She gropes, throws the box on the floor, etc., but fails.

Observation 181.—At 1;6 (23) for the first time Lucienne plays with a doll carriage whose handle comes to the height of her face. She rolls it over the carpet by pushing it. When she comes against a wall, she pulls, walking backward. But as this position is not convenient for her, she pauses and without hesitation, goes to the other side to push the carriage again. She therefore found the procedure in one attempt, apparently through analogy to other situations but without training, apprenticeship, or chance.

In the same kind of inventions, that is to say, in the realm of kinematic representations, the following fact should be cited. At 1;10 (27) Lucienne tries to kneel before a stool but, by leaning against it, pushes it further away. She then raises herself up, takes it and places it against a sofa. When it is firmly set there she leans against it and kneels without difficulty.

Observation 181 repeated.—In the same way Jacqueline, at 1;8 (9) arrives at a closed door—with a blade of grass in each hand. She stretches out her right hand toward the knob but sees that she cannot turn it without letting go of the grass. She puts the grass on the floor, opens the door, picks up the grass again and enters. But when she wants to leave the room things become complicated. She puts the grass on the floor and grasps the doorknob. But then she perceives that in pulling the door toward her she will simultaneously chase away the grass which she placed between the door and the threshold. She therefore picks it up in order to put it outside the door's zone of movement.

This ensemble of operations, which in no way comprises remarkable invention, is nevertheless very characteristic of the intelligent acts founded upon representation or the awareness of relationships.

Observation 182.—At 1;8 (30) Jacqueline has an ivory plate in front of her, pierced by holes of 1–2 mm. in diameter and watches me put the point of a pencil in one of the holes. The pencil remains stuck vertically there and Jacqueline laughs. She grasps the pencil and repeats the operation. Then I hold out another pencil to her but with the unsharpened end directed toward the plate. Jacqueline grasps it but does not turn it over and tries to introduce this end (the pencil is 5 mm. in diameter) into each of the three holes in succession. She keeps this up for quite a while even returning to the smallest holes. On this occasion we make three kinds of observations:

1. When I return the first pencil to Jacqueline she puts it in the hole correctly at once. When I hand it to her upside down, she turns it over even before making an attempt, thus revealing that she is very capable of understanding the conditions for putting it in. On the other hand, when I hold out the second pencil correctly directed (the point down) she also puts it in by the point. But if I offer it to her upside down she does not turn it over and recommences wishing to put it in by the unsharpened end. This behavior pattern remained absolutely constant during thirty attempts, that is to say, Jacqueline never turned the second pencil over whereas she always directed the first one correctly. Everything happens as though the first attempts had given rise to a sensorimotor schema which persisted in acting during the whole series: the two pencils were accordingly conceived as being in contrast to each other, the first being that which one puts into the hole easily and the second that which resists. However, the pencils are of course identical from the point of view of the facility with which they can be put in the hole; the first is merely shorter than the second and is green and the second is brown (both have hard, black lead).

2. Several times Jacqueline, seeing the second pencil will not go in, tries to put it in the same hole as the first one. Hence, not only does she try to put it in by the unsharpened end but also she wants to put it into a hole which is already filled by the other pencil. She

resumed this strange procedure several times despite total failure. This observation shows very well how, in a child of this age, representation of things is still ignorant of the most elementary mechanical and physical laws and so makes it possible to understand why Jacqueline so obstinately tries to put in the second pencil by the wrong end. Ignorant of the fact that two objects cannot occupy the same small opening, there is no reason for her not to try to put an object 5 mm. in diameter in a 1–2 mm. hole.

3. At about the thirtieth attempt, Jacqueline suddenly changes methods. She turns the second pencil over as she does the first and no longer tries a single time to put it in by the wrong end. If the series of these new attempts is compared with the first series, one has the impression of a sudden understanding, as of an idea which arises and which, when it has suddenly appeared, definitively imposes itself. In other words, the second pencil has suddenly been assimilated to the first. The primitive schema (connecting the two pencils by contrast) has dissociated itself and the pencil which one did not turn over has been assimilated to the particular schema of the pencil that one had to turn over. This kind of a process is consequently again capable of making us understand the mechanism of invention.

The respect in which these behavior patterns are original in relation to the preceding ones may thus be seen. The child finds himself in a situation which is new to him, that is to say, the objects arising between his intentions and the arrival at an end demand unforeseen and particular adaptation. It is therefore necessary to find adequate means. Now these means cannot be brought back to the procedures acquired earlier in other circumstances (as in the "application of familiar means to new circumstances"); it is therefore necessary to innovate. If these behavior patterns are compared to all the preceding ones, they resemble most the "discovery of new means through active experimentation." Their functional context is exactly the same. But, contrary to the latter, the present behavior patterns do not appear to operate by groping or apprenticeship, but by sudden invention; that is to say, that instead of being controlled at each of the stages and *a posteriori* by the facts themselves, the searching is controlled *a priori* by mental combination. Before trying them, the child foresees which maneuvers will fail and which will succeed. The control of the experiment therefore bears upon the whole of this deduction and no longer, as before, upon the details of each particular step. Moreover, the procedure con-

ceived as being capable of succeeding is in itself new, that is to say, it results from an original mental combination and not from a combination of movements actually executed at each stage of the operation.

§2. INVENTION AND REPRESENTATION.—The two essential questions raised by such behavior patterns in relation to the preceding ones are those of *invention* and *representation.* Henceforth there exists invention and no longer only discovery; there is, moreover, representation and no longer only sensorimotor groping. These two aspects of systematic intelligence are interdependent. To invent is to combine mental, that is to say, representative, schemata and, in order to become mental the sensorimotor schemata must be capable of intercombining in every way, that is to say, of being able to give rise to true inventions.

How can this transition from directed gropings to invention, and from motor schema to representative schema be explained? Let us begin by reëstablishing the continuity between the extremes in order subsequently to account for the differentiations.

It must be understood, with regard to the first point of view, that the contrast between directed groping and actual *invention* is primarily due to a difference in speed. The structuring activity of assimilation only operates step by step in the course of experimental groping, so that it is not immediately visible, and one is tempted to attribute the discoveries which result from it solely to fortuitous contact with external facts. In invention, on the contrary, it is so rapid that the structurization seems sudden. The structuring assimilatory activity thus once again passes unnoticed at first glance and one is tempted to consider the "structures" as organizing themselves. Thereafter the contrast between the empiricism of simple groping and the intelligence of deductive invention seems to be complete. But if one thinks about the role of intellectual activity peculiar to combined assimilation and accommodation, one perceives that this activity is neither absent from empirical groping nor useless to the structuring of representations. On the contrary, it constitutes the real motor of both,

and the primary difference between the two situations stems from the speed at which the motor goes, a speed slowed down in the first case by the obstacles on the road and accelerated in the second case by the training acquired.

But this continuous increase of speed entails a differentiation in the very procedure of the functioning. At first cut up and visible from the outside, it becomes regularized and seems to be internalized by becoming rapid. In this respect, the difference between empirical groping and invention is comparable to that which separates induction from deduction. The empiricists have tried to reduce the second to the first, thus making induction the only genuine reasoning. Induction being, according to them, only a passive recording of the results of experience, deduction then became a sort of internal replica of this experience, a "mental experience" as Mach and Rignano put it. In contrast to this thesis is that of a certain logicism according to which induction and deduction have nothing in common, the former consisting, according to the empiricists, in a catalogue of statements, and the second in purely formal combinations. At last came the sound logistic analysis which showed the relationship as well as the contrast between these two complementary processes. Both consist in constructions of relationships, induction, thus involving deduction and resting upon its constructive activity. But in the first, construction is ceaselessly controlled from without and so can appeal to those extralogical procedures of anticipation which appeared to the empiricists to constitute the essence of thought, whereas in the second, construction is regulated from within, solely by the play of operations. So also, empirical groping already presupposes the mechanism of invention. As we have seen, there is no pure accommodation, but accommodation is always directed by a play of schemata whose reorganization, if it were spontaneous, would become identified with the constructive deduction of the present behavior patterns. But as this reorganization peculiar to accommodation is unable, when the problem transcends the subject's level, to dispense with a continuous external control, it always works through cumulative assimilation; that is to say, the structuring activity keeps a slow pace and only intercombines the sequential data of assimilation. In

the present case, on the contrary, in which the question raised is addressed to a mind sufficiently furnished with already constructed schemata so that the reorganization of these schemata operates spontaneously, the structuring activity no longer needs always to depend on the actual data of perception and, in the interpretation of these data, can make a complex system of simply evoked schemata converge. Invention is nothing other than this rapid reorganization and representation amounts to this evocation, both thus extending the mechanisms at work in the ensemble of the preceding behavior patterns.

From this point of view let us again take up Observations 177–182, comparing them to the mechanism of empirical groupings. As before, the point of departure of these behavior patterns consists in the impetus given by the schema assigning an end to the action; for instance, in Observation 180, sight of the chain in the matchbox sets in motion the schema of grasping. This schema of the goal immediately arouses a certain number of schemata which the child will utilize as initial means and which he must accommodate, that is to say, differentiate according to the variations of the new situation. In Observation 180, Lucienne tries to turn the box over or to slide her finger into the slit in order to extract the chain. But in utilizing these schemata the child perceives at the same time the difficulties of the present situation. In other words, there occurs here, as in the course of empirical groping, an encounter with the unforeseen fact which creates an obstacle (the slit is too narrow to admit the finger). Now in both cases, this encounter entails a new intervention of earlier schemata. It is due to the latter that these unforeseen facts acquire meaning. The only difference is that, henceforth, such encounters with the obstacle no longer take place in the course of discovery (since the latter is no longer groping and consists in sudden invention) but beforehand, at the moment when the first procedures tried out as hypotheses fail, and when the problem is clarified by virtue of that very failure. In Observation 180, these auxiliary schemata which attribute a meaning to the facts are those that permit the child to understand what the slit is that he sees before him (= sign of a subjacent opening) and how it is troublesome (because it is too narrow). The child often

opens and closes boxes, wants to put his hand through very small openings, etc. Those are the schemata which confer a meaning on the present situation and which at the same time direct the search. They intervene, therefore, as secondary means and hence are subordinated to the initial procedure. It is then that invention comes in, in the form of sudden accommodation of the ensemble of those schemata to the present situation. How does this accommodation work?

It consists, as always, in differentiating the preceding schemata according to the variations of the present situation, but this differentiation, instead of operating through actual groping and cumulative assimilation, results from a spontaneous assimilation, hence more rapid and operating by means of simply representative attempts. In other words, instead of exploring the slit with his finger and groping until he has discovered the procedure which consists in drawing to him the inner side (of the box) in order to enlarge the opening; the child is satisfied to look at the opening, except for experimenting no longer on it directly, but on its symbolic substitutes. Lucienne opens and closes her mouth while examining the slit of the box, proving that she is in the act of assimilating it and of mentally trying out the enlargement of the slit; moreover, the analogy thus established by assimilation between the slit perceived and other openings simply evoked leads her to foresee that pressure put on the edge of the opening will widen it. Once the schemata have thus been spontaneously accommodated on the plane of simple mental assimilation, Lucienne proceeds to act and succeeds right away.

An interpretation of this sort applies to each of our observations. In Observation 179, for example, if Lucienne rolls the chain up into a ball to put it into the box after having noted the failure of the direct method, it is because the schemata acquired in putting the chain into a pail or a necklace into a watering can (Obs. 172) or again in squeezing materials, putting her pillow or handkerchief in her mouth, etc., afford her sufficient assimilation of the new situation. Instead of groping she mentally combines the operations to be performed. But this mental experience does not consist in mnemonic evocation of already manufactured images; it is an essentially constructive process the representation

of which is only a symbolic adjuvant, since genuine invention exists and it never perceived a reality identical to the one it is in the process of elaborating. In Observations 180 and 180 repeated there also exists spontaneous functioning of the schemata of displacement, by analogy, to be sure, with the experiments the child was able to make in reality, but this analogy entails imagination of new combinations. Finally, in Observation 182, we see how an initial schema can be differentiated, without progressive groping, through sudden dissociation and assimilation.[1]

But how can we account for the mechanism of this spon-

[1] In order better to understand the mechanism of this assimilation which has become deductive while remaining on the plane of sensorimotor operations, let us again analyze a case of elementary practical invention observed in an adult and consequently capable of correct introspection. While driving an old automobile I am bothered by oil on the steering wheel which makes it slippery. Lacking time to stop, I take out my handkerchief and dry the spots. When putting it in my pocket I observe that it is too greasy and look for a place to put it without soiling anything. I put it between my seat and the one next to me, as deeply as possible in the crevice. An hour later the rain forces me to close the windshield but the resulting heat makes me try to open it a little. The screws being worn out, I cannot succeed; it only stays wide open or completely shut. I try to hold the windshield slightly open with my left hand, but my fatigue makes me think that some object could replace my hand. I look around me, but nothing is in evidence. While looking at the windshield I have the impression that the object could be put, not at the bottom of the windshield (one pushed it at the bottom to open it), but by wedging it in the angle formed by the right edge of the windshield and the vertical upright of the body of the car. I have the vague feeling of an analogy between the solution to be found and a problem already solved before. The solution then becomes clarified. My tendency to put an object into the corner of the windshield meets a sort of motor memory of having just a few minutes before placed something into a crevice. I try to remember what it was, but no definite representation comes to mind. Then suddenly, without having time to imagine anything, I understand the solution and find myself already in the act of searching with my hand for the hidden handkerchief. Therefore the latter schema directed my search and directed me toward the lateral corner of the windshield when my last idea was a different one.

This trite observation demonstrates very well how a sensorimotor search can arouse schemata previously acquired and make them function independently of internal language and clear representation. The tendency to introduce an object into a slit, in this example, is modeled exactly on a schema remaining in an almost purely motor state, and the conjunction thus produced suffices to insure discovery of a solution. One therefore understands how a sensorimotor deduction is possible in the small child through simple practical evocation of the schemata and independently of a well-defined system of representations.

taneous reorganization of schemata? Take, for example, the construction of the schema of "rolling into a ball" in Observation 179, or that of "widening the slit" in Observation 180; does this construction consist in a sudden structurization of representations or of the perceptual field, or is it the result of assimilatory activities prior to invention? As we have just recalled, a certain number of already acquired schemata direct the search at the moment of invention without, however, any one of them containing in itself the correct solution. For example, before rolling the chain up into a ball to put it in the narrow opening, Lucienne has already: (1) squeezed the material, (2) put the chain in a wide opening, and (3) compared large objects to inadequate openings (as when she tried to bring objects through the bars of her playpen). In Observation 180 she also possesses the earlier schemata we have already emphasized. The question raised is therefore to find out how these schemata will intercoördinate in order to give rise to invention: Is it by a structuring independent of their genesis or due to the very activity which engendered them and which is now pursued without any longer depending on the external circumstances in which it began? One might as well ask whether ideas organize themselves in the course of theoretical invention or whether they are organized as a function of implicit judgments and of the potential intelligent activity they represent. We do not doubt that the second of these two theses is in both cases (in sensorimotor intelligence as well as in reflective thought) much the most satisfying to the mind, the first only consisting in a manner of speaking which veils the dynamism of the facts with static language.

But how is this reorganization of schemata to be conceived if it must fulfill the dual condition of extending their assimilatory activity and of liberating itself from the external circumstances in which this activity began? It is due to the process of reciprocal assimilation but in so far as it is henceforth extended on a plane independent of the immediate action.

Here we rediscover a remark already made in connection with the "application of familiar means to new circumstances"; it is that, in the act of practical intelligence, means are subordinated to ends through a coördination analogous to that of the

heterogeneous schemata in the case of intersensorial coördinations (hearing and sight, etc.), hence through reciprocal assimilation of the schemata. In other words, each schema tends to extend the assimilatory activity that gave rise to it (just as every idea tends to extend the judgments from which it derived), and consequently applies to the ensemble of the situations which lend themselves to it. Thereafter, when there is a watch chain to be put into a narrow opening, the schemata presenting some analogy to the situation and so capable of assimilating the data will enter into activity by themselves. We have constantly met with examples illustrating this process. But, until the present time, the activity thus set in motion has always given rise to real actions, that is to say, to immediate applications ("application of familiar means to new circumstances") or to empirical groping. The novelty of the case of invention consists, on the contrary, in that henceforth the schemata entering into action remain in a state of latent activity and combine with each other before (and not after) their external and material application. This is why invention seems to come from the void. The act which suddenly arises results from a previous reciprocal assimilation instead of manifesting its vicissitudes before everyone. The best example of this process is in footnote 1 of this chapter. Introspection enabled us to observe clearly how the schema of the handkerchief wedged down into a slit assimilated progressively and mentally the schema of the object to be slid into the opening of the windshield and vice versa, this reciprocal assimilation bringing with it the invention of the correct solution. With respect to Lucienne, Observation 180 also shows this explanation to be well founded. The gesture of opening and closing the mouth in the presence of the opening to be widened indicates sufficiently clearly how the internal reorganization of the schemata works through assimilation. G. Tarde's[2] well-known formula illustrates this mechanism: Invention, Tarde said, results from the interference of independent currents of action. The process of this interference could only be, in our language, that of reciprocal assimilation.

In short, invention through sensorimotor deduction is

2 G. Tarde, *Les lois sociales,* Paris, Alcan.

nothing other than a spontaneous reorganization of earlier schemata which are accommodated by themselves to the new situation, through reciprocal assimilation. Until the present time, that is to say, including empirical groping, the earlier schemata only functioned due to real use, that is to say, by actual application to a concretely perceived datum. So it is that in Observation 165 Jacqueline must really see that the rooster is stopped by the bars of the playpen and she must really have noted the possibility of tilting it up when it fell backward by chance, in order that she might acquire the idea of pushing it back systematically before tilting it up and bringing it in between the bars. The earlier schemata thus intervene to give a meaning to these events, but they only intervene when a concretely perceived datum (the rooster's fall, etc.) excites and makes them function. On the contrary, in preventive deduction, the schemata function internally by themselves, without requiring a series of external acts to aliment them continually from without. It is also necessary, of course, that a problem be raised by the facts themselves and that this problem set up, in the capacity of hypothesis, the use of a sensorimotor schema serving as initial means (otherwise we would no longer be in the realm of practical intelligence and would already reach the plane of reflective intelligence). But, once the goal has been set and the difficulties encountered by the use of initial means have been perceived, the schemata of the goal, those of the initial means and the auxiliary schemata (evoked by awareness of the difficulties) organize themselves into a new totality, without there being need of external groping to support their activity.

It is therefore inaccurate to speak, as does the empiric theory of "mental experience," of a simple internalization of earlier actual experiences; what has been internalized is solely the knowledge acquired due to these experiences. But actual or external experience involves from the outset, as does simply mental deduction, an internal assimilatory activity which forms schemata, and it is this activity, internal from the very beginning, which henceforth functions by itself without having any further need of external alimentation. Let us keep the term "mental experience" to designate these primitive deductions. But this is

on condition that we remember that all experience, including empirical groping, presupposes a previous organization of assimilatory schemata and that contact with the facts is nothing, at any level, outside of accommodation to these schemata. The baby who mentally combines the operations to be performed in order to widen the slit of the matchbox is in the same situation as the older child who no longer needs to count apples with his fingers in order to establish that "2 + 2 are 4" and who confines himself to combining the numbers. But the latter "mental experience" would be incomprehensible if, ever since the numeration of concrete objects, an abstract activity did not assimilate the realities not provided by themselves with numerical properties. Mental experience is therefore an assimilation functioning by itself and thus becomes partly formal, in contrast to the initial material assimilation. So also, Lucienne's deduction pertaining to the matchbox results from a spontaneous functioning of her assimilatory schemata when they succeed in intercombining without any immediate perceptive content and only now operating through evocations. Deduction thus appears at its beginnings as the direct extension of earlier mechanisms of assimilation and accommodation, but on a plane which begins to become differentiated from direct perception and action.

Thereafter can one state, as does the theory at the opposite extreme, that invention is due to an immediate structuring of the perceptual field independently of any apprenticeship and of earlier actions? The foregoing observations do not seem to favor so radical a thesis any more than the thesis of "mental experience" of the pure empiricists. The defect of the empirical thesis is that it does not explain the creative element of invention. By making all deduction the internal repetition of external groupings, it ends by negating the existence of a constructive activity, remaining internal (at all levels) and which alone accounts for the progressive purification of reasoning. But the theory of "structures," by emphasizing the originality of invention too much, leads to the same result and, in order to account for novelties without invoking the activity belonging to combined assimilation and accommodation, finds itself obliged to attribute them to a structural preformation. While empirical association-

ism considered all constructive deduction as being the internal replica of external experiences already completely organized, according to the theory of structures it is an outward projection of internal forms also completely prepared in advance (because they are connected with the nervous system, with the *a priori* laws of perception, etc.). But analysis of the assimilatory activity leads us to doubt this. If the schemata of assimilation seem to us, in the case of Observations 177–182, to reorganize themselves spontaneously when confronted by the problem raised by the external environment, that does not mean at all that these schemata, however global and totalizing they may be, are identical to "structures" imposing themselves independently of any intellectual construction. The assimilatory schema is not, in effect, an entity separable from assimilatory and accommodating activity. It is only formed by functioning and it only functions in experience. The essential is, therefore, not the schema in so far as it is a structure, but the structuring activity which gives rise to the schemata. Thereafter if, at a given moment, the schemata reorganize themselves until they bring about inventions through mental combination, it is simply because the assimilatory activity, trained by many months of application to the concrete data of perception, finally functions by itself by only using representative symbols. That does not mean at all, let us repeat, that this purification is a simple internalization of earlier experiences: Gestalt psychology has happily emphasized this point by showing that the reorganization peculiar to invention creates the new. But that means that the reorganization is not produced by itself, as though the schemata were endowed with a structure independently of the assimilatory activity which gave rise to them; the reorganization that characterizes invention simply extends this activity. So it is that, in observing our children (Obs. 177–182) and doubtless every time the history of the subjects examined is known in detail, it is possible to discover which old schemata intervened in the course of invention. Invention is no less creative, for all this, but it also presupposes a genetic process which functioned long before it.

What is now the role of *representation* in these first sensorimotor deductions? At first it seems vital: due to representation,

reciprocal assimilation can remain internal instead of giving rise, from the outset, to empirical gropings. It is therefore due to representation that "mental experience" succeeds actual experimentation and that assimilatory activity can be pursued and purified on a new plane, separate from that of immediate perception or action properly so called. That explains how Köhler, in his research concerning animal intelligence, was led to place all emphasis on the reorganization of the field of perception, as though it were that reorganization which brought intellectual invention after it, and not the reverse.[3] Representation is, in effect, a novelty essential to the formation of the behavior patterns of the present stage; it differentiates these behavior patterns from those of earlier stages. As we have seen, the most complex behavior patterns of the preceding stages, including the "discovery of new means through active experimentation," can dispense with representations, if one defines the latter as the evocation of absent objects. The motor anticipation peculiar to the mobile schemata of assimilation suffices to insure the comprehension of signs and the coördination of means and ends, without need for perception to substitute for representation. Rolling a chain up into a ball to put it in a narrow opening (when the subject never had occasion to roll anything up in such circumstances), combining in advance the positions of a stick before bringing it through bars (when the experiment is new to the child, widening a slit in advance in order to pull a hidden object out of it (when the child comes to grips with such a problem for the first time), all that presupposes that the subject represents the data offered to his sight otherwise than he perceives them directly. In his mind he corrects the thing he looks at; that is to say, he evokes positions, displacements or perhaps even objects without actually contemplating them in his visual field.

But if representation accordingly constitutes an essential acquisition, characteristic of this stage, we should not, however, exaggerate its scope. Representation is surely necessary for in-

[3] This role attributed to visual representation is not essential to Gestalt explanations, as revealed by the applications made by K. Lewin on the theory of form on the activity itself.

vention, but it would be erroneous to consider it the only cause. Furthermore, it can be maintained, with at least as much verisimilitude, that representation results from invention. The dynamic process belonging to the latter precedes the organization of images, since invention arises from spontaneous functioning of the schemata of assimilation. The truth seems to be that between invention and representation there is interaction and not simply a connection. What can be the nature of this interaction?

Things are clarified as soon as, with the theory of signs, one makes of the visual imagery peculiar to representation, a simple symbolism serving as "signifier," and of the dynamic process peculiar to invention the signification itself, in other words, the signified. Representation would thus serve as symbol to inventive activity which takes away nothing from its utility, since the symbol is necessary for deduction, but which relieves it of the burden of the too-difficult role it is sometimes made to play of being the motor of invention itself.

Here it is necessary to distinguish between two cases. The first is that in which the child merely evokes a movement or an operation already performed previously. For instance, when Lucienne perceives that her stick does not enter through the bars and that she tilts it up before trying to bring it through (Obs. 178), it is very possible that, in combining the new movements required by the operation, she visually evokes the movements of the stick previously performed (either just before or during other experiments). In that case representation plays the role of simple visual memory and one might think that invention consists in merely intercombining these image memories. Unfortunately this simple hypothesis, which is all on which the associationist theory of mental experience rests, encounters serious obstacles. Observation does not seem to show at all that during the first year of life the visual image extends action so easily. The observations described in connection with the "invention of new means through active experimentation" (Obs. 148–174) would be unexplainable if visual imagery formed itself by itself as a function of perception. How can it be explained, for example, that in Observation 165 Jacqueline had so much difficulty in profiting from her experiences (iinpossibility of bringing the

rooster through the bars) if an adequate visual representation allowed her to record what she sees? It seems, on the contrary, in such a case as though the apprenticeship were of a motor kind and the image did not yet extend the movement. Thereafter it would seem difficult to interpret invention through mental combination as a simple reorganization of the perceptual field. That reorganization results from the organization of the movements themselves and does not precede it. If images intervene it is therefore in the capacity of symbols accompanying the motor process and permitting the schemata to depend on them to function by themselves, independently of immediate perception. Images are not in this case elements but simply the tools of nascent thought.[4]

With regard to finding out why the image does not intervene at the level of empirical groping and seems necessary for invention through mental combination, that can be explained in accordance with the same hypothesis. The image being a symbol, it does not merely extend movement and perception joined together, and that is why it does not intervene in empirical groping. On the other hand, as soon as the schemata begin to function spontaneously, that is to say, outside of immediate groping and so to combine mentally, by that very fact they confer meaning on the traces left by perception,[5] and thenceforth raise them to the rank of symbols by relation to themselves. The image so constituted therefore becomes the signifier of which the signified is none other than the sensorimotor schema itself.

This leads us to the second case: when representation accompanies invention or mental combination, it happens that the

[4] See the excellent article by I. Meyerson, on Sur les images, in Dumas, *Nouveau Traité de Psychologie,* vol. II.

[5] Perhaps it will be said that these traces constitute images by themselves and that thus the image precedes invention. But, as we shall see ("The genesis of imitation"), perception is only extended in a representative image, that is to say only leaves a durable trace to the extent that it is revealed in imitation and that imitation itself is internalized. Now this internalization of imitation (of things as of persons) only occurs during the sixth stage at the time of the conquest of the mechanism of imitation under the influence of the liberation of schemata in connection with the immediate action. There is an ensemble of united intellectual processes and not at all a simple seriation going from sensation to the image as classic associationism believed.

child does not simply evoke the operations already performed but combines or compares the various images in his imagination. A good example is furnished by Observation 180, in which Lucienne opens her mouth while looking at a slit to be widened and so reveals the representative combinations she is in the process of realizing. But in such a case, the image is *a fortiori* a symbol. To utilize the imagined movements of the mouth in order to think out the operations to perform on an opening given in perception is surely to make the image into a simple "signifier" whose signification is to be sought in the motor operation itself.

In short, the fact that invention is accompanied by representation does not speak in favor either of the associationist theory of mental experience or even of the thesis of a spontaneous reorganization of the perceptual field, the thesis maintained by certain famous works derived from the Gestalt theory. All representation admits of two groups of elements which correspond to words or symbols, on the one hand, and to the concepts themselves, on the other, with regard to theoretical representation: Those are the signs and significations. Now the image is to be classed in the first group, whereas the second group is formed by the schemata themselves whose activity engenders invention. This shows that if invention presupposes representation, the converse is true, because the system of signs could not be elaborated independently of that of significations.

The manner of this advent of the image still remains to be clarified, inasmuch as it is derived from the activity of the schemata. But this is not the place to discuss it, for an important question must first be taken up: the problem of imitation. If it is true that the image does not, from the outset, accompany the movement, an intermediate term must be able to explain the transition from the motor to the representative and the image must in some way be acted before being thought. This intermediate is none other than imitation. Observation 180, in which Lucienne imitates the opening contemplated and imitates it with movements of the mouth—that is to say, with an organ not directly perceived by sight—is an excellent example of this transition. Let us postpone the problem until we can give the history

of the motor schemata from the particular point of view of imitation.

Let us limit ourselves to concluding that the intervention of "representations" in the mechanisms of the present stage implies that of a sixth and last type of "signifiers," that of *symbolic images*. One recalls that, during the fourth stage, the "signals" thitherto connected with the child's very movements begin to break away from the immediate action in the form of "signs," permitting prevision of events independent of the activity itself (Obs. 132–135). In the course of the fifth stage, the character of these "signs" is again accentuated; that is to say, they allow the child to foresee the properties of the objects themselves, thus adapting to the mechanism of the "tertiary circular reactions" (Obs. 175). Now, the development of signs in the dual sense of accommodation to the things themselves and of detachment with respect to the immediate action reaches its conclusion during the sixth stage when the schemata become capable of functioning alone through purely mental combination. On the one hand, due to the progress of accommodation (which, as we shall state later on, is perforce extended in imitation), the signs are modeled increasingly upon the characteristics of things and so tend to form "images." On the other hand, due to the progressive detachment of the signs with respect to the immediate action for the benefit of mental combination, these images are liberated from direct perception and become "symbolic."

One observes this dual movement in the facts of imitation and of play. Imitation characteristic of the sixth stage becomes representative as much because the child begins to imitate new movements by means of parts of his body invisible to him (imitation relating to head movements, etc., which leads to a representation of his own face), as because of the "deferred imitations" which presage symbolism (imitating absent persons, etc.). On the other hand, play, during the same period, becomes symbolic inasmuch as it begins to involve the "as if."

Now, from the point of view of meanings and of intelligence in general, the development of representations is not only predicated on the "invention of new means through mental combination" but on a series of other behavior patterns bearing wit-

ness to the existence of representative images necessary for the evocation of absent objects. Here is one example:

Observation 183.—At 1;6 (8) Jacqueline plays with a fish, a swan and a frog which she puts in a box, takes them out again, puts them back in, etc. At a given moment, she lost the frog. She places the swan and the fish in the box and then obviously looks for the frog. She lifts everything within reach (a big cover, a rug, etc.) and (long after beginning to search) begins to say *inine, inine* (= "*grenouille* = "frog"). It is not the word which set the search in motion, but the opposite. There was therefore evocation of an absent object without any directly perceived stimulus. Sight of the box in which are found only two objects out of three provoked representation of the frog, and whether this representation preceded or accompanied the act is of little importance.

Thus may be seen the unity of the behavior patterns of the sixth stage: Mental combination of schemata with possibility of deduction surpassing actual experimentation, invention, representative evocation by image symbols, so many characteristics marking the completion of sensorimotor intelligence and making it henceforth capable of entering the framework of language to be transformed, with the aid of the social group, into reflective intelligence.

CONCLUSIONS

"Sensorimotor" or "Practical" Intelligence and the Theories of Intelligence

There exists a sensorimotor or practical intelligence whose functioning extends that of the mechanisms of a lower level: circular reactions, reflexes and, still more profoundly, the morphogenetic activity of the organism itself. That is, it seems to us, the main conclusion of this study. It is now fitting to clarify the scope of such an interpretation by trying to supply a view of the whole of this elementary form of intelligence.

First, let us recall the picture of the possible explanations of the different psychobiological processes, in order to be able to insert our description. There are at least five different ways of conceiving the operation of intelligence and they correspond to the conceptions which we have already enumerated with regard to the genesis of acquired associations and of habits (Chap. II, §5) and of biological structures themselves (Introduction, §3).

In the first place, one can attribute intellectual progress to the pressure of the external environment whose characteristics (conceived as being constituted independently of the subject's activity) would impress themselves little by little on the child's mind. Such an explanation, a principle of Lamarckism when applied to hereditary structures, leads to setting up habit as a primary act and to considering mechanically acquired associations as being the source of intelligence. It is, in effect, difficult to conceive of links between environment and intelligence other than those of atomistic association when, like the empiricists, one neglects intellectual activity for the benefit of the constraint of things. The theories which regard the environment as a totality or a collection of totalities are obliged to admit that it is in-

telligence or perception which endows it with this quality (even if the latter corresponds to data independent of ourselves, which then implies a preëstablished harmony between the "structures" of the object and those of the subject). One does not see how, in the empirical hypothesis, the environment, were it regarded as constituting a totality in itself, could impose itself on the mind if not by successive fragments, that is to say, again by association. The primacy accorded to the environment therefore brings with it the *associationist* hypothesis.

In the second place, one can explain intelligence by intelligence itself, that is to say, presuppose the existence of an activity structured since its beginnings by merely applying itself to increasingly rich and complex contents. Accordingly an "organic intelligence" would exist as early as the physiological plane and would be extended into sensorimotor intelligence and, in the end, into truly reflective intelligence. This kind of an interpretation is of course on a par with vitalism in biology. With regard to associations and habits, as we have already seen, it considers them derived from intelligence at its different levels and not as primary facts. We shall call this second solution *intellectualistic.*

In the third place one can, in accordance with the *a priori* conceptions, consider the progress of intelligence as being due not only to a faculty given in a completed state but to the manifestation of a series of structures which are imposed from within on perception and intelligence in proportion to the needs aroused by contact with the environment. The structures would thus express the very contexture of the organism and its hereditary characteristics, which would nullify any reconciliation between intelligence and the associations or habits acquired under the influence of the environment.

In the fourth place, intelligence can be conceived as consisting in series of attempts or gropings, inspired by needs and the implications that result from them but selected by the external environment (as in biology the mutations are endogenous but their adaptation is due to a selection after the event). This pragmatic interpretation of intelligence would be intermediate between the empiricism of the first solution and the apriority of

the third. From the point of view of the relationship between intelligence and association based on habit it results, like the latter, in contrasting these two types of behavior, but less radically, since acquired association plays an essential role in groping.

Finally, in the fifth place, one can conceive of intelligence as the development of an assimilatory activity whose functional laws are laid down as early as organic life and whose successive structures serving it as organs are elaborated by interaction between itself and the external environment. Such a solution differs from the first in that it does not place the accent on experience alone but on the subject's activity making this experience possible. It therefore allies itself principally with the three other solutions. It is distinguished from the second one in that it does not consider intelligence as being ready made and given from the very beginning: Intelligence elaborates itself and only its functional laws are involved in organic organization and assimilation. To the static apriority of the third solution it presents in contrast the idea of a structuring activity, without preformed structures, which engenders the organs of intelligence in proportion to the functioning in contact with experience. Finally, it differs from the fourth solution in that it limits the role of chance in groping in behalf of the idea of directed searching, this direction being explained by the continuity of the assimilatory activity, of the reflex organization and of the elaboration of the most elementary habits to that of the most complex structures of deductive intelligence. But this continuity does not amount to reducing the higher to the lower nor to effecting the inverse reduction; it consists in a gradual construction of organs obeying the same functional laws.

To justify the fifth interpretation, let us first examine the four others possible, limiting ourselves to discussing them in the light of our results.

§ 1. ASSOCIATIONIST EMPIRICISM.—It seems impossible to deny that environmental pressure plays an essential role in the development of intelligence and we cannot follow Gestalt psychology in its effort to explain invention independently of

acquired experience (§3). That is why empiricism is destined to arise perpetually from its ashes and to play its useful role of antagonist of *a priori* interpretations. But the whole problem is to find out how the environment exerts its action and how the subject records the data of experience. It is on this point that the facts demand departure from associationism.

It is permissible to invoke, in favor of empiricism, everything that in the sequence of our stages manifests an influence of the history of the behavior patterns on their present state. The importance of the environment is only evident in an historical unfolding, when additional experiences put the individual series sufficiently in opposition to each other to make it possible to determine the role of external factors. On the contrary, the actual pressure of things on the mind, in an act of comprehension or of invention, for example, can always be interpreted as a function of internal qualities of perception or intellection. Now the role of the history lived by the subject, that is to say, the action of past experiences on present experiences, has seemed to us to be considerable in the course of the successive stages we have studied.

As early as the first stage one notices how the use of a reflex mechanism influences its maturation. What does this mean if not that, from the beginning, environment exerts its action. The use or nonuse of an hereditary arrangement of parts depends principally on external circumstances. During the second stage the importance of experience merely increases. On the one hand, conditioned reflexes, acquired associations and habits whose advent characterizes this period, consist in connections imposed by the external environment. Whatever explanation one adopts with regard to the capacity for establishing such connections (hence relatively to their formal mechanism), it is beyond doubt that their content is empirical. On the other hand, we have observed that certain maturations ordinarily considered as depending solely on internal factors, are in reality controlled, at least partially, by the environment. So it is that coördination between vision and prehension occurs at dates which vary between 0;3 and 0;6 according to the experience acquired by the subject (Obs. 84–93).

The behavior pattern whose apparition characterizes the third stage is, it will be recalled, the secondary circular reaction. Now, here again, no matter what interpretation one may give to the capacity of reproducing interesting results obtained by chance, there is no doubt that the connections acquired by virtue of such behavior patterns are due to empirical comparisons. Secondary circular reactions thus merely extend the primary reactions (to which the first habits are due). Whether acting upon things or his own body, the subject only discovers real connections through continuous exercise whose power of repetition presupposes as substance the data of experience as such.

With the advent of the coördination of schemata belonging to the fourth stage, the child's activity no longer only consists in repeating or conserving but also in combining and uniting. One might then expect the role of experience to diminish to the advantage of *a priori* structurizations. That is not the case. First of all, the schemata always being abridgments of experience, their reciprocal assimilations or combinations, however refined they may be, only express an experimental reality, past or future. Next, if these coördinations of schemata presuppose, as do the circular reactions and the reflexes, an activity peculiar to the subject, they only operate as function of the action, its successes or its failures. The role of experience, far from diminishing from the third to the fourth stage, only increases in importance. During the fifth stage, the utilization of experience spreads still more, since this period is characterized by the "tertiary circular reaction" or "experiment in order to see," and the coördination of schemata extends henceforth into "discoveries of new means through active experimentation."

Lastly, the sixth stage adds one more behavior pattern to the preceding ones: the invention of new means through deduction or mental combination. As with regard to the fourth stage, one can ask oneself if experience is not thereafter held in check by the work of the mind and if new connections, of *a priori* origin, will not henceforth double the experimental relationships. This is not the case, at least with respect to the content of the relations elaborated by the subject. Even in invention itself, which apparently precedes experience, the latter plays its role

as "mental experience." Moreover invention, however free it may be, joins experience and, in the last analysis, subjects it to its findings. This subjection can, it is true, sometimes assume the aspect of immediate and complete harmony, whence the illusion of a structure endogenous in its very content and joining the real through preëstablished harmony. But, in the majority of cases observed by us (in contrast to the facts of the first type cited by W. Köhler) the harmony is only progressive and does not at all exclude a series of indispensable corrections.

In short, at every level, experience is necessary to the development of intelligence. That is the fundamental fact on which the empirical hypotheses are based and which they have the merit of calling to attention. On this question our analyses of the origin of the child's intelligence confirm that point of view. But there is more in empiricism than just an affirmation of the role of experience: Empiricism is primarily a certain conception of experience and its action. On the one hand, it tends to consider experience as imposing itself without the subject's having to organize it, that is to say, as impressing itself directly on the organism without activity of the subject being necessary to constitute it. On the other hand, and as a result, empiricism regards experience as existing by itself and either owing its value to a system of external ready-made "things" and of given relations between those things (metaphysical empiricism), or consisting in a system of self-sufficient habits and associations (phenomenalism). This dual belief in the existence of an experience in itself and in its direct pressure on the subject's mind explains, in the last analysis, why empiricism is necessarily associationist. Every method of recording experience other than association in its different forms (conditioned reflex, "associative transfer," association of images, etc.) presupposes an intellectual activity partaking of the construction of the external reality perceived by the subject.

To be sure, empiricism thus presented is nowadays only a limited doctrine. But certain famous theories of intelligence are still very close to it. For example, when M. Spearman describes his three steps of intellectual progress, the "intuition of experience" (immediate apprehension of the data), the "eduction of

relations," and the "eduction of correlatives," he uses language very different from that of associationism and which seems to indicate the existence of a *sui generis* activity of the mind. But what comprises it, in the particular case? Immediate intuition of experience does not surpass passive awareness of immediate data. Concerning the "eduction" of relations or correlatives, it is only the reading of an already completely formed reality and one which does not clarify the particulars of the mechanism. M. N. Isaacs, subtly continuing the ideas of Spearman, has, it is true, recently tried to analyze this process.[1] The important thing in experience would be "expectation," that is to say, anticipation resulting from earlier observations and destined to be confirmed or belied by present events. When prevision is invalidated by the facts, the subject would yield himself to new anticipations (would form new hypotheses) and finally, in case of failure, would return to himself to modify his method. But, either the schemata thus serving "expectation" and the control of its results only consist in a mnemonic residue of past experiences, and we fall back into an associationism whose sole progress is to be motor and no longer only contemplative, or else they involve intellectual organization properly so called (an active elaboration of the schemata of anticipation due to an assimilatory or constructive mechanism) and we depart from empiricism since, in this case, experience is structured by the subject himself.

Now if we admit the necessity for experience, at all levels, and if, in particular, we can follow Isaacs in all that he affirms (if not in everything that he negates), the facts analyzed in this volume seem to prohibit us from interpreting this experience in the empiricist fashion, that is to say, as a direct contact between things and the mind.

The first reason may seem paradoxical but, if it is carefully weighed, it carries all the others with it. It is that the importance of experience augments rather than diminishes in the course of the six stages we have singled out. The child's mind advances to the conquest of things as if the progress of experience presup-

[1] In S. Isaacs, *The Intellectual Growth in Young Children*, London, Routledge, 1930.

posed an intelligent activity which organizes the latter instead of resulting from it. In other words, contact with things is less direct at the beginning than at the end of the evolution envisaged. Furthermore, it never is direct, but only tends toward becoming so. This is what we have established by showing that experience is only an "accommodation," however exact it may become. Now, it is the essence of empiricism, on the contrary, to put the "thing" or, lacking it, the "immediate datum," that is to say, always the receptive attitude of the mind, at the point of departure of all intellectual evolution, the progress of intelligence simply consisting in constructing abridgments of reactions or increasingly "deferred" reactions, destined to dispense with direct contact to find it only at long intervals.

Let us recall how things occur in the course of our six stages from the point of view of this progressive accommodation with the external environment. During the first stage there exists, of course, no direct contact with experience, since activity is simply reflex. Accommodation to things is therefore confused with reflex use. During the second stage, new associations are formed and so the pressure of experience begins. But these associations are limited, at the beginning, to interconnecting two or more movements of the body or else a reaction of the subject to an external signal. That is certainly an acquisition due to experience. But this "experience" does not yet put the mind in the presence of the "things" themselves; it places it exactly halfway between the external environment and the body itself. Hence accommodation remains undissociated from the activity of repetition, the latter bearing simply on the results acquired fortuitously instead of being due to the development of reflex activity. In the third stage, the acquired associations constitute relations between the things themselves and no longer only between different body movements. But these relations still remain dependent on the action, that is to say, the subject still does not experiment; his accommodation to things remains a simple attempt at repetition, the results reproduced just being more complex than in the preceding stage. In the fourth stage, experience comes still closer to the "object," the coördinations between schemata permitting the child to establish real relations between

things (in contrast to practical, purely phenomenalistic relationships). But it is only in the fifth stage that accommodation is definitively liberated and gives rise to true experience which still develops in the course of the sixth stage.

The mind, then, proceeds from pure phenomenalism whose presentations remain half-way between the body and the external environment, to active experimentation which alone penetrates inside things. What does this mean, if not that the child does not undergo simple external pressure from the environment but tries, on the contrary, to adapt himself to it? Experience, accordingly, is not reception but progressive action and construction: This is the fundamental fact.

Now this first reason for correcting the empirical interpretation entails a second one. If the "object" is not imposed at the beginning of mental evolution but is proposed as highest goal, would this not be because it cannot be conceived independently of an activity of the subject? Examination of the facts seems to us to admit of a decisive answer: "Accommodation" by which we have defined contact with experience is always indissociable from "assimilation" of the data to the activity of the subject himself. Let us choose something which we, as observers, shall consider as an "object" independent of ourselves—which doubtless means that we assimilate it to the mental structures of our adult mind —and let us try to find out how the child progressively adapts himself to it.

During the first two stages, external reality can have only one meaning: Things are only aliments for reflex use (sucking, etc.) or mechanisms in process of being acquired (following with the glance, etc.). If, then, the subject adapts himself empirically to the qualities of the objective, it is only a question of accommodating to it the innate or acquired schemata to which he was assimilated from the outset. Acquisition of schemata of the second type necessitates assimilation. It is by trying to assimilate the objective to an earlier schema that the child accommodates the latter to the former (thus going back to the reflex schemata), and it is by repeating (through "reproductive assimilation") the movement which is successful that the subject performs this operation and constitutes the new schema. Experience cannot there-

fore be, even at the beginning, a simple contact between the subject and a reality independent of himself, since accommodation is inseparable from an act of assimilation which assigns to the objective a meaning relating to the activity itself.

During the third stage it may seem that experience is freed from assimilation. For example, when the child discovers that the movements of his hand in grasping a cord set in motion those of the bassinet hood, it seems that such a phenomenon, whose sudden appearance was not anticipated, constitutes the prototype of pure experience. Nevertheless this spectacle gives rise, in the child, to an immediate attempt at reproduction, that is to say, to an assimilatory reaction, accommodation simply intervening in order to rediscover the movements which led to the desired result. Now this repetition would be unexplainable if, as soon as it was produced, the fortuitous phenomenon had not been assimilated, in one of its aspects, to an earlier schemata, of which it is a differentiation. So it is that, from their earliest manifestations, the movements of the bassinet hood are perceived, not only as things to see, to hear, etc. (primary schemata), but as extensions of the action of the hand (pulling the cord, etc.) or of the whole body (shaking oneself, etc.). Moreover, as soon as these first secondary reactions, through their assimilatory activity, result in the formation of new schemata, the latter assimilate in their turn all the new empirical events which will differentiate them. The first secondary schemata, consequently, derive from the primary schemata by a continuous assimilatory process and engender all the later secondary schemata by differentiation. Never is accommodation free from any assimilation.

In the course of the fourth stage, the coördination of schemata results in attempts which are confirmed or nullified by experience alone. But that coördination being itself the result of a reciprocal assimilation, the accommodation of schemata is consequently again inseparable from their assimilation. In the course of the fifth stage, on the other hand, accommodation tends to free itself in order to give rise to essentially experimental behavior patterns. But with regard to these "tertiary" reactions, two circumstances suffice to demonstrate that they always presuppose assimilation. On the one hand, the tertiary schemata

derive through differentiation from the secondary schemata. It is during the use of the latter that the new fact arises which provokes experimentation. Concerning the latter, it too consists in a circular reaction, that is to say, in active searching, and not in pure reception. However advanced the accommodations to which it gives rise may be, it always has assimilation itself as motor and is limited to differentiating circular reactions in the direction of conquest of the new. On the other hand, the behavior patterns of "discovery of new means through active experimentation" consist in coördinations analogous to those of the fourth stage, but with, in addition, an adjustment to the data of experience due precisely to the method of the tertiary reactions. Hence this means that such behavior patterns are doubly dependent on assimilation. In the course of the sixth stage the same is true *a fortiori,* since the "mental experiences" which then appear attest to the assimilatory power of the schemata which thus intercombine internally.

In conclusion, not only does experience become more active and comprehensive as intelligence matures, but also the "things" on which it proceeds can never be conceived independently of the subject's activity. This second statement reinforces the first and indicates that, if experience is necessary to intellectual development, it cannot be interpreted as being self-sufficient as the empirical theories would have it. It is true that the more active experience is, the more the reality on which it bears becomes independent of the self and consequently "objective." We shall demonstrate this in Volume II in studying how the object is dissociated in proportion to intellectual progress. But, far from substantiating empiricism, this phenomenon seems to us, on the contrary, the better able to characterize the true nature of experience. It is, in effect, to the extent that the subject is active that experience is objectified. Objectivity does not therefore mean independence in relation to the assimilatory activity of intelligence, but simply dissociation from the self and from egocentric subjectivity. The objectivity of experience is an achievement of accommodation and assimilation combined, that is to say, of the intellectual activity of the subject, and not a primary datum imposed on him from without. The role of assimilation is con-

sequently far from diminishing in importance in the course of the evolution of sensorimotor intelligence, by virtue of the fact that accommodation is progressively differentiated. On the contrary, to the extent that accommodation is established as centrifugal activity of the schemata, assimilation fills its role of coördination and unification with growing vigor. The ever-increasing complementary character of these two functions allows us to conclude that experience, far from freeing itself from intellectual activity, only progresses inasmuch as it is organized and animated by intelligence itself.

A third reason is added to the first two reasons to prevent us from accepting the "empirical" explanation of intelligence: It is that contact between the mind and things does not consist, at any level, in perceptions of simple data or in associations of such unities, but always consists in apprehensions of more or less "structured" complexes. That is clear during the first stage, since the elementary perceptions which can accompany reflex use necessarily extend its mechanism. They are therefore organized from the outset. With regard to the second stage, we have tried to establish that the first elementary habits and associations are never made manifest as connections constituted later between isolated terms, but that they result from behavior patterns which have been complex and structured ever since their point of departure. An habitual association is only formed to the extent that the subject pursues a determined end and consequently attributes to the presenting data a meaning relating to this exact end. That results from the already mentioned fact that accommodation to things always depends on an assimilation of these things to schemata already structured (the formation of a new schema always consists in a differentiation of the preceding schemata). It goes without saying that the connections which are established in the later stages (from the third to the sixth), are still less simple, since they derive from secondary and tertiary reactions and from the various reciprocal interassimilations of schemata. Thereafter they can lay claim still less to the quality of pure associations. They are always formed in the midst of totalities which are already organized or are in process of reorganization.

Now, as we have already said, one does not very well see how empiricism could stop being associationist. To say, as does Hume, that spatial and temporal perceptions are from the outset "compound impressions" and to say that the order of the sequence of sounds, in a musical phrase, constitutes a directly perceived "form" is to abandon, on these points, the empirical explanation. To the extent that experience appears to perception as organized from the outset, this is either because perception is itself structured in a corresponding way, or else because it imposes its own structure on the matter perceived. In both cases, contact with experience presupposes an organizing or structuring activity, as experience does not impress itself just as it is on the subject's mind. It is only in the hypothesis of isolated mnemonic traces and of associations due to mechanical repetition (to the repetition of external circumstances) that one understands that there can be pure reception. Every hypothesis surpasses empiricism and attributes to the subject a power of adaptation with all that such a concept allows.

In short, if experience appears to be one of the conditions necessary to the development of intelligence, study of the first stages of that development invalidates the empirical conception of experience.

§2. VITALISTIC INTELLECTUALISM.—If intelligence is not a sum of traces laid down by the environment nor of associations imposed by the pressure of things, the simplest solution consists in making it a force of organization or a faculty inherent in the human mind and even in all animal life of whatever sort.

It is useless to recall here how such an hypothesis, abandoned during the first phases of experimental psychology, reappears today under the influence of preoccupations which are at the same time biological (neovitalism) and philosophical (the revival of Aristotelianism and of Thomism). We are not interested here in the various historical or contemporary forms of intellectualism, but only in the grounds for such an interpretation to the extent that it is applicable to our results. Now it is undeniable that the hypothesis has its merits and that the very reasons which militate

in favor of vitalism in biology are of a kind favoring intellectualism in the psychology of intelligence.

These reasons are at least two in number. The first stems from the difficulty of accounting for intelligence, once it has been achieved, by anything other than its own organization considered as a self-sufficient totality. Intelligence in action is, in effect, irreducible to everything that is not itself and, moreover, it appears as a total system of which one cannot conceive one part without bringing in all of it. From that to making intelligence a power *sui generis* just as vitalism makes the organism the expression of a special force) is only a step. Now, in speaking, as we have done, of an "organization" of schemata and of their spontaneous "adaptation" to the environment, we have constantly skirted this kind of explanation of totalities by themselves in which the vitalistic and spiritualistic interpretation consists. We resist it inasmuch as we call neither organization nor assimilation forces, but only functions; we yield, on the other hand, as soon as we substantiate those functions; that is to say, as soon as we conceive them as being mechanisms with a completed and permanent structure.

Whence the arguments of the second group which are genetic in kind. Granted that intelligence constitutes a self-explanatory mechanism, the organization which characterizes it is immanent in the most primitive stages. Intelligence thus germinates in life itself, either because the "organic intelligence" which is at work on the physiological plane potentially contains the highest realizations of abstract intelligence, or because it progressively rouses them by conducing toward them as a necessary end. Now let us not try to conceal the fact that, despite the difference in vocabulary, our interpretations also lead to establishing behavior which is between the vital and the intellectual and, to this extent, they can make use of vitalistic inspiration. We have always emphasized the profound unity of the phenomena of organization and adaptation on the morphological reflex plane of systematic intelligence itself. Intellectual adaptation to the external environment and the internal organization it involves thus extend the mechanisms which one can follow from the beginnings of the elementary vital reactions. The creation of intelli-

gent structures is related to the elaboration of forms which characterizes life as a whole. In a general way, it is difficult not to make the relationships between knowledge and reality the ideal balance to which all biological evolution leads because they alone completely blend assimilation and accommodation thitherto more or less mutually antagonistic. Nothing, therefore, would be easier than to translate our conclusions into vitalistic language, to appeal to the hierarchy of vegetative, sensitive and rational souls to express the functional continuity of development and to oppose in principle life and unorganized matter in order metaphysically to justify the activity of the intelligent subject.

But if vitalism has the merit, unceasingly renewed, of underlining the difficulties or above all the gaps in the positive solutions, it is only too clear that its own explanations reveal the disadvantages of their simplicity and their realism; that is to say, they are always threatened by the progress of biological analysis as well as by that of the reflection of intelligence on itself. Now our ambition being precisely to make the dual light of the biological explanation and the critique of knowledge converge on the development of reason, it would be paradoxical if that union were to result in a reinforcement of the vitalistic thesis. In reality, three essential divergences distinguish the description we have adopted from the system we are now examining: The first pertains to the realism of intelligence as a faculty, the second to that of organization as a vital force, and the third to the realism of adaptive knowledge.

In the first place, it is of the essence of vitalistic intellectualism to consider intelligence a faculty, that is to say, a mechanism complete in its structure and its operation. Now, an essential distinction obtrudes in this connection. If epistemological analysis, whether it is simply reflexive or bears upon scientific knowledge, results in considering intellection an irreducible act, in the latter case it is only a question of knowledge itself, inasmuch as it is obedient to the ideal norms of truth and translates itself in thought in the form of different states of consciousness *sui generis*. But from this intimate experience of intellection nothing can be drawn concerning factual conditions, that is to say, psychological and biological ones, of the intellectual mechanism: proof

that, without speaking of metaphysical theories of knowledge, harmony is far from being realized, in the scientific field, between the various logical-mathematical analyses of rational truth, between the multiple theories of the psychology of intelligence, nor *a fortiori* between these two groups of research. Now, intellectualism claims to draw from the fact of intellection the conclusion that there exists a psychic faculty which is easy to know, which would be intelligence itself. It is therefore not intellection as such that this doctrine posits as irreducible, it is a certain reification of that act in the form of a given mechanism in the completely formed state.

Now, it is from this point on that we can no longer follow. From the fact that the living being achieves knowledge and that the child is one day destined to master science, we certainly believe the conclusion must be drawn that there is a continuum between life and intelligence. Furthermore we infer, from the fact that the most complex operations of logical thought seem prepared from the time of the elementary sensorimotor reactions, that this continuum can be already observed in the transition from the reflex to the first acquired adaptations and from the latter to the simplest manifestations of practical intelligence. But the entire question remains of finding out what is permanent in the course of this evolution and what remains characteristic of each level under consideration.

The solution to which our observations lead is that only the functions of the intellect (in contradistinction to the structures) are common to the different stages and consequently serve as connecting link between the life of the organism and that of the intelligence. So it is that at every level the subject assimilates the environment, that is to say, incorporates it to the schemata while maintaining the latter through this use and by means of a constant generalization. At every level accommodation is hence simultaneously accommodation of the organism to objects and assimilation of objects to the organism's activity. At each level this adaptation is accompanied by a search for coherence which unifies the diversity of experience by intercoördinating the schemata. In short, there exists an operation common to all stages of sensorimotor development and of which the operation

of logical intelligence seems to be the extension (the formal mechanism of concepts and relationships extending the organization of the schemata and adaptation to experience following accommodation to the environment). Moreover, this sensorimotor operation in turn extends that of the organism, the working of schemata being functionally comparable to that of the organs, whose "form" results from an interaction of the environment and the organism.

But it is apparent that one could not draw from this permanence of functioning the proof of the existence of an identity of structures. The fact that the working of reflexes, of circular reactions, mobile schemata, etc., is identical to that of the logical operations does not prove at all that concepts are sensorimotor schemata nor that the latter are reflex schemata. It is therefore necessary, beside the functions, to make allowance for the structures and admit that the most varied organs can correspond to the same function. The psychological problem of intelligence is just that of the formation of those structures or organs and the solution of this problem is in no way prejudiced by the fact that one acknowledges a permanence of the functioning. This permanence does not at all presuppose the existence of a ready-made "faculty" transcending genetic causality.

Could one not, however, raise the objection that a permanence of functions necessarily implies the idea of a constant mechanism, of a "function" conserving itself, in short, whether one likes it or not, of a "faculty" of invariant structure? So it is that, in current psychological parlance, the word "function" is sometimes used as a synonym for "faculty," and under cover of this terminology is hidden a virtual collection of entities. Memory, attention, intelligence, will, etc., are thus too often called "functions" in a sense which hardly has anything "functional" in it any longer and which tends to become structural or pseudo-anatomical (as though one said "the circulation" no longer thinking of the function but only of the assemblage of organs involved). That being the case, have we the right to concede the existence of a permanent intellectual function without recognizing the existence of an intelligence faculty? Here the comparisons with biology seem decisive. There are functions whose abso-

lute invariability is accompanied by considerable structural variations from one group to the other (nutrition, for instance). It can even be said that the most important and general functions by means of which one can try to define life (organization, assimilation in the wide sense of the word, etc.) do not correspond to any particular organ but have as their structural instrument the whole organism. The permanency of these functions is on a par with a still greater variability of the organ. Consequently, to assert that a permanent intellectual function exists is in no way to prejudge the existence of an invariable structural mechanism. Perhaps it exists, just as a circulatory system is necessary to the circulation. But perhaps also intelligence is confused with the whole of behavior[2] or with one of its general aspects without there being need of isolating it in the form of a particular organ endowed with powers and conservation. Moreover, if it characterizes the behavior as a whole, it is not therefore necessary to call it a faculty or the emanation of a substantial soul, for the same reasons.

Biological realism to which the vitalistic interpretation refers is exactly parallel to intellectualistic realism which we have rejected; just as the permanency of intellectual functions may seem to imply the existence of an intelligence faculty, so also the fact of vital organization leads improperly to the hypothesis of a "force" of organization. The vitalistic solution is the same in both cases. From the function one proceeds to the structural interpretation and thus one "realizes" the functional totality in the form of a single and simple cause. Now, with respect to this second point too, we are unable to follow vitalism. From the fact that the organization of the living being implies a power of adaptation which leads to intelligence itself, it does not follow at all that these various functions are unexplainable and irreducible. But the problems of organization and of adaptation (including that of assimilation) surpass psychology and presuppose a biological interpretation of the whole.

These first two expressions of vitalistic realism lead to a realism of adaptation itself, in relation to which the contrast

[2] H. Piéron, *Psychologie Expérimentale,* Paris, 1927, pp. 204–208.

between the results of our studies and the system of interpretation we are now examining seems to us still more marked. Inasmuch as it considers life as irreducible to matter, and intelligence as a faculty inherent in life, vitalism conceives knowledge as an adaptation *sui generis* of that faculty to an object given independently of the subject. In other words, this adaptation, while remaining mysterious because of these very contrasts, is reduced in fact to what common sense has always envisaged as being of the essence of knowledge: a simple copy of things. Intelligence, we are told, tends to conform to the object and possess it due to a sort of mental identification; it "becomes the object" in thought. Thus vitalism always joins empiricism on the field of knowledge as such with this faint difference that, from our present point of view, intelligence subordinates itself to the thing instead of being subordinated to it from without. There is voluntary imitation and not simple reception.

But that epistemological realism, it seems to us, clashes with the fundamental fact we have emphasized in the course of our analyses. That is, that adaptation—intellectual and biological, hence adaptation of intelligence to "things" as well as of the organism to its "environment"—always consists in a balance between accommodation and assimilation. In other words, knowledge could not be a copy, since it is always a putting into relationship of object and subject, an incorporation of the object to the schemata which are due to activity itself and which simply accommodate themselves to it while making it comprehensible to the subject. To put it still differently, the object only exists, with regard to knowledge, in its relations with the subject and, if the mind always advances more toward the conquest of things, this is because it organizes experience more and more actively, instead of mimicking, from without, a ready-made reality. The object is not a "known quantity" but the result of a construction.

Now this interaction of intelligent activity and experience finds its counterpart, on the biological plane, in a necessary interaction between the organism and the environment. To the extent that one refuses to explain life, as does vitalism, by a force *sui generis* of organization, one is obliged to consider living beings simultaneously as being conditioned by the physicochemical uni-

verse and as resisting it by assimilating it. Consequently there is interdependence between the organism and the entire universe, on the one hand, objectively, because the former results from the latter while completing and transforming it, on the other hand, subjectively, because the adaptation of the mind to experience presupposes an activity which enters as a component into the play of objective relationships.

In short, the biological interpretation of intellectual processes based upon the analysis of assimilation does not result at all in the epistemological realism of vitalistic intellectualism. Even by making knowledge a singular case of organic adaptation one ends, on the contrary, with the conclusion that true reality is neither an organism isolated in its entelechy, nor an external environment capable of subsisting if one abstracts life and thought from it. Concrete reality is the ensemble of the mutual relationships of the environment and the organism, that is to say, the system of interactions which unify them. Once those relationships have been posited, one can try to elucidate them either by the biological method starting with a ready-made environment to try to explain the organism and its properties, or by the psychological method, starting with mental development to try to find out how environment is constituted for the intelligence. Now, if adaptation does consist, as we have asserted, in a balance between accommodation of the schemata to things and the assimilation of things to the schemata, it is self-evident that these two methods are complementary: but provided one no longer believes in ready-made intelligence or a vital force independent of the environment.

§3. APRIORITY AND THE PSYCHOLOGY OF FORM. —If intellectual development results neither from constraint exerted by the external environment nor from the progressive affirmation of a ready-made faculty for knowing that environment, perhaps it should be conceived as being the gradual explanation of a series of preformed structures in the psycho-physiological constitution of the subject himself.

Such a solution obtrudes in the history of philosophical theories of knowledge when, disappointed by both English em-

piricism and classical intellectualism (and above all by the Wolffian theory of the rational faculty), Kantianism resorts to the *a priori* hypothesis to explain the possibility of science. In biology, moreover, apriority arose when the difficulties related to the problem of heredity of acquired characteristics led it to reject Lamarckian empiricism. Some people tried to return to vitalism, whereas others tried to account for evolution and adaptation by the hypothesis of the preformation of genes. Finally, in the psychological field, a solution of the same kind took the place of associationist empiricism and intellectual vitalism. It consists in explaining every invention of intelligence by a renewed and endogenous structuring of the perceptual field or of the system of concepts and relationships. The structures which thus succeed each other always constitute totalities; that is to say, they cannot be reduced to associations or combinations or empirical origin. Moreover, the Gestalt theory to which we allude, appeals to no faculty or vital force or organization. As these "forms" spring neither from the things themselves nor from a formative faculty, they are conceived as having their root in the nervous system or, in a general way, in the preformed structure of the organism. In this regard we can consider such a solution *"a priori."* Doubtless, in most cases, the Gestalt psychologists do not clarify the origin of the structures and confine themselves to saying that they are necessarily imposed on the subject in a given situation. This doctrine is reminiscent of a sort of Platonism of perception. But, as Gestalt psychology always returns to the psycho-physiological constitution of the subject himself when it is a question of explaining this necessity for forms, such an interpretation certainly consists in a biological apriority or a variety of preformation.

Now, the theory of form, far from confining itself to the enunciation of general principles, has furnished a series of fundamental works for understanding the mechanism of intelligence: those of Wertheimer on the psychological nature of the syllogism, of Köhler on intelligence and invention, of K. Lewin on the "field" theory, etc. These studies all result in explaining by structures of the field of conception or perception what we attribute to assimilation. It is therefore essential to compare closely this system of explanation with that which we have used and

even, in order the better to conclude this comparison, to interpret our results in terms of "Gestalt." On at least two essential points we agree with the "theory of form."

In the first place, it is quite true that every intelligent solution and even every behavior pattern in which the comprehension of a given situation intervenes (however wide the meaning attributed to the word "comprehension") appear as totalities and not as associations or syntheses of isolated elements. In this respect the "schema" whose existence we have always acknowledged can be compared to a "form" or "Gestalt." The system, composed of determined and completed movements and perceptions, reveals the dual character of being structured (hence of itself structuring the field of perception or comprehension) and of constituting itself from the outset inasmuch as it is a totality without resulting from an association or synthesis between elements separated earlier. Without speaking of the reflex schemata which are the more totalized and structured since they were already assembled at birth, one can observe these characteristics as early as the first nonhereditary schemata, due to the primary circular reactions. The simplest habits, as well as the so-called acquired "associations" do not result from true associations, that is to say, interuniting the terms given as such, but also result from connections implying a structured totality from the outset. Only the global signification of the act (the place of assimilation which connects the result to the need to be satisfied) insures, in effect, the existence of relations which, from the outside may seem to be "associations." The "secondary schemata," moreover, also always constitute systems of the whole analogous to "Gestalts." It is only to the extent that a child tries to reconstruct a spectacle which he has just witnessed or created that he connects one movement with another. Perceptions and movements are hence only associated if their significations are already related to each other and if this system of mutual relations itself implies a meaning of the whole given in the initial perception. With regard to the coordinations between schemata characteristic of the fourth stage, one could not consider them associations either. Not only do the coördinations work by reciprocal assimilation—that is to say, due to a process which stems more from global reorganization

than from simple association—but also this reorganization results at once in the formation of a new schema revealing all the characteristics of a new and original totality. With the "experiments in order to see" and the acts of intelligence which flow from them (fifth stage) we are surely outside the realm of pure "Gestalt." But the theory of form has never claimed to suppress the existence of groping searching; it has only tried to remove it from the realm of actually intelligent behavior patterns in order to consider it a substitute for structuring and place it in the intermediate periods between two structures. At the sixth stage we again find, on the contrary, authentic "structures." The invention of new means through mental combination manifests all the characteristics of these rapid or even instantaneous regroupings by means of which Köhler has described the true act of intelligence.

Taken as a whole, except with regard to groping—whose role is really constant but is revealed particularly during the first experimental behavior patterns (fifth stage)—the schemata whose existence we have recognized manifest the essence of the characteristics of a structured totality by means of which the theory of form opposed the "Gestalts" to classic associations.

A second point of convergence between the two systems of interpretations is the rejection of every faculty or special force of organization. W. Köhler emphasizes the fact that his criticism of associationism frequently rejoins analogous objections already formulated by vitalism. But, he adds, with reason, one cannot deduce from this agreement that the "forms" are to be interpreted as the product of a special energy of organization; Vitalism too quickly reaches the conclusion that totalities exist from the hypothesis of a vital principle of unification. We therefore sympathize completely with the effort of Gestalt psychology to find the roots of intellectual structures in the biological processes conceived as systems of relationships and not as the expression of substantial forces.

Having thus defined these common traits, we find ourselves more free to show how the hypothesis of assimilation tries to surpass the theory of forms and not to contradict it, and how the "schema" is a "Gestalt" made dynamic and not a concept destined

to react against the progress of the Gestalt movement. To resume our comparison between the theory of form and epistemological apriority, the "Gestalt" manifests the same advantages over association as were formerly made manifest by Kantian apriority over classical empiricism, but only to arrive at parallel difficulties: having conquered static realism without, apriority finds it again within the mind and runs the risk, in the last analysis, of ending in a restored empiricism. In effect, the theory of form, like former epistemological apriority, wanted to defend the internal activity of perception and intelligence against the mechanism of external associations. It has therefore located the principle of organization in ourselves and not outside us, and, the better to shelter it from empirical experience, it rooted it in the preformed structure of our nervous system and our psycho-physiological organism. But, by trying thus to guarantee the internal activity of organization against the immixtures of the external environment, it finally withdrew it from our personal power. It consequently enclosed it in a static formalism conceived as preexisting or as being elaborated outside of our deliberate intention. This formalism certainly marks great progress over associationism because it predicates the existence of syntheses or totalities instead of remaining atomistic, but it is precarious progress. To the extent that "forms," like the categories of yore, antecede our intentional activity, they revert to the rank of inert mechanisms. That is why, in the theory of form, intelligence ends by disappearing to the advantage of perception, and the latter, conceived as being determined by ready-made internal structures —that is to say, consequently as preformed from within—ends by becoming more and more confused with "empirical" perception, conceived as preformed from without: In both cases, in effect, the activity disappears to the advantage of the elaborated whole.

Our critique of the theory of form must therefore consist in retaining all that is positive which it opposes to associationism—that is to say, all the activity it discovers in the mind—but in rejecting everything in it which is only restored empiricism—that is to say, its static apriority. In short, to criticize Gestalt psychology is not to reject it but to make it more mobile and consequently to replace its apriority with a genetic relativity.

The analysis of a primary divergence will permit us to define these positions from the outset. A "Gestalt" has no history because it does not account for earlier experience, whereas a schema embodies the past and so always consists in an active organization of the experience lived. Now this point is fundamental: the analysis of three children, almost all of whose reactions were observed, from birth to the acquisition of speech, has convinced us of the impossibility of detaching any behavior pattern whatever from the historical context of which it is a part, whereas the hypothesis of "form" makes the history useless and the Gestalt psychologists deny the influence of acquired experience on the solution of new problems.[3]

So it is, to begin at the end, that we have never, even during the sixth stage, observed "intelligent" reorganizations, even unforeseen and sudden ones, unless the invention or mental combination which determined them was prepared, however little, by earlier experience. With regard to the theory of form, on the contrary, an invention (like that of the ladder of boxes made by Köhler's chimpanzees) consists in a new structuring of the perceptual field, which nothing in the subject's past can explain: whence the hypothesis according to which this structure would come solely from a certain degree of maturation of the nervous system or the organs of perception so that nothing from without, that is, no present or past experience, would cause its formation (present experience is confined to starting or necessitating structuring, but without explaining it). True, some of our observations of the sixth stage seem at first to corroborate this outlook. Thus if Jacqueline and Laurent gradually discovered the use of the stick due to empirical groping, Laurent, whom we let go much longer without putting him in the same situation, understood at once the signification of that instrument. Everything occurs as though a structure not yet mature in the case of the two former were imposed ready-made on Laurent's perception. So also Lucienne immediately found the solution to the problem of the watch chain, whereas Jacqueline groped laboriously. But before

3 See Claparède, La Genèse de l'hypothèse, *Archives de Psychol.*, XXIV, the summary (pp. 53–58) of the works of K. Duncker and N. R. F. Maier, intended to demonstrate the uselessness of acquired experience.

concluding that there is radical novelty in such mental combinations, and consequently before having recourse, in order to explain them, to the emergence of endogenous structures having no root in the individual's past experience, it is necessary to make two remarks. The first is that, in default of external groping one cannot exclude the possibility of a "mental experience" which would occupy the moments of reflection immediately preceding the act itself. The most sudden inventions which we can introspect always reveal at least a beginning of search or internal groping outside of which ideas and perceptions cannot regroup themselves all alone. It goes without saying and we have emphasized that this "mental experience" is not simply the passive extension of states lived through earlier and that it consists, like actual experience, in real action. But the fact remains that, even without visible groping from the outside, the subject's thought can always yield internally to experimental combinations, however rapid. Sudden reorganization can therefore be conceived as an extreme case of mental combination. Now, and this second remark is essential, these mental experiences, even if the known quantities of the problem are entirely new, can always apply to the present situation earlier schemata utilized in more or less analogous cases, whether these schemata merely apply to some aspect of this situation or simply inspire the method to be followed in order to solve the problem. Thus, if Lucienne never rolled up a watch chain in order to put it in a small opening, she was able to execute similar movements when rolling up pieces of material, cords, etc. So also Laurent, without ever having utilized the stick, is able to apply to the new situation the schemata drawn from the use of other intermediates ("supports," strings, etc.). Between simple prehension and the idea that one solid can cause displacement of another is found a series of imperceptible transitions.

The idea is therefore conceived that the sudden inventions characteristic of the sixth stage are in reality the product of a long evolution of schemata and not only of an internal maturation of perceptive structures (the existence of the latter factor having of course to be reserved). This is revealed by the existence of a fifth stage, characterized by experimental groping and situated be-

tween the fourth (coördination of schemata) and the sixth (mental combinations). If, with respect to the theory of form, groping search constitutes an activity in the margin of the maturation of structures and without influence on what maturation, we, on the other hand, believe we have established that the sudden invention of new structures, which characterizes the sixth stage, only appears after a phase of experimentation or of "tertiary circular reaction." What does this mean if not that the practice of actual experience is necessary in order to acquire the practice of mental experience and that invention does not arise entirely preformed despite appearances?

Furthermore, the entire sequence of stages, from the first to the two last, is there to attest to the reality of the evolution of schemata and consequently to the role of experience and of history. A complete continuity exists between the behavior patterns characteristic of the different stages. The primary circular reactions thus prolong the activity of the reflex schemata by systematically extending their sphere of application. The secondary circular reactions, moreover, derive from the primary reactions to the extent that every discovery historically entails a series of others. Thus coördination between vision and prehension induces the child to grasp the objects which hang from the hood of his bassinet and the manipulation of these objects leads him to act upon the hood itself, etc. Thereafter, once the secondary schemata have been formed as function of the historical development of the circular reactions, a coördination of schemata is established during the fourth stage, which itself results from earlier activities. The act of pushing back the obstacle, for example, coördinates the cycles of prehension with the schemata such as striking, etc., and it has seemed to us impossible to explain the advent of such coördinations without knowing the subject's past, in each particular case. Concerning the discovery of new means through active experimentation (fifth stage), it constitutes a coördination of schemata extending that of the preceding stage, with only this difference that the coördination no longer occurs in an immediate way but necessitates a more or less laborious readjustment, that is to say, an experimental groping. Now this groping is itself

prepared by the exploratory behavior patterns inherent in assimilation by mobile schemata.

In short, the new behavior patterns whose appearance marks each stage are always revealed as developing those of the preceding stages. But two interpretations can be given of this same fact. One could see in it, first of all, the expression of a purely internal maturation, such as that the formal structure of perceptions and of acts of intelligence develops by itself without use as a function of experience or transmission of the contents from one stage to another. One can, on the contrary, conceive of this transformation as being due to an historical evolution such as that the use of the schemata is necessary to their structuring and that the result of their activity is thus transmitted from one period to the other. Now this second interpretation alone seems reconcilable with the particulars of the individual facts. In comparing the progress of intelligence in three children, day after day, one sees how each new behavior pattern is formed by differentiation and adaptation of the preceding ones. One can follow the particular history of each schema through the successive stages of development, as the formation of structures cannot be dissociated from the historical development of experience.

The schema is therefore a Gestalt which has a history. But how does it happen that the theory of form came to dispute this role of past experience? From the fact that one refuses to consider the schemata of behavior as being the simple product of external pressures (like a sum of passive associations) it clearly does not necessarily follow that their structure is imposed by virtue of preëstablished laws, independent of their history. It is enough to acknowledge an interaction of form and content, the structures thus being transformed gradually as they adapt themselves to increasingly varied conditions. For what subtle reasons do writers as informed as the Gestalt psychologists reject an interaction which seems so obvious?

A second divergence should be noted here: A "schema" is applied to the diversity of the external environment and hence is generalized as a function of the contents it subsumes, whereas a "Gestalt" is not generalized and even is "applied" less than it is imposed in an immediate way and internally to the situation

perceived. The schema, as it appeared to us, constitutes a sort of sensorimotor concept or, more broadly, the motor equivalent of a system of relations and classes. The history and development of a schema therefore consist primarily in its generalization, through application to increasingly varied circumstances. Now a "Gestalt" manifests itself quite differently. Take two objects, for example an objective and its "support" at first perceived without interrelationships, then suddenly "structured"; and let us say that the subject, after having "understood" the connection which links them, subsequently understands a series of analogous relations. In order to explain this, the theory of form does not maintain either that the "Gestalt" which intervenes here is generalized, or even that it is "applied" successively to various objects. If perception, at first not structured, suddenly acquires a "form," this is because at any degree of maturation whatever it is impossible for the subject to see things differently, given the ensemble of the situation. The "form" thus constitutes a sort of ideal necessity or immanent law which is imposed on perception, and when the Gestalt psychologists describe the thing from the phenomenological point of view, they speak of this form as Platonists speak of an "idea" or logicians of a "subsistent" being: The Gestalt is simply established by virtue of its "pregnance." When the same writers speak as physiologists, they add that this internal value is related to the subject's nervous constitution. In both cases it is always a question of an immediate necessity which can be renewed at the time of each perception but which does not require the existence of a generalizing schematism. This is what the Gestalt psychologists again express by invoking the *Einsicht* or the total comprehension which arises according to the variations of the goal pursued and by specifically stating, as does Duncker,[4] that "reasoning is a battle which creates its own weapons." If we say that the theory of form constitutes a sort of apriority, this is simply because, according to this doctrine, structuring results from an intrinsic necessity and not at all from experience, and is thus connected with the conditions of the subject himself; the criterion of the *"a priori"* has always been necessity as such. The "Gestalten" do not therefore consist in mobile frames successively

4 Quoted by Claparède (above-mentioned article), p. 53.

applied to various contents: structuring is simply a predetermined process, that is to say, obtruding necessarily, sooner or later, and thereafter this process can repeat itself every time the situation demands, but without involving the activity of schemata supplied with a history and capable of generalization.

How does genetic observation determine the choice between necessary preformation and generalizing activity? It is apparent that, to the extent that one attributes a history to the structures, one is obliged to admit an element of generalization, that is to say, one is led to detach the structures from structured situations to make active schemata of them which are due to a structuring assimilation. As nearly as the use of hereditary reflexes, one has the impression that the subject searches for aliments for his activity and that thus the latter is generalizing: so it is that the baby sucks, looks, listens in a growing number of given situations. But if, during this first period, just as during that of the primary circular reactions, it is difficult to dissociate active generalization from simple structuring, the contrast becomes striking from the third stage on, that is to say, with the advent of the secondary circular reactions. From the moment when the child truly acts upon the external world each of his conquests gives rise, not only to immediate repetition, but also to a generalization which is henceforth very obvious. Thus after having grasped a cord hanging from his bassinet hood and having by chance discovered the results of this traction, the child applies this behavior to all hanging objects. Now it is very hard not to interpret the thing as a generalization since the child is not satisfied to shake the hood in a different way but will go so far as to use the same means to make interesting spectacles last, whatever the distance which separates him from them may be. This perpetual extension which we have noted of the secondary schemata into "procedures to make interesting spectacles last" is the best proof of their generalizing power. Concerning the fourth stage, it is characterized by a greater mobility of schemata than before, that is to say, by new progress in generalization. In effect, not only is the coördination of certain schemata due to their reciprocal assimilation, that is to say, to a generalizing process, but in addition the power of generalization belonging to the mobile schemata is confirmed

by certain special behavior patterns which we have called "exploration of new objects." These behavior patterns, which extend the generalizing assimilations of the third stage, consist in applying to new objects all the familiar schemata in succession, so as to "understand" these objects. It seems evident, in such a case, that the effort to generalize obtains over all preformed structures, since there is a laborious adjustment of the familiar to the unfamiliar and since this search presupposes a series of choices. So also, during the fifth stage, the series of gropings which lead the child to discovery of the use of supports, of strings and of sticks is directed by the ensemble of the earlier schemata which give meaning to the present search. This application of the known to the unknown also presupposes constant generalization. Finally, we have considered generalization as being indispensable to the mental combinations of the sixth stage.

If one follows, then, the development of schemata, stage after stage, either in general or taken each individually, one proves that this history is one of continuous generalization. Not only is all structuring capable of reproducing itself when confronted by events which caused its appearance, but also it is applied to new objects which differentiate it in case of need. This correlative generalization and differentiation reveal, it seems to us, that a "form" is not a rigid entity to which perception leads as though under the influence of predetermination, but a plastic organization, just as frames adapt themselves to their contents and so depend partially on them. This means that "forms," far from existing before their activity, are rather comparable to concepts or systems of relationships whose gradual elaboration works when they are generalized. Observation forces us, therefore, to detach them from pure perception and raise them to the rank of intellectual schemata. Only a schema, in effect, is capable of real activity, that is to say, of generalization and differentiation combined.

This leads us to examine a third difficulty in the theory of structure. To the extent that the "forms" do not have a history or a generalizing power, the very activity of intelligence is preterit to the advantage of a more or less automatic mechanism. The "Gestalten" do not have any activity in themselves. They arise

at the time of the reorganization of the perceptual fields and are imposed as such without resulting from any dynamism anterior to themselves; or, if they are accompanied by an internal maturation, the latter is itself directed by preformed structures which it therefore does not explain.

Now, it is there that the facts envisaged in their historical continuity prevent us all the more from accepting unreservedly the theory of form, whatever the static analogy which can exist between the "Gestalt" and the schemata may be. In effect, the schemata have always seemed to us to be not antonomous entities but the products of a continuous activity which is immanent in them and of which they constitute the sequential moments of crystallization. As this activity is not external to them, it therefore does not constitute the expression of a "faculty," and we have already seen why. It only forms one whole with the schemata themselves, as the judgment's activity is made manifest in the formation of concepts; but, just as the concepts become dissociated from the continuous chain of the judgments which gave rise to them, so also the schemata detach themselves little by little from the organizing activity which engendered them and with which they became confused at the time of their formation. More precisely, the schemata, once formed, serve as instruments for the activity which engendered them, just as the concepts, once they have been derived from the judicatory act, are the point of departure of new judgments.

What, then, is this organizing activity, if it is not external but immanent in the schemata, without however consisting in a simple maturation? As we have constantly repeated, the organization of schemata is only the internal aspect of their adaptation which is simultaneously accommodation and assimilation. The primary fact is therefore the assimilatory activity itself without which no accommodation is possible, and it is the combined action of assimilation and accommodation which accounts for the existence of the schemata and consequently of their organization.

In effect, however high one may raise the advent of the first psychological "behavior pattern," they are revealed in the form of mechanisms leading to the satisfaction of a need. That means that the behavior patterns are from the outset the func-

tion of the general organization of the living body. Every living being constitutes a totality which tends to conserve itself and consequently to assimilate to itself the external elements it needs. From the biological point of view, assimilation and organization are on a par, without its being possible to consider organized forms as being anterior to assimilatory activity or the contrary. Need, the satisfaction of which is insured by the reflexes subordinated to the whole of the organism, is thus to be considered as the expression of an assimilatory tendency simultaneously dependent on the organization and fit to conserve it. But, from the subjective point of view, this same need, however complex the reflex organization of which it is the expression may be, appears, in its primitive form, as a global tendency simple to satisfy, that is to say, as barely differentiated from states of consciousness proceeding from desire to satisfaction and from satisfaction to the desire to conserve or recommence. From the psychological point of view, the assimilatory activity, which is immediately extended in the form of reproductive assimilation, is consequently the primary fact. Now this activity, precisely to the extent that it leads to repetition, engenders an elementary schema—the schema being formed by active reproduction—then, due to this nascent organization, becomes capable of generalizing and recognitory assimilations. Moreover, the schemata thus constituted accommodate themselves to external reality to the extent that they try to assimilate it and so become progressively differentiated. Thus, on the psychological as well as on the biological plane, the schematism of the organization is inseparable from an assimilatory and accommodating activity, whose functioning alone explains the development of successive structures.

It can now be understood how the fact of considering the "forms" as having no history and of conceiving their continuous reorganizations as independent of active generalization amounts sooner or later to neglecting the activity of the intelligence itself. To the extent that the schemata are regarded as subtended by an activity at once assimilatory and accommodating, only then do they appear as being capable of explaining the later progress of systematic intelligence in which conceptual structures and logical relationships are superposed on simple sensorimotor mecha-

nisms. To the extent, on the contrary, that static "form" excels activity, even if this "form" is endowed with an undefined power of maturation and of reorganization, one does not understand why intelligence is necessary and is dissociated from simple perception. Here we probably touch on the essential point of divergence: In the theory of form, the ideal is to explain intelligence by perception, whereas for us perception itself must be interpreted in terms of intelligence.

There is undoubtedly a continuity of mechanism between perception and intelligence. All perception appeared to us to be the elaboration or the application of a schema, that is to say, a more or less rapid organization of sensorial data as function of an ensemble of acts and movements, explicit or simply outlined. Moreover, intelligence, which in its elementary forms involves an element of search and groping, ends, in the course of the sixth stage, in sudden reorganizations consisting in extreme cases, in almost immediate "perceptions" of the correct solution. Hence it is accurate to emphasize, with the theory of form, the analogy of perception and of practical intelligence. But this identification can have two meanings. According to the first, perceptions are self-sufficient, and search only constitutes a sort of accident or interlude revealing the absence of organized perception. According to the second, on the contrary, all perception is the product of an activity whose most discursive or groping forms are only the clarification. Now, it is in this way that things are always presented to us: All perception is an accommodation (with or without regrouping) of schemata which have required, for their construction, systematic work of assimilation and organization; and intelligence is only the progressive complication of this same work, when immediate perception of the solution is not possible. The reciprocating motion between direct perception and searching does not warrant considering them as essentially opposites; only differences of speed and complexity separate perception from comprehension or even from invention.

These remarks lead us to examine a fourth objection to the theory of form. How is it possible to explain the mechanism of the reorganizations essential to the act of intelligence, and more precisely, to account for the discovery of "good forms" in contra-

distinction to those which are less good? When it is only a question of static perception (for example, perceiving a figure formed by scattered points on a white sheet of paper) and of a high mental level, one often finds that such a form is more satisfactory than that which immediately succeeds it. Thus, after having perceived the points as constituting a series of juxtaposed triangles one suddenly perceives a polygon. Thereafter one has the impression that forms succeed one another according to a "law of pregnance," the good forms, which end by prevailing, being those which fulfill certain *a priori* conditions of simplicity, cohesion and completion (those which are "closed," etc.). Thence the supposition that the act of comprehension consists in reorganizing the perceptual field by replacing the inadequate forms with more satisfactory ones and that in general the progress of intelligence is due to an internal maturation directed toward the best forms. But, in our hypothesis, the perceptions of completed structure constitute the coming to a head of complex elaborations, in which experience and intellectual activity intervene and cannot therefore be chosen as representative in the problem of the discovery of "good" forms. As soon as one goes beyond the particular case of static perceptions to analyze how the perceptions are structured once they are placed in intelligent activity in which they bathe as in their natural environment, one observes that the "good forms" do not arise all by themselves, but always as a function of previous searching which, far from being lost in simple maturation or use, constitutes real searching, that is to say, involving experimentation and control.

Groping, let us repeat, appears in the theory of form as an extraintelligent activity, destined to replace by the empiricism of fortuitous discoveries the reorganizations too difficult to accomplish systematically. If we have often recognized the existence of disordered gropings, corresponding in part to this conception and proceeding from the fact that the problem involved was too far above the subject's level, we have constantly emphasized, on the other hand, the existence of another type of groping which is directed and manifests this activity whose completed structures constitute the result. This second groping would therefore be

the very expression of the reorganization in progress and of the dynamism of which the schemata are the static product.

In effect, if at all stages, the schemata seemed to us to emanate from assimilatory activity, the latter was always revealed as a functional exercise before resulting in the various structures. As early as the first stage it is apparent that a certain use is necessary to make the reflex mechanisms function normally, this use of course allowing an element of groping. During the second and third stages, the primary and secondary reactions result from a reproductive assimilation, whose gropings are therefore necessary to the formation of the schemata. The same is true of the coordinations belonging to the fourth stage. With regard to the behavior patterns of the fifth stage, they reveal still better than the preceding ones the connection that exists between groping and the organization of the schemata. Far from manifesting itself as a passive recording of fortuitous events, the search characteristic of this type of behavior is directed simultaneously by the schemata assigning a purpose to the action, by those which serve by turns as means and by those which attribute a meaning to the vicissitudes of experience. In other words, groping of the second type is primarily gradual accommodation of the schemata to the conditions of reality and to the exigencies of coördination. Whether it is external as during the fifth stage or becomes internal at the same time as the behavior patterns of the sixth stage, it thus presupposes a permanent process of active correction or control.

Now this question of control of schemata is fundamental. By very reason of its hypothesis of pregnance (or theory of best shape), the theory of form has been led to neglect almost entirely the role of correction. The good forms, it is true, are supposed to eliminate those that are less good, not only to the extent that the latter are not very satisfactory in themselves, but also to the extent that they are inadequate to the whole of the given situation. But the process of reorganization, although set in motion by a sort of global control, remains, in its intimate mechanism, independent of this very control. On the contrary, every reorganization of the schemata has always seemed to us to constitute a correction of earlier schemata, through progressive differentiation,

and every organization in the process of becoming has manifested itself to us as a balance between the assimilatory tendency and the exigencies of accommodation, hence as controlled use.

Thus as early as the first stage, reflex use is corrected by its very effects; it is reinforced or inhibited according to circumstances. During the second and third stages, the formation of the circular reactions presupposes a development of this control; in order to rediscover the interesting results obtained by chance, one must correct the searching according to its success or its failures. The coördination of the schemata, characteristic of the fourth stage, also only works by being sanctioned by its results. From the fifth stage on, the control operations become differentiated still more; the child no longer limits himself to suffering an automatic sanction from facts, he tries to foresee, by a beginning experimentation, the object's reactions and so submits his search for novelty to a sort of active control. Finally, during the sixth stage, the control is internalized in the form of mental correction of the schemata and their combinations. It can therefore be said that control exists from the very beginning and becomes more and more established in the course of the stages of sensorimotor development. To be sure, it always remains empirical, in the sense that it is always the success or failure of the action which constitutes the sole criterion, the search for truth as such only beginning with reflective intelligence. But control suffices to insure an increasingly active correction of the schemata and so to explain how the good forms take the place of the less satisfactory ones through a gradual accommodation of the structures to experience and to each other.

We have hitherto taken up four main divergences between the hypothesis of forms and that of schemata. A fifth difference results, it would seem, from the four preceding ones and even sums them up in a certain way. In a word, it can be said that the "forms" exist by themselves whereas the schemata are only systems of relationships whose developments always remain interdependent.

The various extensions of the theory have adequately demonstrated that the "Gestalten" are conceived as existing by themselves. To the writers who have confined themselves to the

analysis of the psychological fact of perception or intellection, the forms are, it is true, simply given in the same capacity as any relationships whatever and the very concept of "form" thus involves no realism. But, to the extent that one refuses to try to explain the genesis of these forms, they tend to become entities of which perception or intellection partake (in the Platonic way). Then one passed from this phenomenological "subsistence" to the hypothesis of their *a priori* character. One is thus tempted to account for their necessity by the innate psychobiological structure of the organism, which makes them definitely anterior to experience. Finally there comes a third step: the "forms" become the condition for all possible experience. This is how, on the plane of scientific thought, Köhler has described "physical forms" as though they conditioned the phenomena of the external world and imposed themselves on the electromagnetic, chemical or physiological systems.

Now, if the foregoing reservations are taken into account, nothing warrants our believing in the existence of "structures" in themselves. As far as their external existence is concerned, in the first place, it is self-evident that, to the extent that phenomena are structurable in conformity to the framework of our mind, it can be explained by an assimilation of the real to the forms of intelligence as well as by the realistic hypothesis. With regard to the forms of intelligence they cannot be considered, either, as "subsisting" by themselves, inasmuch as they have a history and reveal an activity. In so far as they are mobile, the forms are therefore only good or bad relatively to each other and to the data to be systematized. Relativity, here as always, must temper a constantly renascent realism.

Undoubtedly such relativity presupposes the existence of some invariants. But the latter are functional and not structural. Thus a "form" is the better the more it satisfies the dual exigency of organization and adaptation of thought, organization consisting in an interdependence of given elements and adaptation consisting in a balance between assimilation and accommodation. But if this dual postulate excludes chaotic forms, the coherence it claims can no doubt be attained by means of an infinity of diverse structures. Thus the principle of contradiction does not

teach us whether two concepts contradict each other, and that two propositions can appear to be mutually compatible for a long time and subsequently be disclosed to be contradictory (the reverse also being possible).

§4. THE THEORY OF GROPING.—According to a famous hypothesis due to Jennings and taken up by Thorndike, an active method of adaptation to new circumstances exists—the method of groping: on the one hand, a succession of "trials" admitting, in principle, of "errors" as well as of fortuitous success, on the other hand a progressive selection operating after the event according to the success or failure of these same trials. The theory of "trials and errors" thus combines the *a priori* idea, according to which the solutions emanate from the subject's activity and the empirical idea according to which adoption of the right solution is definitely due to the pressure of the external environment. But, instead of acknowledging, as we shall do (§5), an indissoluble relation between subject and object, the hypothesis of trials and errors makes distinction between two terms: the production of trials which are due to the subject since they are fortuitous in relation to the object, and their selection, due to the object alone. Apriority and empiricism are here juxtaposed, in a way, and not outstripped. Such is the dual inspiration of the pragmatic system in epistemology and the mutational system in biology. Intellectual or vital activity remains independent in origin from the external environment, but the value of its products is determined by their success in the midst of the same environment.

In his well-known theory of intelligence[5] Claparède takes up Jennings' hypothesis, but by generalizing and inserting it in a concerted conception of adaptational acts. Intelligence, according to Claparède, constitutes a mental adaptation to new circumstances or, more precisely, "the capacity to resolve new problems by thought." Hence every complete act of intelligence presupposes three periods: the question (which orients searching), the hypothesis (or actual searching), and control. Moreover, intelli-

5 Claparède, La Psychologie de l'intelligence, *Scientia*, XI (1917), pp. 353–367.

gence would not be derived from the adaptations of a lower order, the reflex or hereditary adaptation and the habitual association or adaptation acquired in circumstances which repeat themselves, but it would arise at the time of the insufficiencies of the reflex and of habit. What happens when the novelty of the situation overflows the framework of instinct or of acquired associations? The subject does not remain passive but, on the contrary, manifests the behavior pattern emphasized by Jennings: He gropes and abandons himself to a series of "trials and errors." That is, according to Claparède, the origin of intelligence. Before the "systematic intelligence" characterized by the internalization of the processes of search is elaborated, intelligence is revealed in an empirical form which prepares the higher forms and constitutes their practical or sensorimotor equivalent. Thus to the "question" corresponds the need aroused by the new situation in which the subject finds himself. To the "hypothesis" groping corresponds, the series of trials and errors being nothing but the successive suppositions assumed by action before being assumed by thought. Finally, to the "control" corresponds the selection of trials which results from the pressure of things before the awareness of relations enables thought to control itself by mental experience. Empirical intelligence would consequently be explained by groping and it would be the progressive internalization and systematization of these processes which would subsequently account for intelligence properly so called.

The generality of the phenomenon of groping in all the stages we have outlined can be cited in favor of such a solution. In the first place, the "correction" of schemata by progressive accommodations, which we have just emphasized in connection with the "Gestalt," constitutes a primary example of this groping. Now it is stated, on the one hand, that groping is internalized during the sixth stage in the form of a sort of experimental reflection or mental experience (as when Lucienne, in Observation 180, opens her mouth in front of the opening which must be enlarged in order to reach the contents of the match box), and, on the other hand, that before this internalization, the same groping is revealed externally during all the fifth stage during which it constitutes the basis for the "tertiary circular reactions"

and the "discovery of new means through active experimentation." Then it is easy to observe that this groping, which is so evident in the fifth stage, is itself prepared by a series of analogous processes discoverable as early as the first stage. Ever since reflex accommodation we have noted the groping of the newborn child seeking the nipple. Since the acquisition of the first habits, moreover, one finds the importance progressively augmenting with the formation of the secondary schemata and the later coordination of those schemata. In short, the history of groping is nothing but that of accommodation with its successive complications and, in this respect, it seems as though much truth must be ascribed to the theory which identifies intelligence with searching which proceeds by active groping.

But there are two ways of interpreting groping. Either one asserts that groping activity is directed, from the outset, by a comprehension related to the external situation and then groping is never pure, the role of chance becomes secondary, and this solution is identified with that of assimilation (groping being reduced to a progressive accommodation of the assimilatory schemata); or else one states that there exists a pure groping, that is to say, taking place by chance and with selection, after the event, of favorable steps. Now, it is in this second sense that groping was at first interpreted and it is this second interpretation that we are unable to accept.

It is true that certain facts seem to substantiate Jennings. It happens that groping really does develop by chance, that the right solutions are fortuitously discovered and become fixed by simple repetition before the subject has been able to understand their mechanism. Thus the child sometimes prematurely discovers the solutions which transcend his level of understanding, this discovery being only due to happy fortuitous circumstances and not to a directed search (proof of which is that the acquisitions are often lost and later give rise to an intelligent rediscovery). But that is because, as we have already said, two kinds of groping exist, or rather, two extreme terms between which a whole series of intermediates extends. The one emerges when a problem, while being at the subject's level, does not give rise to an immediate solution, but to a directed search; the other ap-

pears when the problem transcends the intellectual level or the subject's knowledge and then the search operates by chance. It is only to the second of these two situations that Jennings' schema applies, whereas the other interpretation applies to the first case. The whole question, therefore, is to ascertain which relationship unites these two kinds of groping: Are they independent or does the one derive from the other and which?

Now, in order to resolve this question, nothing is more instructive than to examine the evolution of Claparède's doctrine which, from 1917 to 1933, has been thoroughly examined and, influenced by excellently analyzed facts relating to the "genesis of the hypothesis,"[6] has resulted in an exact delimitation of the role of groping.

From the very beginning of his studies, Claparède has distinguished between the two kinds of groping which we have just reviewed:

> I had then established two kinds or degrees of groping: *nonsystematic* groping, which is purely fortuitous and the "trials" at which would be selected, chosen mechanically, as by a sieve, by external circumstances; and *systematic* groping, guided and controlled by thought, especially by the awareness of relationships. Nonsystematic groping would characterize what I call "empirical intelligence"; the other belongs to intelligence properly so called.[7]

But, between the study of 1917 and that of 1933, a reversal of meaning is observable with regard to the relationships of these two kinds of gropings. In 1917 nonsystematic groping was considered the primitive act of intelligence which is supposed to explain the development of systematic groping, through the progressive contact with experience to which it gives rise and the awareness of relationships which springs from it: "The act of intelligence consists essentially in groping which derives from the groping manifested by animals of the lowest order when they find themselves in a new situation."[8] On the contrary, in the study made in 1933, three innovations in reality result in reversing the

[6] Claparède, La Genèse de l'hypothèse, *Arch. de Psychol.*, XXIV, 1933, pp. 1–155.

[7] *Ibid.*, p. 149.

[8] *Ibid.*, p. 149.

order of that filiation: (1) The two kinds of groping are no longer conceived "as two entirely separate types, but as the two extremities of a chain which comprises all the intermediates";[9] (2) Nonsystematic groping itself is already relatively directed:

> No groping is altogether incoherent because its function always is to attain some end, to satisfy some need, it is always oriented in some direction. . . . In the lower forms of thought, this direction is still very vague, very general. But the higher the mental level of the seeker becomes, the more the awareness of relationships is strengthened and, consequently, the more specific becomes the direction in which the search for the problem's solution must take place. . . . Thus every new groping tightens a little the circle within which the next gropings will occur. . . . Groping, *at first guided by the awareness of relationships—relationships between certain acts to be performed and a certain end to be attained*[10]—is consequently the agent which permits the discovery of new relationships.[11]

(3) Finally, and most important, not only does nonsystematic groping presuppose, as has just been shown, awareness of certain relationships which direct it from the outset, but furthermore those elementary relationships themselves stem from a fundamental mechanism of adjustment to experience which Claparède wisely emphasizes in his 1933 article, and which like the logicians he calls "implication": "Implication is a process indispensable to our needs in adjusting. *Without it we would not know how to profit from experience.*"[12] Hence implication is a primitive phenomenon which does not result from repetition, as does association, but on the contrary, from the outset introduces a necessary link between the terms which are implied. Thus implication is rooted in organic life: "The organism seems to us, as early as its most reflex manifestations, to be a machine for implication."[13] It is also the source of the conditioned reflexes and the circular reactions. Moreover, from its inception it directs even nonsystematic groping. "To imply is to expect, and that is

[9] *Ibid.*, p. 149.
[10] Our italics.
[11] *Ibid.*, pp. 149–150.
[12] *Ibid.*, p. 104 (italicized in the text).
[13] *Ibid.*, p. 106.

to aim toward what one expects,"[14] to the extent that the expectation is not disappointed, groping is useless, but to the extent that it is disappointed, groping, oriented by expectation, is directed toward the goal by the implications which connect it with the need experienced.

Having said this, we should now like to show why the hypothesis of a pure groping should not be retained and how the corrections made by Claparède in his last interpretation not only completely square with what we have observed regarding the origins of intelligence in children but also seem to us to involve the theory of schemata and assimilation in general.

The hypothesis of pure groping conceived as being the point of departure of intelligence cannot be justified because, either this nonsystematic groping appears in the margin of directed groping and often even after it, or else it precedes directed groping, but then it is either without influence on the latter or else it is itself relatively directed and consequently already systematic.

In a general way, let us recall first of all that the difference between nonsystematic and directed groping is simply a matter of proportion and that the situations in which these two types of behavior are made manifest only differ from one another in degree and not in quality. Systematic groping is, in effect, characterized by the fact that the successive trials condition each other with cumulative effect, that, in the second place, they are illuminated by earlier schemata conferring a meaning on the fortuitous discoveries and, last, they are directed by the schemata assigning a purpose to the action and by those serving as initial means and whose groping attempts constitute the differentiations or gradual accommodations (see Obs. 148–174). Systematic groping is therefore triply or quadruply directed, according to whether the end and the initial means form a whole or are separate. On the contrary, in nonsystematic groping like that of Thorndike's cats, the successive trials are relatively independent of each other and are not directed by experience acquired earlier. In this sense groping is fortuitous and the discovery of the solution is due to chance. But, from the time when the gropings, even nonsystematic, are always oriented by the need experienced,

[14] *Ibid.*, p. 102.

hence by the schema assigning a purpose to the action (Thorndike himself recognizes that the selection is made among the attempts due to annoyance at failure)—it is apparent that earlier experience plays a role in spite of everything and that the system of schemata already elaborated is not unconnected with the subject's seemingly most disorganized behavior. The successive gropings are only relatively independent of each other and the results to which they lead, while largely fortuitous, only acquire meaning from hidden but active schemata which illuminate them. The difference between nonsystematic gropings and directed search is therefore only of degree and not of kind.

That being true, it is evident that very often, far from preceding the directed search, nonsystematic groping only appears in the margin of this search, or after it, and that, when it precedes it in appearance it is either already oriented by it or without influence on it. In effect, the relationship between the two extreme types of behavior is determined by the situations in which they are made manifest. There is directed groping whenever the problem is sufficiently adapted to the intellectual level and knowledge of the subject for the latter to seek the solution by means of an adjustment of his habitual schemata, whereas there is groping when the problem is too far above the subject's level and when a simple readjustment of the schemata does not suffice to solve it. Groping, consequently, is the more directed when the situation approximates the first kind, and the less systematic when it approaches the second.

There are two possible cases with regard to the sequence of the two kinds of groping. Concerning the first, the subject only adopts the method of pure groping by "trials and errors" after having exhausted the resources of directed searching. This mode of sequence is observable even in the adult. When an automobile breaks down, the intellectual, who is not a mechanic, begins by trying to utilize various bits of knowledge pertaining to the carburetor, the spark plugs, the ignition, etc. That constitutes a search directed by earlier schemata, hence a systematic groping. Then, having achieved nothing, he tries everything haphazardly, touches parts of whose function he is ignorant and thus succeeds in repairing his motor by a purely chance maneuver: this is non-

systematic groping. In such a case, it is clear that pure groping extends the directed search. It is the fact of having tried an increasing number of solutions that induces the subject to generalize that behavior and it is to the extent that he understands the data of the problem less and less that he goes from directed groping to nonsystematic groping. In this first case, groping is the most slack form, so to speak, of intellectual searching, and not the point of departure of the act of intelligence.

But there is a second case: that in which the problem is absolutely new to the subject and in which nonsystematic groping seems to appear before directed searching. For example, an animal seeking food can get involved by chance in a series of successive ways without being capable of perceiving the relationships, or a child, in order to attain an object half hidden by various obstacles, can manage to extricate it without understanding the relation "situated under or behind." But then, one of two things: either the role of chance is considerable in the successful trial and the nonsystematic gropings thus crowned with success remain unconnected with intelligence and do not engender by themselves and as such later directed searches, or else nonsystematic groping is already sufficiently directed so that one may attribute success to that direction and then it is this beginning system which explains the later systematic searching. In the example of the child who wants to grasp a half-hidden object, it is possible, clearly, that the subject may achieve his ends without knowing how; but in this case, the nonsystematic groping which led to this fortuitous result does not at all pave the way for the directed searching which will later enable the child to discover the relations "placed upon," "situated below," etc. Nonsystematic groping is, then, only a sporadic behavior pattern appearing in the margin of intelligence and prolonging the attitude of groping search common to all stages (reflex use, circular reaction, etc.). It is only the extreme limit of accommodation, when the latter is more controlled by assimilation. On the contrary, it is possible that the search for the half-hidden object, while not yet involving more than knowledge of the relation "situated below" and thus permitting a great deal of haphazard groping, may nevertheless be directed by certain general sche-

mata, such as those of pushing back the obstacle, of utilizing a mobile object to bring a distant objective to oneself (in the case of the toys hanging from the bassinet hood, etc.). In this case, nonsystematic groping certainly paves the way for later directed searching (which will permit the child really to understand the relation "situated below"); but this groping is already itself directed although in a vague and general way. The difference between these two possibilities is easily discerned by the fact that, in the first, the child's fortuitous discovery is not followed by any lasting utilization, whereas in the second, it gives rise to various uses (to circular reactions or acts of reproductive assimilation with gradual accommodation) and to a more or less continuous progress.

Thus it may be seen that, even when nonsystematic groping seems to appear before the directed search, it does not explain the latter but is already explained by it, since from the very beginning it allows a *minimum* of direction. Without rejecting the idea of groping, we do not consider it adequate, in itself alone, to explain the mechanism of intelligence. Now, this is precisely what Claparède, in his most recent study, has pointed out with great wisdom. Led to reject the hypothesis of pure groping, he has come to acknowledge that, if needs and awareness of an end to be attained orient even the most elementary gropings, that is because an elementary implication of the acts and interests constitutes the primary datum presupposed by the groping itself. We should now like to show how this implication necessarily comprises assimilation and the system of schemata.

As far as reflective intelligence is concerned, it is self-evident that implication presupposes a system of concepts and consequently the assimilatory activity of judgment. To say that A implies B (for example that the fact of being "right angle" implies that a triangle satisfies Pythagoras' theorem), is to affirm that one is in possession of a certain concept C (for example that of the "right angle triangle"), in which A and B are united by logical necessity or by definition. The implication is thus the result of the judgments which have engendered the concepts C, A and B, and the necessity of the implication results from previous assimilation operated by those judgments.

The same is true of sensorimotor intelligence, including its preparatory phases constituted by the acquisition of the first habitual associations (second stage). Claparède, who rightly considers implication to be the condition of experience ("without it, we could not profit from experience"), shows in some very suggestive pages that the conditioned reflex is a phenomenon of implication. In effect, he says, "B is implied in A when, A being given, the subject behaves toward it as he would behave toward B." "If the dog sees a pink-colored A presented at first with the meal B, this will provoke the salivary and gastric reaction set in motion by meal B. The dog reacts to A as if B were contained, were implied in A." "If there were simple association and not implication, the color pink would simply evoke in the dog's memory the recollection of the meal but without being followed by any reaction signifying that the color pink *is taken for* the meal, *functions like* the meal."[15] But how can it be explained why, according to this excellent description, the color "is taken for" the meal? Claparède emphasizes the fact that the necessity for such connections appears from the beginning: "Far from its being the repetition of a pair of elements which creates between them a link of implication, the implication already arises at the first meeting of the two elements of this pair. And experience only intervenes to break this rapport of implication where it is not legitimate." And again: "The necessity for a connection tends, accordingly, to appear at the very beginning. If the necessity did not exist at the very beginning, one does not see when it will ever appear, for habit is not necessity."[16] But the problem is only extended. How is it possible to explain this necessity which appears at the first meeting of two terms thitherto foreign to each other so that they immediately appear to the subject as implying each other?

In the same way Mr. Claparède interprets the classic analogy of perception and reasoning in the light of implication: "If the operation which constitutes perception is identical to that which forms the backbone of reasoning, *it is because this operation is an implication.* If we notice the sweet flavor in the col-

[15] *Ibid.*, pp. 105–106.
[16] *Ibid.*, p. 105.

ored spot which the orange forms to our eye, this is not solely by virtue of association but due to implication. It is because this sweet flavor is *implied* in the other characteristics of the orange. . . ."[17] But, here again, how can it be explained why the given qualities in the sensation immediately assume a deeper signification and invoke an ensemble of other qualities necessarily interconnected?

The only possible answer is that schemata exist (that is to say, the organized totalities whose internal elements are mutually implied) as well as a constitutive operation of these schemata and of their implications, which is *assimilation*. In effect, without this formative operation of implications which is the sensorimotor equivalent of judgment, anything at all would imply anything at all, at the mercy of the fortuitous comparisons of perception. Implication would be governed by William James' "law of coalescence" according to which the data perceived simultaneously form a totality as long as they have not been dissociated by experience. "The law of coalescence," Claparède also says, engenders implication on the plane of action and syncretism on the plane of representation."[18] But then one can ask oneself if the idea of implication still keeps its value and if the necessity of which the implying relationships admit is not illusory. Claparède's interpretation is much more profound when he connects implication with his "law of reproduction of the similar" and when he adds: "Implication is rooted in the being's motor strata. It might be said that life implies implication."[19] But then a connecting link between the motor organization and the implication is lacking, and this connecting link is assimilation. But, in effect, assimilation explains how the organism tends simultaneously to reproduce the actions which have been profitable to it (reproductive assimilation)—which suffices to constitute schemata, not only due to the repetition of external conditions, but also and primarily due to an active reproduction of earlier behavior patterns as function of those conditions—and to incorporate, in the schemata thus formed, data capable of

[17] *Ibid.*, p. 107.
[18] *Ibid.*, p. 105.
[19] *Ibid.*, pp. 104–105.

serving them as aliments (generalizing assimilation). But, consequently, assimilation explains how active reproduction engenders implication. On the one hand, in effect, in order to reproduce the interesting behavior patterns, the subject constantly assimilates to the schemata of these behavior patterns the familiar objects already utilized in similar circumstances, that is to say, he confers a signification upon them; in other words, he inserts them in a system of implications. So it is that the doll hanging from the bassinet hood implies to the baby the quality of being pulled or struck, shaken, etc., because every time he perceives it, it is assimilated to the schemata of pulling, etc. Moreover, the new objects are themselves assimilated, due to their apparent characteristics or their situation, to familiar schemata, whence new networks of significations and implications. Thus in Observation 136 the cigarette case examined by Jacqueline is successfully sucked, rubbed, shaken, etc. Reproductive (and recognitory) assimilation, on the one hand, and generalizing assimilation, on the other, are hence the source of the implication which could not be explained without them, and these implications, far from resulting from simple "coalescences," are from the outset directed and organized by the system of schemata.

In the conditioned reflex, to resume Claparède's examples, the pink A is implied in the meal A because, according to the author's terms, that color "is taken for" the meal. What does this mean, if not that the color is assimilated to the meal itself, or that it receives a signification as a function of this schema? Here as everywhere, the implication results from a previous assimilation. So also, in perception, the orange's sweet flavor is implied in the color perceived at first, because this color is immediately assimilated to a familiar schema. In short, without assimilation, this implying "necessity" which Claparède places "at the source" and which he rightly distinguishes from habit due to passive repetition (which is quite different from active reproduction), remains unexplainable and implication remains without organic basis. Inasmuch as implication is really rooted in the organism, which seems to us indisputable, this is because all sensorimotor activity develops by functioning (reproductive assimilation) and through generalizing assimilation utilizes the ob-

jects capable of serving it as aliments. Thereafter every external datum is perceived as function of the sensorimotor schemata and it is this incessant assimilation which confers on all things meanings permitting implications of every degree. Through that very fact it can be understood why all groping is always directed, however little: Groping proceeds necessarily by accommodation of earlier schemata and the latter become assimilated or tend to assimilate to themselves the objects on which the former operates.

So it is that, corrected by Claparède's remarks concerning the controlling role of need or the issue and concerning the priority of implication in relation to "trials and errors," the theory of groping joins that of assimilation and the schemata.

§5. THE THEORY OF ASSIMILATION.—Two conclusions seem to us to derive from the foregoing discussions. The first is that intelligence constitutes an organizing activity whose functioning extends that of the biological organization, while surpassing it due to the elaboration of new structures. The second is that, if the sequential structures due to intellectual activity differ among themselves qualitatively, they always obey the same functional laws. In this respect, sensorimotor intelligence can be compared to reflective or rational intelligence and this comparison clarifies the analysis of the two extreme terms.

Now, whatever the explanatory hypotheses between which the main biological theories oscillate, everyone acknowledges a certain number of elementary truths which are those of which we speak here: that the living body presents an organized structure, that is to say, constitutes a system of interdependent relationships; that it works to conserve its definite structure and, to do this, incorporates in it the chemical and energetic aliments taken from the environment; that, consequently, it always reacts to the actions of the environment according to the variations of that particular structure and in the last analysis tends to impose on the whole universe a form of equilibrium dependent on that organization. In effect, contrary to unorganized beings which are also in equilibrium with the universe but which do not assimilate the environment to themselves, it can be said that the living being assimilates to himself the whole universe, at the same time

that he accommodates himself to it, since all the movements of every kind which characterize his actions and reactions with respect to things are regulated in a cycle delineated by his own organization as well as by the nature of the external objects. It is therefore permissible to conceive assimilation in a general sense as being the incorporation of any external reality whatever to one part or another of the cycle of organization. In other words, everything that answers a need of the organism is material for assimilation, the need even being the expression of assimilatory activity as such; with regard to the pressures exerted by the environment without their answering any need, they cannot give rise to assimilation as long as the organism is not adapted to them but, as adaptation consists precisely in transforming constraints into needs, everything, in the last analysis, can lend itself to being assimilated. The functions of relationship, independently even from psychic life which proceeds from them, are thus doubly the sources of assimilation: On the one hand, they help the general assimilation of the organism, since their use is essential to life; but, on the other hand, each of their manifestations presupposes a particular assimilation since this use or exercise is always related to a series of external conditions which are peculiar to them.

Such is the context of previous organization in which psychological life originates. Now, and this is our whole hypothesis, it seems that the development of intelligence extends that kind of mechanism instead of being inconsistent with it. In the first place, as early as the reflex behavior patterns and the acquired behavior patterns grafted on them, one sees appear processes of incorporation of things to the subject's schemata. This search for the functional aliment necessary to the development of behavior and this exercise stimulating growth constitute the most elementary forms of psychological assimilation. In effect, this assimilation of things to the schemata's activity, although not yet experienced by the subject as an awareness of objects and though consequently not yet giving rise to objective judgments, nevertheless constitutes the first operations which, subsequently, will result in judgments properly so called: operations of reproduction, recognition and generalization. Those are the operations which,

already involved in reflex assimilation, engender the first acquired behavior patterns, consequently the first nonhereditary schemata, the schema resulting from the very act of reproductive and generalizing assimilation. Thus every realm of sensorimotor reflex organization is the scene of particular assimilations extending, on the functional plane, physicochemical assimilation. In the second place, these behavior patterns, inasmuch as they are grafted on hereditary tendencies, from the very beginning find themselves inserted in the general framework of the individual organization; that is to say, before any acquisition of consciousness, they enter into the functional totality which the organism constitutes. Thus they contribute immediately to insuring and maintaining that equilibrium between the universe and the body itself, an equilibrium which consists in an assimilation of the universe to the organism as much as in an accommodation of the latter to the former. From the psychological point of view, that means that the acquired schemata form, from the outset, not only a sum of organized elements, but also a global organization, a system of interdependent operations, at first virtually due to their biological roots, then actually due to the mechanism of the reciprocal assimilation of the presenting schemata.

In short, at its point of departure, intellectual organization merely extends biological organization. It does not only consist —as accepted by a reflexology entirely impregnated with empirical associationism—in an ensemble of responses mechanically determined by external stimuli and in a correlative ensemble of conductions connecting the new stimuli with old responses. On the contrary, it constitutes a real activity, based upon an appropriate structure and assimilating to the latter a growing number of external objects.

Now, just as sensorimotor assimilation of things to the subject's schemata extends biological assimilation of the environment to the organism, so also it presages the intellectual assimilation of objects to the mind, such as is proven to exist in the most evolved forms of rational thought. In effect, reason simultaneously manifests a formal organization of the ideas it utilizes and an adaptation of those ideas to reality—an organization and adaptation which are inseparable. Now, the adaptation of reason

to experience presupposes an incorporation of objects to the subject's organization as well as an accommodation of the latter to external circumstances. Translated into rational terminology, it can therefore be said that organization is formal coherence, accommodation is "experience" and assimilation the act of judgment inasmuch as it unites experimental contents to logical form.

Now these comparisons, which we have often emphasized, between the biological plane, the sensorimotor plane and the rational plane, make it possible to understand how assimilation, from the functional point of view, constitutes the primary fact whence analysis must proceed regardless of the real interdependence of the mechanisms. On each of the three planes, in effect, accommodation is only possible as function of assimilation, since the very formation of the schemata called upon to accommodate themselves is due to the assimilatory process. Concerning the relationships between the organization of those schemata and assimilation, one can say that the latter represents the dynamic process of which the former is the static expression.

On the biological plane one could, it is true, raise the objection that every assimilatory operation presupposes a previous organization. But what is an organized structure, if not a cycle of operations of which each one is necessary to the existence of the others? Assimilation is hence the very functioning of the system of which organization is the structural aspect.

On the rational plane, this primacy of assimilation is expressed by the primacy of judgment. To judge is not necessarily to identify, as has sometimes been said, but it is to assimilate; that is to say, to incorporate a new datum in an earlier schema, in an already elaborated schema of implications. Hence rational assimilation always presupposes a previous organization. But whence comes that organization? From assimilation itself, for every concept and every relationship demands a judgment in order to be formed. If the interdependence of judgments and concepts thus demonstrates that of assimilation and organization, at the same time it emphasizes the nature of that interdependence. Assimilatory judgment is the active element of the process of which the organizing concept is the result.

Finally, on the sensorimotor plane which is that of elementary intellectual life, we have always emphasized the assimilatory mechanism which gives rise to the schemata and to their organization. Psychological assimilation in its simplest form is nothing other than the tendency of every behavior pattern or of every psychic state to conserve itself and, toward this end, to take its functional alimentation from the external environment. It is this reproductive assimilation that constitutes the schemata, the latter acquiring their existence as soon as a behavior pattern, however small its complexity, gives rise to an attempt at spontaneous repetition and thus becomes schematized. Now, this reproduction which, by itself and to the extent that it is not encased in an earlier schematism does not involve any organization, necessarily leads to the formation of an organized whole. In effect, the successive repetitions due to reproductive assimilation at first bring with them an extension of assimilation in the form of recognitory and generalizing operations. To the extent that the new objective resembles the old one, there is recognition and, to the extent that it differs from it, there is generalization of the schema and accommodation. The very repetition of the operation hence entails the formation of an organized totality, the organization resulting merely from the continuous application of an assimilatory schema to a given diversity.

In short, in every realm, assimilatory activity appears simultaneously as the resultant and the source of organization; that is to say, from the psychological point of view which is necessarily functional and dynamic, it constitutes a veritable primary act. Now, if we have shown, stage after stage, how the progress of the assimilatory mechanism engenders the various intellectual operations, it remains to be explained, more synthetically, how the initial act of assimilation takes into account the essential characteristics of intelligence, or the combined workings of mental construction leading to deduction and to actual or representative experience.

The principal problem to be resolved for an interpretation based upon assimilation as well as for every theory of intelligence entailing the biological activity of the subject himself is, it seems to us, the following: If the same process of assimilation of the

universe to the organism occurs from the physiological to the rational plane, how is it possible to explain why the subject comes to understand external reality sufficiently to be "objective" and to place himself in it? Physiological assimilation is, in effect, entirely centered on the organism. It is an incorporation of the environment to the living body, and the centripetal character of this process is so advanced that the incorporated elements lose their specific nature to be transformed into substances identical to those of the body itself. Rational assimilation, on the contrary, as revealed in judgment, does not at all destroy the object incorporated in the subject since, by manifesting the latter's activity, it subordinates it to the reality of the former. The antagonism of these two extreme terms is such that one would refuse to attribute them to the same mechanism if sensorimotor assimilation did not bridge the gap between them. At its source, in effect, sensorimotor assimilation is as egocentric as physiological assimilation, since it only uses the object to aliment the functioning of the subject's operations, whereas at its end the same assimilatory impetus succeeds in inserting the real in frameworks exactly adapted to its objective characteristics, so that the frameworks are ready to be transported to the plane of language in the form of concepts and logical relationships. How can this transition from egocentric incorporation to objective adaptation be explained, a transition without which the comparison of biological assimilation with intellectual assimilation would only be a play on words?

An easy solution would consist in attributing this evolution to the progress of accommodation alone. One recalls that accommodation, at first reduced to a simple global adjustment, gives rise, at the time of the coördination of the secondary schemata and above all of the tertiary circular reactions, to directed groupings and increasingly accurate experimental behavior patterns. Would it not suffice, then, to explain the transition from distorted assimilation to objective assimilation, to invoke the concomitant factor of accommodation?

It is undoubtedly the progress of accommodation which marks the increasing objectivity of the schemata of assimilation. But to be satisfied with such an explanation would be tantamount

either to answering the question by the question itself, or else to saying that the assimilation of things to the subject loses in importance in proportion as intelligence develops. In reality, at each step assimilation conserves the same essential role and the true problem, which is to find out how the progress of accommodation is possible, can only be resolved by having recourse once again to analysis of the assimilatory mechanism.

In effect, why is the accommodation of the schemata to the external environment, which becomes so accurate in the course of development, not given at the outset? Why does the evolution of sensorimotor intelligence appear as a progressive extraversion instead of the elementary operations being turned toward the external environment from the very beginning? In reality this gradual externalization, which seems at first to be the essential characteristic of the sequence of our six stages, only constitutes but one of the two aspects of that evolution. The second movement, exactly complementary and necessary to the explanation of the first, is none other than the process of growing coördination marking the progress of assimilation as such. Whereas the initial schemata are only interconnected due to their reflex and organic substructure, the more evolved schemata, at first primary, then secondary and tertiary, become organized little by little into coherent systems due to a process of mutual assimilation which we have often emphasized and which we have compared to the increasing implication of concepts and relationships. Now, not only is this progress of assimilation correlative to that of accommodation, but also it makes possible the gradual objectification of intelligence itself.

In effect, it is the nature of a schema of assimilation to apply itself to everything and to conquer the whole universe of perception. But by being thus generalized it is necessary for it to become differentiated. This differentiation does not only result from the diversity of the objects to which the schema must accommodate itself. Such an explanation would lead us back to the solution already rejected which is too simple because nothing forces the child to take into consideration the multiplicity of reality so long as his assimilation is distorted, that is to say, so long as he utilizes objects as simple functional aliments. The dif-

ferentiation of schemata works to the extent that objects are assimilated by several schemata at once and that their diversity thus becomes sufficiently worthy of interest to be imposed on accommodation (for example, visual images are differentiated by prehension, sucking, hearing, etc.). Undoubtedly, even without being coördinated with other schemata, each of them gives rise to spontaneous differentiations, but they remain of small importance and it is the infinite variety of combinations possible among schemata which is the big factor in differentiation. This explains how the progress of accommodation is correlative to that of assimilation: It is to the extent that the coördination of schemata leads the subject to interest himself in the diversity of reality that accommodation differentiates the schemata, and not by virtue of an immediate inclination toward accommodation.

Now this coördination and differentiation of schemata suffice to account for the growing objectification of assimilation without need to break the unity of this process to explain the transition from the egocentric incorporation at the beginnings to judgment properly so called. For example, let us compare the attitude of the baby toward an object he swings or a body he throws to the ground to that which is presupposed by the judgments "this is a hanging object" or "bodies fall." These judgments are certainly more "objective" than the active corresponding attitudes in the sense that the latter are confined to assimilating perceived data to a practical activity of the subject, whereas the formulated propositions insert them, no longer into a single, elementary schema, but into a complex pattern of schemata and relationships. The definitions of the hanging object or the fall of bodies presuppose an elaboration of things in hierarchical classes joined by multiple relationships, the schemata and relationships comprising, from far or near, all the subjects present and past experience. But, apart from this difference in complexity, hence in differentiation and coördination of schemata (without, of course, speaking of their symbolic translation to the plane of language and the regrouping that this verbal construction and socialization presuppose), these judgments do nothing other than incorporate perceived qualities into a system of schemata definitively resting upon the subject's action. There would be no diffi-

culty in showing that the hierarchical classes and relationships implied by these judgments in the last analysis apply to the sensorimotor schemata subjacent to any active elaboration. Thus the qualities of position, form, movement, etc., perceived in the hanging object or the falling bodies are in themselves neither more nor less objective than the more global qualities enabling the baby to recognize the object to be swung or the object to be thrown. It is coördination itself; that is to say, the multiple assimilation constructing an increasing number of relationships between the compounds "action X object" which explains the objectification. This is what we shall see in detail in Volume II, when studying the construction of the object and the objectification of space, of causality and time during the first two years of childhood.

Hence it is one and the same process of assimilation which leads the subject who is in process of incorporating the universe in himself to structure that universe according to the variations of its own organization and finally to place his activity among the things themselves. But if this gradual reversal of the meaning of assimilation is not due to experience alone, the role of accommodation to experience is no less necessary to it, and it is fitting to recall it now. The current theories tend either to overestimate the role of experience, as does neoassociationist empiricism, or to underestimate it, as does the psychology of form. In reality, as we have just seen, accommodation of the schemata to experience develops to the very extent of the progress of assimilation. In other words, the relationships between the subject and his environment consist in a radical interaction, so that awareness begins neither by the awareness of objects nor awareness of the activity itself, but by an undifferentiated state and from this state proceed two complementary movements, the one of incorporation of things to the subject, the other of accommodation to the things themselves.

But in what does the subject's share consist and how to discern the object's influence? In the beginning the distinction remains illusory: the object as functional aliment and the activity itself are radically confused. On the other hand, inasmuch as accommodation is differentiated from assimilation, one can say

that the subject's role is essentially established in the elaboration of forms whereas it is up to experience to provide them with a content. But, as we have already observed, form cannot be dissociated from matter. The structures are not preformed within the subject but are constructed gradually as needs and situations occur. Consequently they depend partly on experience. Inversely, experience alone does not account for the differentiation of schemata since through their very coördinations the schemata are capable of multiplications. Assimilation is therefore not reduced to a simple identification, but is the construction of structures at the same time as the incorporation of things to these structures. In short, the dualism of subject and object is brought back to a simple progressive differentiation between a centripetal pole and a centrifugal pole in the midst of the constant interactions of organism and environment. Also, experience is never simply passive receptiveness: it is active accommodation, correlative to assimilation.

This interaction of accommodation to experience and of organizing assimilation makes it possible, it would seem, to supply an answer to the crucial question of every theory of intelligence: How can the union of the fecundity of intellectual construction with its progressive rigor be explained? It must not be forgotten that if, in the order of the sciences, psychology stems from the biological disciplines, it is nevertheless on the former that the formidable task devolves of explaining the principles of mathematics—for, given the interdependence of subject and object, the sciences themselves constitute a circle and, if the physicochemical sciences which furnish biology with their principles are based on the mathematical sciences, the latter, in turn, stem from the activity of the subject and are based on psychology and, consequently, on biology. So it is that geometers resort to psychological data in order to explain the formation of space and of solid objects and that we shall see, in Volume II, how the laws of sensorimotor intelligence account for the beginning of "groups of displacements" and for the permanence of the object. It is therefore necessary, in every theory of intelligence, to think of the generality of problems that it raises, which was understood by

Wertheimer, for example, when he tried to apply Gestalt theory to the question of the syllogism.

As far as the fecundity of reasoning is concerned, one can conceive of the act of intellectual construction in a great number of ways varying from the discovery of a ready-made external reality (empiricism) to the making explicit of a preformed internal structure (theory of form). But, in the first case, if the work of intelligence leads to indefinitely fruitful results, since the mind is called upon to discover little by little a universe which is already completely structured and constructed, this work does not admit of any internal principle of construction and consequently of any principle of deductive rigor. In the second case, on the contrary, it is from the subject as such that intellectual progress stems, but if the internal maturation of the structures is capable of explaining their progressive coherence, this is at the expense of fecundity, because what reason have we to think that forms, however numerous they may be, sprung only from the subject's structure without his experience playing any part, will suffice to embrace reality *in toto*? Now, inasmuch as one acknowledges the necessary interdependence of accommodation to experience and of assimilation to activity itself, fecundity becomes correlative to coherence. In effect, all the intermediates are then made manifest between simple empirical discovery which results from the purely fortuitous insertion of a new datum in a schema, and the internal combination of schemata resulting in mental combination. In the most empirical discovery (like that which results from the tertiary circular reactions), an element of assimilation already intervenes which, in the case of active repetition and of the intellectual need for conservation, presages the judgment of identity, just as in the most refined internal combination (such as mathematical constructions) a factor again intervenes to which thought must accommodate itself. Consequently, there is naturally no antithesis between discovery and invention (no more than between induction and deduction), both simultaneously revealing activity of the mind and contact with reality.

Thereafter will it be said that the assimilatory organization does not show any fecundity in itself and is limited to work of identification, novelty always resulting from assimilated external

reality? But the interaction of subject and object is precisely such, given the interdependence of assimilation and accommodation, that it is impossible to conceive of one of the terms without the other. In other words, intelligence is the construction of relationships and not only identification. The elaboration of schemata involves a logic of relationships as well as a logic of classes. Consequently, intellectual organization is fecund in itself, since the relationships engender each other, and this fecundity is one with the richness of reality, since relationships cannot be conceived independently of the terms which connect them, any more than the reverse.

With regard to the rigor or the coherence thus obtained, it is in direct proportion to the fecundity, to the extent that the coordination of schemata equals their differentiation. Now, as it is precisely this growing coördination which permits accommodation to the diversity of reality, and as the coördination is obtained not only by identifying fusion but also by any system whatever of reciprocal relationships, there is certainly a correlation between the unity of the system of schemata and its richness. In effect, the rigor of the operations does not necessarily result from identification, but from their reciprocity in general. The reciprocal assimilation which accounts for the coördination of the schemata is therefore the point of departure of this reversibility of operations which, at all levels, appears as the criterion of rigor and coherence.

In short, the problem of invention, which in many respects constitutes the central problem of intelligence, does not, in the hypothesis of the schemata, require any special solution because the organization which assimilatory activity reveals is essentially construction and so is, in fact, invention, from the outset. That is why the sixth stage, or the stage of invention through mental combination, seemed to us the crowning of the five preceding ones and not the beginning of a new period. As early as the empirical intelligence of the fourth and fifth stages and even as early as the construction of primary and secondary schemata, this power of construction is in bud and is revealed in each operation.

In conclusion, assimilation and accommodation, at first antagonistic to the extent that the first remains egocentric and the

second is simply imposed by the external environment, complete each other to the extent they are differentiated, the coördination of the schemata of assimilation favoring the progress of accommodation and vice versa. So it is that, from the sensorimotor plane on, intelligence presupposes an increasingly close union of which the exactitude and the fecundity of reason will one day be the dual product.

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